

IMPERIAL

MATHEMATICAL METHODS FOR PHYSICISTS

IMPERIAL COLLEGE LONDON

DEPARTMENT OF PHYSICS

Comprehensive Lecture Summaries on Mathematical Methods for Physicists

Based on Lectures by Prof. Fay Dowker

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These summaries draw upon the lecturer's outstanding presentations and are further enhanced by my own insights and carefully curated materials from respected sources, including:

- Mathematical Methods for Physics and Engineering: A Comprehensive Guide by Reily, Hobson, & Bence
- Mathematical Methods for Physicists by Arfken, Weber, & Harris
- Mathematical Methods in the Physical Sciences by Mary L. Boas

I hope these compiled summaries prove beneficial and insightful for students who aspire to learn MMP.

– Yoogeun Song.

Ch. 1. Vector Spaces & Tensors

The 2 key properties of vectors on a vector space V are:

- (1) they can be added
- (2) they can be scaled (by a scalar either in $\mathbb{R}$ or $\mathbb{C}$)

If $\underline{u}, \underline{v} \in V$, then

(i) $\underline{u} + \underline{v} \in V$,

& (ii) $\lambda \underline{u} \in V$ for any $\lambda \in \mathbb{R}$

if V is a real vector space,

OR $\lambda \in \mathbb{C}$ if V is a complex vector space.

* Linear Transformation

$$\mathcal{O} : V \rightarrow W$$

$\mathcal{O}$ is a linear transformation if

$$\begin{aligned} \mathcal{O}(a\underline{u} + b\underline{v}) &\rightarrow \text{Some vector in } W \\ &= a\mathcal{O}(\underline{u}) + b\mathcal{O}(\underline{v}) \end{aligned}$$

These are everywhere in physics.

§1.4. Vector Spaces

In this chap., only finite dimensions, e.g.,

$$3D \text{ space} \leftrightarrow \mathbb{R}^3 = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \mid a, b, c \in \mathbb{R} \right\}$$

↑
dim=3

$\mathbb{R}^3$ is a real vector space

(EX) * 2-dim. complex space $\mathbb{C}^2 = \left\{ \begin{pmatrix} \mu \\ \nu \end{pmatrix} \mid \mu, \nu \in \mathbb{C} \right\}$

§1.4.2. Definition, (Linear Independence)

A set of n vectors $\{\underline{u}_1, \dots, \underline{u}_n\}$ is linearly

independent if $\sum_{i=1}^n a_i \underline{u}_i = 0 \Rightarrow a_i = 0$
for $i=1, \dots, n$.

$$\underline{u}_1 = \sum_{i=2}^n b_i \underline{u}_i \rightarrow \sum_{i=1}^n b'_i \underline{u}_i = 0 \quad (2)$$

$$\text{where } b_1 = 1, \quad b'_i = -b_i \quad (i = 2, \dots, n)$$

So, if $\{\underline{u}_1, \dots, \underline{u}_n\}$ is linearly independent, no $\underline{u}_i$ equals a linear combination of the others.

The dimension of vector space V is the Maximum number of vectors in a linearly independent set.

DEF. 1.4.3. A subspace W of vector space V is a subset of V , which is itself a vector space.
($W \subseteq V$)

$$(EX1) \quad \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \right\} \subset \mathbb{R}^3 \rightarrow \text{is a subset}$$

$\hookrightarrow$ is NOT a vector subspace of $\mathbb{R}^3$.

$$(EX2) \quad \left\{ \begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} \mid a \in \mathbb{R} \right\} \text{ is a vector subspace of } \mathbb{R}^3.$$

$$\begin{pmatrix} a \\ 0 \\ 0 \end{pmatrix} \in \mathbb{R}^3$$

Application: a special dir. in $\mathbb{R}^3$ like Mag fld. $\vec{B}$.

$$(EX3) \quad \left\{ \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \lambda \mid \lambda \in \mathbb{R} \right\} \text{ is a vector subspace of } \mathbb{R}^3.$$

(HW) If a vec. sp. V has $\dim. n$, & a vector subspace W also has $\dim. n$,

prove that $W = V$.

(Proof.)

The dimension of a vec. sp. or subspace is defined as the number of vectors in any of its bases.

If V is a vec. sp. w/ $\dim. n$, then any basis of V contains n linearly independent vectors.

Similarly, if W is a subsp. of V & has $\dim. n$, then W has a basis of n linearly independent vectors.

• Check the basis of W .

Let $\{w_1, w_2, \dots, w_n\}$ be a basis of W .

By definition of a subspace, these basis vectors are also elements of V . Since W has $\dim. n$, the basis $\{w_1, w_2, \dots, w_n\}$ is a set of n linearly independent vectors.

• The $\dim.$ of V is also n , so any linearly indep. set of vectors in V can have at most n vectors.

Since the basis of W is a linearly independent set of n vectors & $W \subseteq V$, the basis of W must also form a basis for V .

(3)

If the basis of W is also a basis of V , it spans V . $\therefore W = V$.

§1.4.3. Basis Vectors

Let $\dim(V)$ be n . Then, any linearly indep. set

$\{\underline{u}_1, \dots, \underline{u}_n\}$ is a basis for V ,

i.e., any vector $\underline{v} \in V$ can be written as a

unique linear combination of the basis vectors:

$$\underline{v} = \sum_{i=1}^n \lambda_i \underline{u}_i \quad \text{for} \quad \begin{cases} \lambda_i \in \mathbb{R} & (\text{real } V) \\ \lambda_i \in \mathbb{C} & (\text{complex } V) \end{cases}$$

§1.4.4. Inner Product

Real vector space V

$$\langle \underline{u}, \underline{v} \rangle \in \mathbb{R}$$

for all $\underline{u}, \underline{v} \in V$

Complex vector space V

$$\langle \underline{u}, \underline{v} \rangle \in \mathbb{C}$$

$\forall \underline{u}, \underline{v} \in V$

$$(1) \quad \langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle \quad \forall \underline{u}, \underline{v} \in V$$

$$\langle \underline{u}, \underline{v} \rangle = \langle \underline{v}, \underline{u} \rangle^* \quad \text{c.c.}$$

$$(2) \quad \langle \underline{u}, a_1 \underline{v}_1 + b_1 \underline{v}_2 \rangle = a_1 \langle \underline{u}, \underline{v}_1 \rangle + b_1 \langle \underline{u}, \underline{v}_2 \rangle$$

Same

Linearity

$$(3) \quad \|\underline{v}\|^2 \equiv \langle \underline{v}, \underline{v} \rangle$$

Norm (or size)

" $\equiv$ ": is defined to be

$$\|\underline{v}\|^2 \geq 0 \quad \forall \underline{v}$$

Same

$$(4) \quad \|\underline{v}\| = 0$$

$$\Rightarrow \underline{v} = \underline{0}$$

Same

N.B.: 1 & 2 $\Rightarrow$ Real Inner Product is linear in its 1st argument.

1 & 2 $\Rightarrow$ Complex I.P. is anti-linear in its 1st argument.

④

Complex vec. sp.

$$\langle a \underline{u}_1 + b \underline{u}_2, \underline{v} \rangle$$

$$= \langle a \underline{u}_1, \underline{v} \rangle + \langle b \underline{u}_2, \underline{v} \rangle$$

c.c.'ed

c.c.'ed

$$= a^* \langle \underline{u}_1, \underline{v} \rangle + b^* \langle \underline{u}_2, \underline{v} \rangle$$

Doing
Science = Collaboration

Learn from peers as much as you learn from the lecturer.

Practice explaining concepts in your brain in plain language. → Become a good communicator.
Beneficial to both the person who is explaining & the person who is listening.

The most that you will get from the course is from the discussions with each other.

Examples

In $\mathbb{R}^3$: Standard I.P. (or scalar product) :

$$\left\langle \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \right\rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 = \underline{a} \cdot \underline{b}$$

In $\mathbb{C}^2$: Standard I.P.:

$$\left\langle \begin{pmatrix} \lambda_1 \\ \lambda_2 \end{pmatrix}, \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \right\rangle = \lambda_1^* \mu_1 + \lambda_2^* \mu_2$$

§1.4.5.

DEF. 2 vectors $\underline{u}$ & $\underline{v}$ are orthogonal if

$$\langle \underline{u}, \underline{v} \rangle = 0.$$

DEF. a set of vectors $\{\underline{u}_1, \dots, \underline{u}_n\}$ is orthonormal if

$$\langle \underline{u}_i, \underline{u}_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} = \delta_{ij}$$

Kronecker's
delta symbol

§1.5. Matrices & Index Notation

An $m \times n$ matrix is an array of numbers ($\mathbb{R}$ or $\mathbb{C}$) w/ m rows & n columns:

$$\underline{\underline{A}} = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 5 & 3 \\ 7 & 9 \end{pmatrix} \end{matrix}$$

§1.5.3. A_{ij} is the entry in the i th row & j th column.

Let $\underline{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ & v_i are entries $v_1=1, v_2=2$.

$$\underline{\underline{A}} \underline{v} = \begin{pmatrix} 5 & 3 \\ 7 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5+6 \\ 7+18 \end{pmatrix} = \begin{pmatrix} 11 \\ 25 \end{pmatrix} \equiv \underline{u}$$

$$(\underline{\underline{A}} \underline{v})_i = \sum_{j=1}^2 A_{ij} v_j = A_{ij} v_j$$

Einstein summation convention & sum over j is implied.

Then, $u_i = A_{ij} v_j$.

Free index Dummy index

↓
takes each value in turn

$$\sum_{j=1}^2 A_{ij} v_j \equiv \sum_{k=1}^2 A_{ik} v_k$$

Dummy index

EX. $A_{ij} v_j \rightarrow$ wrong
bc Only 2 repeated dummy indices are allowed.

EX. $B_{ij} C_{jk} D_{ki} = \sum_{j=1}^2 \sum_{k=1}^2 B_{ij} C_{jk} D_{ki}$

Free indices must match.

EX. $A_{ij} v_j \neq u_k \rightarrow$ Wrong

$\begin{cases} A_{ij} v_j = u_i \\ A_{ki} v_i = u_k \end{cases} \rightarrow$ different ways to write the same set of eqns.

#1. Repeated indices are summed over & only 2 repeated indices are allowed.

#2. Free indices must match.

(3)

§1.5.4. Special $n \times n$ matrices

The most important matrix: Unit or identity matrix

$$\underline{\underline{I}} = \underline{\underline{1}} = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} \Rightarrow I_{ij} = \delta_{ij}$$

(EX)

$$\delta_{ij} = \delta_{ji}, \quad \delta_{ij} v_j = v_i \rightarrow \delta_{ij} A_{abcj} d = A_{abci} d$$

δ_{ij} "picks out" $j \rightarrow i$

Contracted!

$$\delta_{ii} = n$$

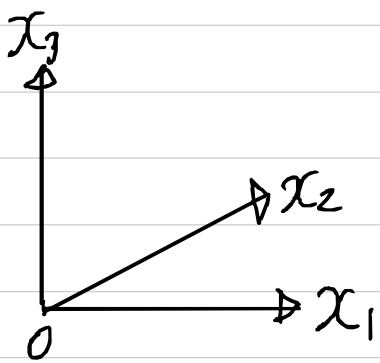
$$\sum_i \delta_{ii} = \underbrace{1 + \dots + 1}_n = n$$

For 3D Euclidean space & 3-component vectors:

$$\langle \underline{a}, \underline{b} \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3 = a_i b_i \delta_{ii} = a_i b_i$$

§1.6. Scalars, Vectors, & Tensors in 3D space.

All our indices run from 1, 2, 3.



• Scalar has magnitude only & no indices.

• Vector has magnitude & direction & has 1 index

• Rank-2 Tensor has 2 indices.

(EX) Conductivity

Electrical conductivity:
(σ_{ij})

$$\overset{\text{Current density}}{J_i} = \underset{\substack{\text{Elec. Conductivity} \\ \uparrow}}{\sigma_{ij}} \overset{\text{vec. (E.F.)}}{E_j}$$

If $\sigma_{ij} = \sigma \delta_{ij}$, then $J_i = \sigma E_i$.

Thermal Conductivity (κ_{ij})

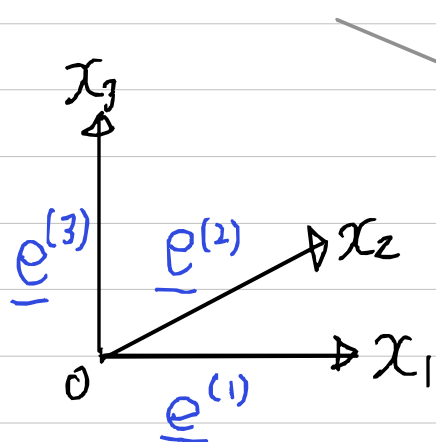
$$\underset{\substack{\text{Heat flow} \\ \text{vector}}}{h_i} = - \kappa_{ij} \underbrace{\partial_j T}_{\text{vector}}$$

T = temperature

Rank- n tensor : $T_{i_1 i_2 \dots i_n}$

§1.7. Transformations : Constant Rotations

Rotate axes of coordinates $\longleftrightarrow$ symmetries of 3D space



Rotation of basis of vectors

$$\{x_1, x_2, x_3\} \rightarrow \{x'_1, x'_2, x'_3\}$$

$$\{\underline{e}^{(1)}, \underline{e}^{(2)}, \underline{e}^{(3)}\} \rightarrow \{\underline{e}'^{(1)}, \underline{e}'^{(2)}, \underline{e}'^{(3)}\}$$

$x'_i = L_{ij} x_j \rightarrow L_{ij}$ is an orthogonal matrix in $SO(3)$.

$$L_{ij} L_{kj} = \delta_{ik}$$

$$L_{lm} L_{ln} = \delta_{mn}$$

Under a rotation,

Scalar: $\phi(x) = \phi'(x')$

Some pt. p

p has coord-s $\{x_i\}$

p also has coord-s $\{x'_i\}$.

Vector :

$$v'_i(x') = L_{ij} v_j(x)$$

Tensor : $\sigma'_{ij}(x') = L_{im} L_{jn} \sigma_{mn}(x)$

* PHYSICAL THINGS

Physical things do NOT change when we change a coordinate system.

Tensors $\rightarrow$ Express physics in a way that is covariant, i.e., the same in all coord. sys.

(EX) $(x_1, x_2, x_3) \rightarrow$ The value of the x_1 -coordinate is Non-physical. $\hookrightarrow$ dependent on the coord. sys.

In Cartesian coords, the rotation:

$$x_i \rightarrow x'_i = L_{ij} x_j$$

$$\left. \begin{aligned} L_{ij} L_{ik} &= \delta_{jk} \\ L_{ij} L_{jk} &= \delta_{ik} \end{aligned} \right\} \begin{array}{l} \text{a Rotation w/} \\ \text{Orthogonal Matrix } L \end{array}$$

• Transformation Rules:

Scalar (or rank-0 tensor) : $\phi'(x') = \phi(x)$

Vector (or rank-1 tensor) : $v'_i(x') = L_{ij} v_j(x)$

p
↓

Some pt. p has coordinates $\{x_i\}$ and $\{x'_i\}$.

$\{x'_i\}$ w/ new components along the new coordinate axes.

Rank-2 tensor : $\sigma'_{ij}(x') = L_{ik} L_{jl} \sigma_{kl}(x)$

Rank- n tensor : $T'_{i_1 i_2 \dots i_n}(x') = L_{i_1 j_1} L_{i_2 j_2} \dots L_{i_n j_n} T_{j_1 j_2 \dots j_n}(x)$

Fact: $\underline{e}^{(i)} \rightarrow \underline{e}'^{(i)} = L_{ij} \underline{e}^{(j)}$

Exercise 1. Prove that $\underline{e}^{(i)} = L_{ji} \underline{e}'^{(j)}$

$$\underline{e}^{(i)} = \sum_j L_{ji} \underline{e}'^{(j)}$$

the "old" basis vectors the "new" basis vectors.

$$\underline{L}^T \underline{L} = \underline{L} \underline{L}^T = \underline{I} = \begin{pmatrix} 1 & 0 \\ 0 & \ddots \\ 0 & \ddots & 1 \end{pmatrix}$$

• Dot product : $\underline{e}^{(i)} \cdot \underline{e}^{(k)} = \sum_j L_{ji} \underline{e}'^{(j)} \cdot \sum_l L_{lk} \underline{e}'^{(l)}$

$$= \sum_{j,l} L_{ji} L_{lk} \delta_{jl} = \sum_j L_{ji} L_{jk}$$

Orthonormality of the new basis : $\underline{e}'^{(i)} \cdot \underline{e}'^{(k)} = \delta_{ik}$

$$\underline{e}^{(i)} \cdot \underline{e}^{(k)} = \delta_{ik}$$

→ orthonormality is preserved.

Exercise 2. Prove:

$$\underline{v} = v_i(x) \underline{e}^{(i)} = v'_i(x') \underline{e}'^{(i)}$$

Using $\underline{e}^{(i)} = L_{ji} \underline{e}'^{(j)}$,

$$\begin{aligned} \underline{v} &= v_i(x) L_{ji} \underline{e}'^{(j)} = v'_j(x') \underline{e}'^{(j)} \\ &= v'_i(x') \underline{e}'^{(i)} \end{aligned}$$

$$\rightarrow \underline{v}'_i(x') = L_{ij} v_j(x)$$

(2)

Physics is all about "change".

It is all about how things vary in space,
& vary in time.

So, we use differential equations to describe the laws of physics. (e.g., EOMs)

§1.9. Tensor Calculus

Scalar, Vector, Tensor

$$\text{Gradient Op. : } \underline{\nabla} = \left(\frac{\partial}{\partial x^1}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3} \right) \\ = (\partial_1, \partial_2, \partial_3)$$

$$\left\{ \begin{array}{l} \nabla_i = \partial_i = \frac{\partial}{\partial x^i} \\ \nabla'_i = \partial'_i = \frac{\partial}{\partial x'^i} \end{array} \right\}, \quad \text{Laplacian} \quad \nabla^2 = \underline{\nabla} \cdot \underline{\nabla} \\ = \left(\frac{\partial}{\partial x^1} \right)^2 + \left(\frac{\partial}{\partial x^2} \right)^2 + \left(\frac{\partial}{\partial x^3} \right)^2$$

$$\nabla'_i = \frac{\partial}{\partial x'^i} = \frac{\partial x^j}{\partial x'^i} \frac{\partial}{\partial x^j} = L_{ij} \frac{\partial}{\partial x^j} = L_{ij} \nabla_j$$

$$x_i \longrightarrow x'_i = L_{ij} x_j \implies L_{ik} x'_i = L_{ik} L_{ij} x_j \\ \text{Let's hit it with } L_{ik} \implies = \delta_{kj} x_j \\ = x_k$$

$$\therefore x_j = L_{ij} x'_i$$

Exercise. Prove that $\nabla'^2 = \nabla^2$.

$$\nabla^2 = \delta^{ij} \nabla_i \nabla_j$$

$$\text{Using } \nabla'_i = L_{ij} \nabla_j$$

$$\nabla'^2 = \delta^{ij} \nabla'_i \nabla'_j = \delta^{ij} (L_{ik} \nabla_k) (L_{jl} \nabla_l) \\ = \delta^{ij} L_{ik} L_{jl} \nabla_k \nabla_l = \underbrace{L_{ik} L_{jl} \delta^{ij}}_{\delta_{kl}} \nabla_k \nabla_l$$

$$= \delta_{kl} \nabla_k \nabla_l = \nabla_l \nabla_l = \underline{\nabla} \cdot \underline{\nabla} = \nabla^2$$

$$\therefore \nabla'^2 = \nabla^2$$

§1.5.5. Epsilon Tensor (Levi-Civita Symbol)

(3)

$$\varepsilon_{ijk} = \begin{cases} 0 & , \text{ if 2 indices are equal} \\ +1 & , \text{ if } ijk \text{ is an even permutation of } 1, 2, 3. \\ -1 & , \text{ if } ijk \text{ is an odd permutation of } 1, 2, 3. \end{cases}$$

$$\varepsilon_{ijk} = -\varepsilon_{jik}$$

$$\varepsilon_{ijk} = \varepsilon_{jki} = \varepsilon_{kij}$$

$$\underline{c} = \underline{a} \times \underline{b} \quad \& \quad c_i = \varepsilon_{ijk} a_j b_k$$

$$(\nabla \varphi)_i = \nabla_i \varphi = \partial_i \varphi \quad \text{is a vector.}$$

Exercise. Prove $\partial_i \varphi$ is a vector.

Calculate $\partial'_i \varphi'(x')$ in terms of $\partial_j \varphi$.

Proof.

The scalar fn. φ satisfies $\varphi'(x') = \varphi(x)$.

$$\partial'_i \varphi'(x') = \frac{\partial \varphi'(x')}{\partial x'_i} = \frac{\partial x_j}{\partial x'_i} \frac{\partial \varphi'(x')}{\partial x_j}$$

$$= \frac{\partial x_j}{\partial x'_i} \underbrace{\frac{\partial \varphi(x)}{\partial x_j}}_{\partial_j \varphi} = \frac{\partial x_j}{\partial x'_i} \partial_j \varphi(x)$$

↓ Invert

$$\partial_j \varphi(x) = \underbrace{\frac{\partial x'_i}{\partial x_j}}_{\text{Jacobian}} \partial'_i \varphi'(x')$$

Since the gradient $\partial_j \varphi(x)$ transforms according to the rule:

$$\partial_j \varphi(x) \rightarrow \frac{\partial x'_i}{\partial x_j} \partial'_i \varphi'(x')$$

∴ $\partial_i \varphi(x)$ is a vector under coord. transformations.

□

(4)

Exercise. Prove: $\nabla \cdot \underline{F} = \partial_i F_i$ is a scalar.

$$F'^i(x') = \frac{\partial x'^i}{\partial x^k} F^k(x)$$

$$\partial'_i = \frac{\partial x^j}{\partial x'^i} \partial_j$$

$$\text{Div } \underline{F}' = \partial'_i F'^i = \left(\frac{\partial x^j}{\partial x'^i} \partial_j \right) \left(\frac{\partial x'^i}{\partial x^k} F^k(x) \right)$$

operator

$$= \underbrace{\frac{\partial x^j}{\partial x'^i} \frac{\partial x'^i}{\partial x^k}}_{= \delta^j_k} \partial_j F^k(x) + F^k \partial_j \underbrace{\left(\frac{\partial x^j}{\partial x'^i} \frac{\partial x'^i}{\partial x^k} \right)}_{= \partial_j \delta^j_k = 0}$$

$$= \partial_k F^k(x) = \partial_j F^j$$

$$\therefore \partial'_i F'^i = \partial_j F^j$$

$\partial_i F^i$ is invariant under coord. transformations.

$\therefore \partial_i F^i$ is a scalar.

(5)

Exercise.

$$(\nabla \times \underline{F})_i = \epsilon_{ijk} \partial_j F_k \text{ is a vector.}$$

Proof.

$$\partial'_j = \frac{\partial x^k}{\partial x'^j} \partial_k \quad \& \quad F'_k = \frac{\partial x^l}{\partial x'^k} F_l$$

$$\epsilon'_{ijk} = \det(L) L_{ip} L_{jq} L_{kr} \epsilon_{pqr}$$

$$L_{ij} = \frac{\partial x'^i}{\partial x^j} \text{ is the Jacobian of the transformation.}$$

$$\det L = \det \text{ of } L_{ij}.$$

$$(\nabla \times \underline{F}')_i = \epsilon'_{ijk} \partial'_j F'_k$$

$$= \det(L) L_{ip} L_{jq} L_{kr} \epsilon_{pqr} \underbrace{\left(\frac{\partial x^m}{\partial x'^j} \partial_m \right)}_{\text{chain rule}} \underbrace{\left(\frac{\partial x^n}{\partial x'^k} F_n \right)}_{\text{coord transformed}}$$

$$= \det(L) L_{ip} L_{jq} L_{kr} \epsilon_{pqr} \underbrace{\frac{\partial x^m}{\partial x'^j} \frac{\partial x^n}{\partial x'^k} \partial_m F_n}_{= \partial'_j F'_k \rightarrow \text{scalar so } \partial'_j F'_k = \partial_j F_k}$$

$$= \det(L) L_{ip} \epsilon_{pqr} \partial_j F_k L_{jq} L_{kr}$$

$$= L_{ip} \epsilon_{ijk} \partial_j F_k = L_{ip} (\nabla \times \underline{F})_i$$

↑
Plug back in ϵ_{ijk} .

$$= \frac{\partial x'^i}{\partial x^p} (\nabla \times \underline{F})_i$$

Exercise. Show: $A_i B_i$ is a scalar, i.e.,

$$A'_i(x') B'_i(x') = A_i(x) B_i(x)$$

$$A'_i = L_{ij} A_j \quad \& \quad B'_i = L_{ij} B_j$$

$$L_{ij} L_{ik} = \delta_{jk}$$

$$\underline{A'_i B'_i} = L_{ij} A_j L_{ik} B_k = L_{ij} L_{ik} A_j B_k = \delta_{jk} A_j B_k$$

$$= A_j B_j$$

$$= \underline{A_i B_i}$$

→ Invariant under coord-transformation.

i.e., a scalar.

Exercise.

6

Prove: (i) $\underline{\nabla} \cdot (\varphi \underline{u}) = \underline{u} \cdot \underline{\nabla} \varphi + \varphi (\underline{\nabla} \cdot \underline{u})$

(ii) $\underline{\nabla} \times (\underline{u} \times \underline{v}) = (\underline{\nabla} \cdot \underline{v}) \underline{u} - (\underline{\nabla} \cdot \underline{u}) \underline{v} + (\underline{v} \cdot \underline{\nabla}) \underline{u} - (\underline{u} \cdot \underline{\nabla}) \underline{v}$

$\left(\underline{\nabla} \times (\underline{u} \times \underline{v}) \right)_i = \left((\underline{\nabla} \cdot \underline{v}) \underline{u} - (\underline{\nabla} \cdot \underline{u}) \underline{v} + (\underline{v} \cdot \underline{\nabla}) \underline{u} - (\underline{u} \cdot \underline{\nabla}) \underline{v} \right)_i$

Use $\epsilon_{ijk} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$

Proof.

(i) $[\underline{\nabla} \cdot (\varphi \underline{u})]_i = \partial_i (\varphi u_i)$

$= u_i (\partial_i \varphi) + \varphi (\partial_i u_i) = [\underline{u} \cdot \underline{\nabla} \varphi]_i + [\varphi \underline{\nabla} \cdot \underline{u}]_i$

Product Rule

(ii) $[\underline{\nabla} \times (\underline{u} \times \underline{v})]_i = \epsilon_{ijk} \partial_j (\epsilon_{klm} u_l v_m)$

(using $[\underline{a} \times \underline{b}]_i = \epsilon_{ijk} a_j b_k$)

$[\underline{\nabla} \times (\underline{u} \times \underline{v})]_i = \epsilon_{ijk} \epsilon_{klm} \partial_j (u_l v_m)$

(since ϵ_{klm} is a const.)

$= (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) \partial_j (u_l v_m)$

$= \partial_j (u_i v_j) - \partial_j (u_j v_i)$

$= u_i \partial_j v_j + v_j \partial_j u_i - u_j \partial_j v_i - v_i \partial_j u_j$

(using product rule)

$= [\underline{u} (\underline{\nabla} \cdot \underline{v})]_i + [(\underline{v} \cdot \underline{\nabla}) \underline{u}]_i - [(\underline{u} \cdot \underline{\nabla}) \underline{v}]_i - [\underline{v} (\underline{\nabla} \cdot \underline{u})]_i$

$= [\underline{u} (\underline{\nabla} \cdot \underline{v}) - \underline{v} (\underline{\nabla} \cdot \underline{u}) + (\underline{v} \cdot \underline{\nabla}) \underline{u} - (\underline{u} \cdot \underline{\nabla}) \underline{v}]_i$

(7)

A coordinate system is arbitrary.

Exercise. Prove $J_i = \sigma_{ij} E_j \Rightarrow J'_i = \sigma'_{ij} E'_j$

Proof.

$$J'_i = L_{ik} J_k \quad \& \quad E'_j = L_{je} E_e$$

$$L_{ij} L_{ik} = \delta_{jk}$$

Plug in: $J_k = \sigma_{km} E_m$

$$\sigma'_{ij} = L_{ip} L_{jq} \sigma_{pq}$$

↓

$$J'_i = L_{ik} \sigma_{km} E_m$$

$$J'_i = L_{ik} \sigma_{km} L_{mj} E'_j$$

$$E_m = L_{mj} E'_j$$

$$= L_{mj} L_{je} E_e = \delta_{me} E_e$$

If we rewrite σ'_{ij} as

$$\sigma'_{ij} = L_{ik} L_{jm} \sigma_{km} = L_{ik} \sigma_{km} L_{mj},$$

then

$$J'_i = \sigma'_{ij} E'_j.$$

□

§1.8. The representation quadric.
(Recap)

Let T_{ij} be a symmetric crystal tensor.

$T_{ij} x_i x_j = 1$ is the eqn. of a 2D surface in 3D space.

EX.

if $T_{ij} = T \delta_{ij}$ where $T > 0$.

$$T(x_1^2 + x_2^2 + x_3^2) = 1 \rightarrow x_1^2 + x_2^2 + x_3^2 = \frac{1}{T}$$

Then,

the rep. quadric is a sphere.

$$= \left(\frac{1}{\sqrt{T}} \right)^2 = (\text{const.})^2$$

Any symmetric real tensor can be diagonalized by an orthogonal matrix. (e.g., T_{kl})

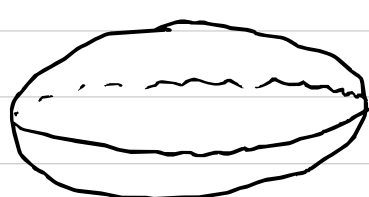
→ ∃ an orthogonal matrix L_{ij} s.t.

$$L_{ik} L_{je} T_{kl} = \text{diag}(\lambda_1, \lambda_2, \lambda_3) = \begin{pmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \lambda_3 \end{pmatrix}$$

Rotate $x \rightarrow x'_i = L_{ij} x_j$

$$\therefore T_{ij} x_i x_j = T'_{ij} x'_i x'_j = \lambda_1 (x'_1)^2 + \lambda_2 (x'_2)^2 + \lambda_3 (x'_3)^2$$

If $\lambda_i > 0$, → this is an ellipsoid.
($i=1,2,3$)



The x'_i axes are the principal axes.

If $\lambda_1 = \lambda_2 = \lambda_3$ → a sphere.



If $\lambda_1 \neq \lambda_2 \neq \lambda_3$ → a squashed sphere.

§ 1.10. Tensors for crystals

§ 1.10.1. There are many tensors that encode properties of crystals.

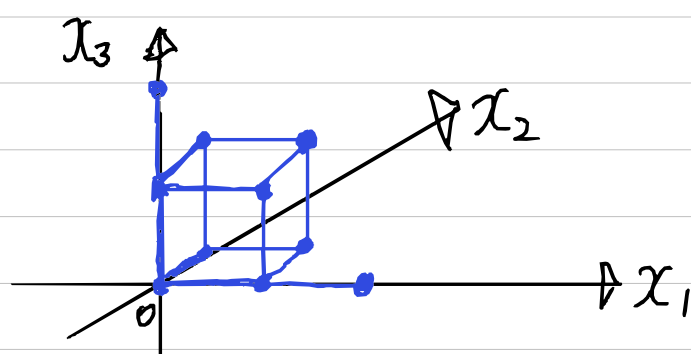
§ 1.10.2. Neumann's Principle.

Any physical property of a crystal must be symmetric under any symmetries of the crystal.

EX. Cubic Lattice

Any matter tensor

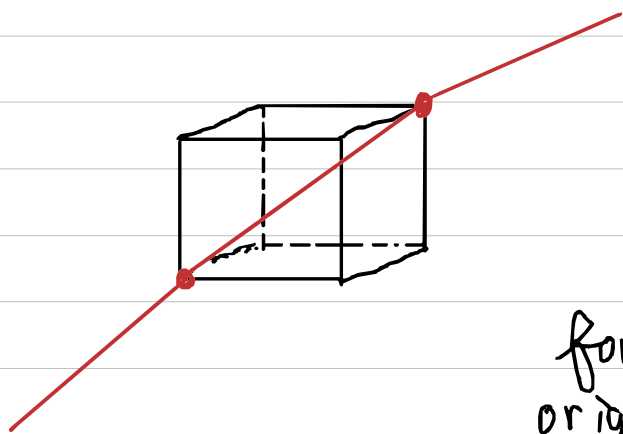
σ_{ij} must be symmetric



under any rotation that takes lattice to itself.

Also, any rep. quadric of a matter tensor σ_{ij}

must be symmetric under any rotation that takes lattice to itself.



Rotation by 120° around this "body diagonal" is a symmetry,

for each of the 4 cubes at the origin in the upper half of the lattice.

The only ellipsoid that is symmetric under all 4 of these rotations is a sphere. $\Rightarrow \sigma_{ij} = \sigma \delta_{ij}$

The matter tensor is isotropic.

Ch. 2. Green functions (or Green's fn.s)

The simplest case:

1D Differential Equations (1D ODE)

The overall arena in which Green fn.s arise is

linear differential eqn.s.

§ 2.1.3. Intro.

Green fn.s in 1D.

$$x \in [\alpha, \beta]$$



 α, β may be infinite. $y(x)$ satisfies ODE, 2nd-order, inhomogeneous.

$$\frac{d^2 y}{dx^2} + p(x) \frac{dy}{dx} + q(x) y = f(x) \quad \text{--- (*)}$$

$$\left[\frac{d^2}{dx^2} + p \frac{d}{dx} + q \right] y = f$$

 L_x $\longleftarrow$ Linear differential operator (2nd-order)

$$L_x (a y_1(x) + b y_2(x)) = a L_x y_1 + b L_x y_2$$

 a, b : numbers(*) is called homogeneous if $f(x) = 0$.Notation: $\frac{d}{dx} y(x) = y'(x)$

$$\frac{d^2}{dx^2} y(x) = y''(x)$$

Prime means derivative w.r.t. argument

 $g(k)$ is a fn.

$$g'(k) = \frac{d}{dk} g(k)$$

In case of time $y(t)$, then

$$\frac{d}{dt} y(t) = \dot{y}(t)$$

$$\frac{d^2}{dt^2} y(t) = \ddot{y}(t)$$

$$(*) \rightarrow y'' + p y' + q y = f$$

The gen. sol. to (*) : $y(x) = y_0(x) + a y_1(x) + b y_2(x)$ where (i) y_1 & y_2 are linearly indep. soln.s of homogeneous eqn. : $L_x y = 0$.(ii) a & b are numbers,(iii) $y_0(x)$ is called a particular integral of (*). $y_0(x)$ is any soln. of $L_x y = f(x)$.

Then, solve for a & b by imposing boundary conditions (B.C.s), there must be 2 of them.

N.B. : If $y_0(x)$ itself satisfies B.C.s of the problem, then $a=b=0$ & $y_0(x)$ is the soln you seek.

Homogeneous initial B.C.s :
$$\left. \begin{aligned} y(x) &= 0 \\ y'(x) &= 0 \end{aligned} \right\}$$

Inhomogeneous initial B.C.s :
$$\left. \begin{aligned} y(x) &= c_1 \\ y'(x) &= c_2 \end{aligned} \right\}$$

We can convert this into a problem w/ homog. B.C.s.

$Y(x) = y(x) - g(x)$ where $g(x)$ is any fn. that satisfies $g(x) = c_1$, $g'(x) = c_2$.

Then, $\mathcal{L}_x[Y] = F(x)$,

where $F(x) = f(x) - (g'' + pg' + qg)$.

$$\left\{ \begin{aligned} Y(x) &= 0 \\ Y'(x) &= 0 \end{aligned} \right\}$$

Homogenous fixed pt. (2-pt.) B.C.s :
$$\left\{ \begin{aligned} y(\alpha) &= 0 \\ y(\beta) &= 0 \end{aligned} \right\}$$

§1.4. Dirac Delta Fn.

The defining property of $\delta(x - \tilde{x})$ is

$$\int_{\alpha}^{\beta} \delta(x - \tilde{x}) f(\tilde{x}) d\tilde{x} = \begin{cases} f(x) & \text{if } x \in (\alpha, \beta) \\ 0 & \text{if } x \notin [\alpha, \beta] \end{cases}$$

$\tilde{x}$: integration variable

$x \in (\alpha, \beta)$ is equiv. to $\alpha < x < \beta$.

Round brackets

$$\alpha < x < \beta$$

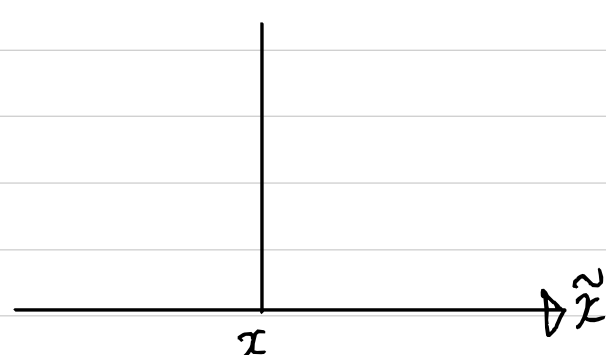
$$\begin{array}{ccc} \circ & + & \circ \\ \alpha & x & \beta \end{array}$$

This integral does NOT care what the fn. $f(\tilde{x})$ is:

$$\int_{\alpha}^{\beta} \delta(x - \tilde{x}) f(\tilde{x}) d\tilde{x}$$

Heuristic for $\delta(x - \tilde{x})$.

"infinite spike" at $\tilde{x} = x$.



As a fn. of $\tilde{x}$:

$$\delta(x - \tilde{x}) = \begin{cases} 0 & , \tilde{x} \neq x \\ \text{"}\infty\text{"} & , \tilde{x} = x \end{cases}$$

(3)

§1.4.1. Properties of δ

$$\bullet \delta(x) = \delta(-x) \implies \delta(x - \tilde{x}) = \delta(\tilde{x} - x)$$

$$\bullet \delta(x)^* = \delta(x) : \text{Real}$$

$$\bullet \int_{\alpha}^{\beta} \delta(x - \tilde{x}) d\tilde{x} = 1 \quad \text{if } x \in (\alpha, \beta),$$

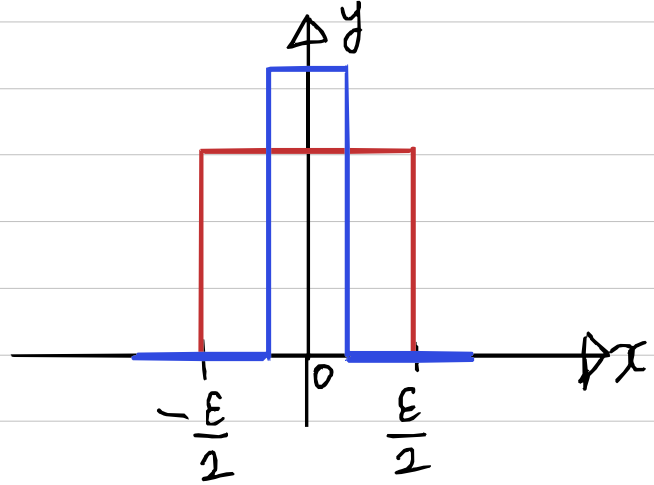
i.e., $\alpha < x < \beta$.

• Fourier Transform is const.:

$$\begin{aligned} \tilde{\delta}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} e^{-ikx} \Big|_{x=0} \\ &= \frac{1}{\sqrt{2\pi}} \end{aligned}$$

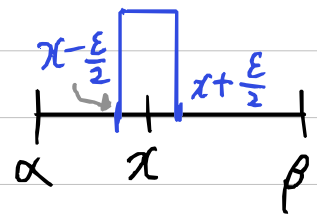
§1.4.2. δ as limit of "top hat" fn.s

$$H_{\epsilon}(x) = \begin{cases} \epsilon^{-1} & \text{if } |x| \leq \frac{\epsilon}{2} \\ 0 & \text{otherwise.} \end{cases}$$



$$\int_{-\infty}^{\infty} H_{\epsilon}(x) dx = 1$$

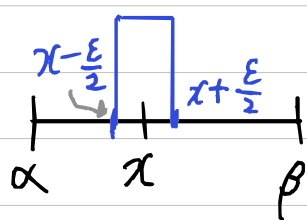
$$\delta(x) \equiv \lim_{\epsilon \rightarrow 0} H_{\epsilon}(x)$$



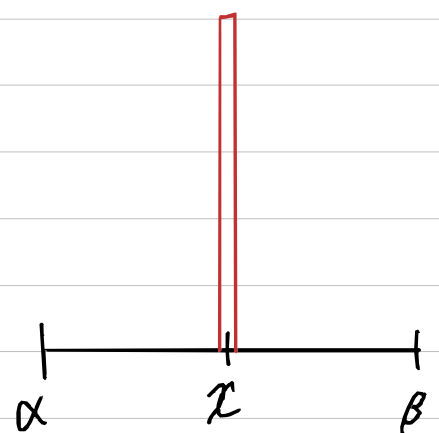
Consider: $\lim_{\epsilon \rightarrow 0} \int_{\alpha}^{\beta} H_{\epsilon}(x - \tilde{x}) f(\tilde{x}) d\tilde{x}$

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \int_{x-\frac{\epsilon}{2}}^{x+\frac{\epsilon}{2}} f(\tilde{x}) d\tilde{x}$$

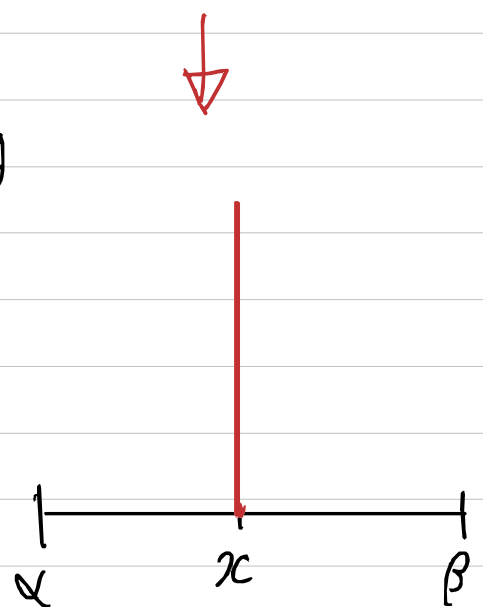
$\rightarrow 0$



$$\text{As } \epsilon \rightarrow 0$$



Eventually,
it can only
pick out
the value
of x .



Exercise.

Work out $I = \int_{\alpha}^{\beta} \delta(x - \tilde{x}) f(\tilde{x}) d\tilde{x}$ for $x = \alpha$.

Proof.

We analyze whether the delta fn's argument $x - \tilde{x} = \alpha - \tilde{x}$ satisfies the conditions within the integration bounds $\alpha \leq \tilde{x} \leq \beta$.

The delta fn. "activates" only when $\tilde{x} = \alpha$, provided that α lies within the range $[\alpha, \beta]$, which is true here.

$$I = \int_{\alpha}^{\beta} \delta(\alpha - \tilde{x}) f(\tilde{x}) d\tilde{x}$$

$\delta(\alpha - \tilde{x})$ picks out the value of $f(\tilde{x})$ at $\tilde{x} = \alpha$

$$\therefore I = \int_{\alpha}^{\beta} \delta(\alpha - \tilde{x}) f(\tilde{x}) d\tilde{x} \Big|_{\tilde{x}=\alpha} = f(\alpha)$$

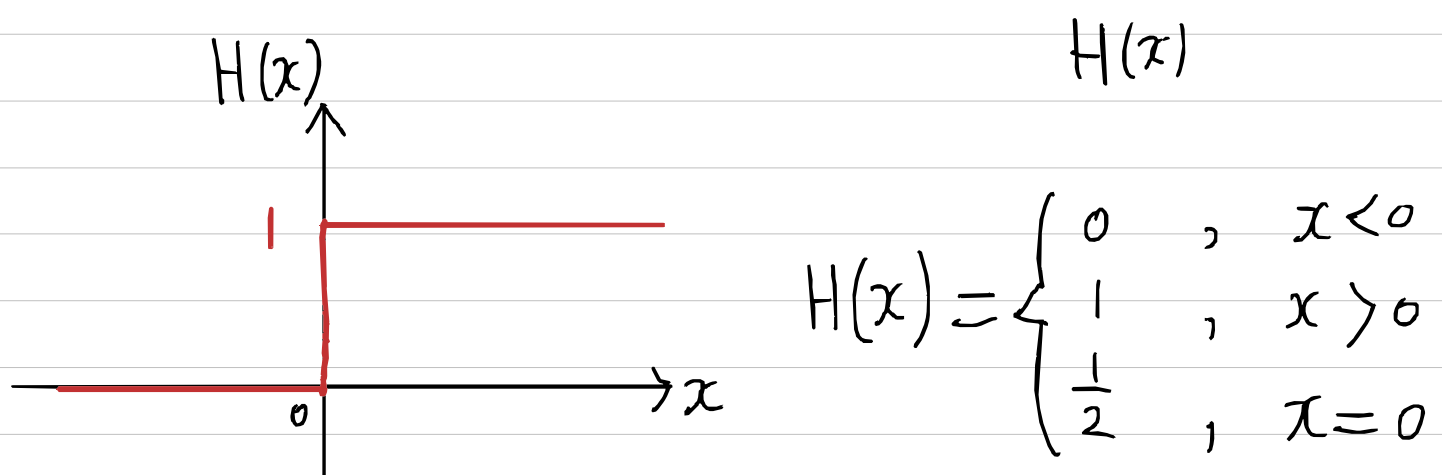
• Dirac δ -fn.

$$\int_{\alpha}^{\beta} \delta(x-\tilde{x}) f(\tilde{x}) d\tilde{x} = \begin{cases} f(x) & \text{for any well-behaved } f \\ & \text{if } x \in (\alpha, \beta) \\ 0 & \text{if } x \notin [\alpha, \beta] \end{cases}$$

• $\delta(x) = 0$ for $x \neq 0$

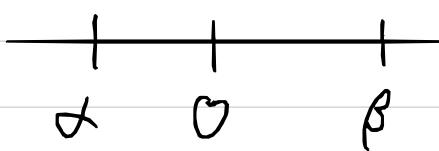
• $\delta(x) = \delta(-x)$

§ 2-4.5. Dirac delta fn. $\delta(x)$ as a derivative of the step fn. (or Heaviside-step fn.)



* Claim. $\frac{d}{dx} H(x) = \delta(x)$

Proof. $\int_{\alpha}^{\beta} H'(\tilde{x}) f(\tilde{x}) d\tilde{x} = 0$ if $0 \notin [\alpha, \beta]$.



Let $0 \in (\alpha, \beta)$.

$$I = \int_{\alpha}^{\beta} H'(\tilde{x}) f(\tilde{x}) d\tilde{x}$$

$$= [H(\tilde{x}) f(\tilde{x})]_{\alpha}^{\beta} - \int_{\alpha}^{\beta} H(\tilde{x}) f'(\tilde{x}) d\tilde{x}$$

IbP. $= f(\beta) - \int_0^{\beta} f'(\tilde{x}) d\tilde{x} = f(\beta) - [f(\tilde{x})]_0^{\beta}$

$$= f(\beta) - (f(\beta) - f(0)) = \underline{f(0)}.$$

□

Corollary. $\frac{\partial}{\partial x} H(x-\tilde{x}) = \delta(x-\tilde{x})$

If function jumps by $\begin{cases} +1 \\ -1 \\ 2 \end{cases}$ across $x=a$,

its derivative contains a δ -function $\begin{cases} +\delta(x-a) \\ -\delta(x-a) \\ 2\delta(x-a) \end{cases}$,

etc.

(2)

§2.5. Green Functions

$$\mathcal{L}_x = \frac{d^2}{dx^2} + p(x) \frac{d}{dx} + q(x)$$

$$\mathcal{L}_x y(x) = f(x) \quad \text{--- Eq. (*)}$$

Subject to B.C.s.

A Green's fn. $G(x, \tilde{x})$ satisfies

2 variables

$$\mathcal{L}_x G(x, \tilde{x}) = \delta(x - \tilde{x}) \quad \text{--- Eq. (+)}$$

defines a Green's fn. for $\mathcal{L}_x$.

There is NOT a unique soln.: If Y satisfies

$$\mathcal{L}_x Y = 0, \text{ then } G(x, \tilde{x}) + Y(x) \text{ satisfies (+)}$$

(+) and 2 B.C.s on $G(x, \tilde{x})$ has a unique soln.

$$\text{Let } y_0(x) = \int_{\alpha}^{\beta} G(x, \tilde{x}) f(\tilde{x}) d\tilde{x}. \quad \text{--- (**)}$$

Claim. $y_0(x)$ is a particular integral of (*).

Proof.

$$\begin{aligned} \mathcal{L}_x y_0 &= \mathcal{L}_x \left[\int_{\alpha}^{\beta} G(x, \tilde{x}) f(\tilde{x}) d\tilde{x} \right] \\ &= \int_{\alpha}^{\beta} \left[\mathcal{L}_x G(x, \tilde{x}) \right] f(\tilde{x}) d\tilde{x} = \int_{\alpha}^{\beta} \delta(x - \tilde{x}) f(\tilde{x}) d\tilde{x} \end{aligned}$$

$$\mathcal{L}_x y_0(x) = f(x)$$

□

$$\text{General Soln: } y(x) = y_0(x) + a y_1(x) + b y_2(x)$$

In many cases, you can find $G(x, \tilde{x})$ that already satisfies B.C.s & then $a=b=0$,

$$\text{Ex. (a) If } G(\alpha, \tilde{x}) = 0 \text{ \& } \frac{\partial G}{\partial x}(\alpha, \tilde{x}) = 0,$$

then

$$y_0(\alpha) = 0 \text{ \& } y_0'(\alpha) = 0.$$

Where $y_0 = (**)$. $\rightarrow$ Exercise — Check this.

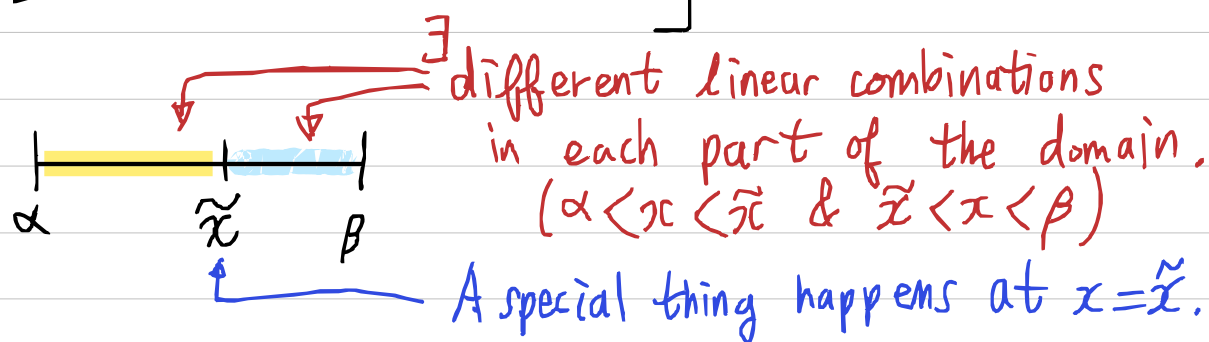
(b)

Prove the same for $G(\alpha, \tilde{x}) = 0$ & $G(\beta, \tilde{x}) = 0$.

$$\Rightarrow y_0(\alpha) = y_0(\beta) = 0.$$

(3)

$$(+)\quad \left[\frac{\partial^2}{\partial x^2} + p(x) \frac{\partial}{\partial x} + q(x) \right] G(x, \tilde{x}) = \delta(x - \tilde{x})$$



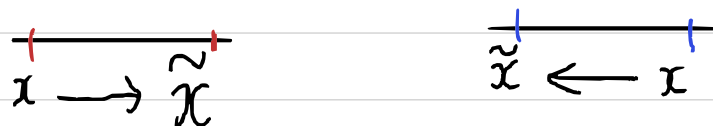
(i)
• When $x < \tilde{x}$ or $x > \tilde{x}$, $G(x, \tilde{x})$ is a soln. of the homogeneous eqn.

• 2nd-order derivative $\frac{\partial^2 G}{\partial x^2} \sim \frac{\partial \delta}{\partial x}$ do not exist.

→ Discontinuity (i.e., Jump Condition)

(ii)
• $G(x, \tilde{x})$ is continuous at $x = \tilde{x}$:

$$\lim_{x \rightarrow \tilde{x}^-} G(x, \tilde{x}) = \lim_{x \rightarrow \tilde{x}^+} G(x, \tilde{x}) = G(\tilde{x}, \tilde{x})$$



(iii)
• $\frac{\partial}{\partial x} G(x, \tilde{x})$ has a unit discontinuity (or discontinuity of +1) at $x = \tilde{x}$, i.e.,

$$\lim_{x \rightarrow \tilde{x}^+} \frac{\partial}{\partial x} G(x, \tilde{x}) = 1 + \lim_{x \rightarrow \tilde{x}^-} \frac{\partial}{\partial x} G(x, \tilde{x})$$

Equivalently,

$$\lim_{x \rightarrow \tilde{x}^+} \frac{\partial}{\partial x} G(x, \tilde{x}) - \lim_{x \rightarrow \tilde{x}^-} \frac{\partial}{\partial x} G(x, \tilde{x}) = 1$$

These 3 properties are equivalent to (+).

(4)

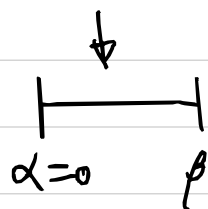
§2.5.2. $y(\alpha) = 0, y'(\alpha) = 0$

$$L_x y = f(x)$$

In some problems,

$$\alpha = 0.$$

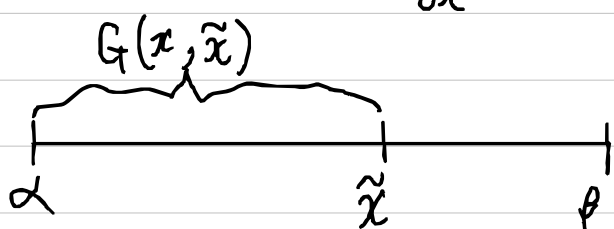
$$f(x) = \int_{\alpha}^{\beta} f(\tilde{x}) \delta(x - \tilde{x}) d\tilde{x}$$



$$y_0(x) = \int_{\alpha}^{\beta} f(\tilde{x}) G(x, \tilde{x}) d\tilde{x}$$

Seek $G(x, \tilde{x})$, assuming we have $y_1(x)$ & $y_2(x)$ linearly indep. solns of $L_x y = 0$.

$$\& G(\alpha, \tilde{x}) = \frac{\partial}{\partial x} G(\alpha, \tilde{x}) = 0$$



$$\rightarrow G(x, \tilde{x}) = 0$$

for $\alpha < x < \tilde{x}$.

assuming

G satisfies

$$G(\alpha, \tilde{x}) = \frac{\partial}{\partial x} G(\alpha, \tilde{x}) = 0$$

Let $x > \tilde{x}$. $G(x, \tilde{x}) = A(\tilde{x}) y_1(x) + B(\tilde{x}) y_2(x)$

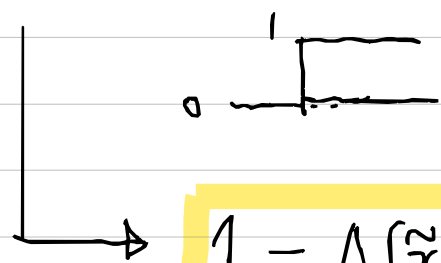
$G(x, \tilde{x})$ is continuous at $x = \tilde{x}$.

$$\Rightarrow 0 = A(\tilde{x}) y_1(\tilde{x}) + B(\tilde{x}) y_2(\tilde{x})$$

Continuity

$$x > \tilde{x}: \frac{\partial}{\partial x} G(x, \tilde{x}) = A(\tilde{x}) y_1'(x) + B(\tilde{x}) y_2'(x)$$

$\frac{\partial G}{\partial x}$ has a unit jump at $x = \tilde{x}$
(i.e., discontinuous)



$$1 = A(\tilde{x}) y_1'(\tilde{x}) + B(\tilde{x}) y_2'(\tilde{x})$$

Discontinuity

(EX) If a 2nd-order homogen. DE, L_x .

$\rightarrow$ 2 B.C.s

$$L_x y = 0$$

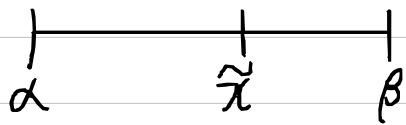
Its soln. $\rightarrow$ satisfies 2 B.C.s.

If the DE is 3rd-order, then we need

3 B.C.s.

$\rightarrow$ 3 B.C.s

5

Recap.

Green Fn. Eq. : $\mathcal{L}_x G(x, \tilde{x}) = \delta(x - \tilde{x})$
(1D)

$$\rightarrow y_0(x) = \int_{\alpha}^{\beta} G(x, \tilde{x}) f(\tilde{x}) d\tilde{x}$$

is a particular integral (or soln.) of $\mathcal{L}_x y = f$.

$$y(x) = y_0(x) + a y_1(x) + b y_2(x)$$

If $G(x, \tilde{x})$ already satisfies homogeneous (initial or 2-pt-) B.C.s then so does $y_0(x)$ & $a = b = 0$.

Homogeneous initial conditions $y(\alpha) = y'(\alpha) = 0$.

$$G(x, \tilde{x}) = 0 \text{ for } \alpha \leq x < \tilde{x}.$$

$$x > \tilde{x}: G(x, \tilde{x}) = A(\tilde{x}) y_1(x) + B(\tilde{x}) y_2(x)$$

$$\begin{cases} A(\tilde{x}) y_1(\tilde{x}) + B(\tilde{x}) y_2(\tilde{x}) = 0 \\ A(\tilde{x}) y_1'(\tilde{x}) + B(\tilde{x}) y_2'(\tilde{x}) = 1 \end{cases}$$

Good to solve them via matrix notation:

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hit both sides by the inverse of the matrix.

$$W = \det \begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} = y_1(\tilde{x}) y_2'(\tilde{x}) - y_2(\tilde{x}) y_1'(\tilde{x})$$

Wronskian

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix}^{-1} = \frac{1}{W} \begin{pmatrix} y_2' & -y_2 \\ -y_1' & y_1 \end{pmatrix}$$

$$\begin{pmatrix} A \\ B \end{pmatrix} = \frac{1}{W(\tilde{x})} \begin{pmatrix} y_2' & -y_2 \\ -y_1' & y_1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \frac{1}{W} \begin{pmatrix} -y_2 \\ y_1 \end{pmatrix}$$

(2)

$$\Rightarrow A(\tilde{x}) = -\frac{y_1(\tilde{x})}{W(\tilde{x})}$$

$$B(\tilde{x}) = \frac{y_1(\tilde{x})}{W(\tilde{x})}$$

$$x > \tilde{x}: G(x, \tilde{x}) = \frac{-y_2(\tilde{x})y_1(x) + y_1(\tilde{x})y_2(x)}{W(\tilde{x})}$$

$$x < \tilde{x}: G(x, \tilde{x}) = 0$$

$$\begin{aligned} \therefore \text{The soln. : } y(x) &= \int_{\alpha}^{\beta} G(x, \tilde{x}) f(\tilde{x}) d\tilde{x} \\ &= \int_{\alpha}^x G(x, \tilde{x}) f(\tilde{x}) d\tilde{x} \end{aligned}$$

$$+ \int_x^{\beta} G(x, \tilde{x}) f(\tilde{x}) d\tilde{x}$$

$$(x < \tilde{x})$$

$$y(x) = \int_{\alpha}^x \frac{y_1(\tilde{x})y_2(x) - y_2(\tilde{x})y_1(x)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x}$$

Exercise.

Check this satisfies the B.C.s.

$$y(\alpha) = 0, \quad y'(\alpha) = 0.$$

$$\begin{aligned} \text{Plug in } x = \alpha: y(\alpha) &= \int_{\alpha}^{\alpha} \frac{y_1(\tilde{x})y_2(\alpha) - y_2(\tilde{x})y_1(\alpha)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x} \\ &= 0 \end{aligned}$$

$$y'(x) = \frac{d}{dx} \int_{\alpha}^x \frac{y_1(\tilde{x})y_2(x) - y_2(\tilde{x})y_1(x)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x}$$

$$= \left[\frac{y_1(x)y_2(x) - y_2(x)y_1(x)}{W(x)} \right] f(x)$$

$$= 0$$

$$y'(\alpha) = 0.$$

The $y(x)$ satisfies the B.C.s.

(3)

$$\text{If } G(\alpha, \tilde{x}) = 0 \text{ \& } \frac{\partial}{\partial x} G(\alpha, \tilde{x}) = 0,$$

$$\text{then } \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \frac{1}{W} \begin{pmatrix} -y_2 \\ y_1 \end{pmatrix}$$

$$\text{both } A = 0, \\ B = 0.$$

$$\text{Homog. 2-pt. B.C.'s. } \underline{y(\alpha) = y(\beta) = 0}$$

$$\alpha \quad \tilde{x} \quad \beta$$

$$\text{Assume } y_1(\alpha) = 0$$

$$y_2(\beta) = 0.$$

Why can't we set y_1 to be zero at both α & β ?

A: then, y_1 would be zero, if it's zero at both pts.

→ Useless to find the Green's fn.s,

$$G(x, \tilde{x}) = A(\tilde{x})y_1(x) + B(\tilde{x})y_2(x) \quad (x < \tilde{x})$$

$$G(x, \tilde{x}) = C(\tilde{x})y_1(x) + D(\tilde{x})y_2(x) \quad (x > \tilde{x})$$

$$\alpha \quad \tilde{x} \quad \beta$$

$$\text{We want } G(\alpha, \tilde{x}) = 0.$$

$$\rightarrow 0 = A(\tilde{x})y_1(\alpha) + B(\tilde{x})y_2(\alpha)$$

$$\Rightarrow B(\tilde{x}) = 0$$

$$\text{We also want } G(\beta, \tilde{x}) = 0.$$

(chosen as a B.C.)

$$\rightarrow 0 = C(\tilde{x})y_1(\beta) + D(\tilde{x})y_2(\beta)$$

$$\Rightarrow C(\tilde{x}) = 0$$

* Continuity of G implies

$$A(\tilde{x})y_1(\tilde{x}) = D(\tilde{x})y_2(\tilde{x})$$

$$\underline{A(\tilde{x})y_1(\tilde{x}) - D(\tilde{x})y_2(\tilde{x}) = 0} \quad \dots (1)$$

* Discontinuity (i.e., $\partial_x G$ jumps by 1)

$$\underline{D(\tilde{x})y_2'(\tilde{x}) - A(\tilde{x})y_1'(\tilde{x}) = 1} \quad \dots (2)$$

$$\text{Eq.s (1) \& (2)} \Rightarrow \begin{pmatrix} y_1 & -y_2 \\ -y_1' & y_2' \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Hit both sides w/ the inverse of $\begin{pmatrix} y_1 & -y_2 \\ -y_1' & y_2' \end{pmatrix}$:

$$\begin{pmatrix} y_1 & -y_2 \\ -y_1' & y_2' \end{pmatrix}^{-1} = \frac{1}{W} \begin{pmatrix} y_2' & y_2 \\ y_1 & y_1' \end{pmatrix}$$

$$\det \begin{pmatrix} y_1 & -y_2 \\ -y_1' & y_2' \end{pmatrix} = W = y_1 y_2' - y_2 y_1'$$

Then,

$$\begin{pmatrix} y_1 & -y_2 \\ -y_1' & y_2' \end{pmatrix}^{-1} \begin{pmatrix} y_1 & -y_2 \\ -y_1' & y_2' \end{pmatrix} \begin{pmatrix} A \\ D \end{pmatrix} = \frac{1}{W} \begin{pmatrix} y_2' & y_2 \\ y_1 & y_1' \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} A \\ D \end{pmatrix} = \frac{1}{W} \begin{pmatrix} y_2 \\ y_1 \end{pmatrix} \Rightarrow \left\{ \begin{array}{l} A = \frac{y_2}{W} \\ D = \frac{y_1}{W} \end{array} \right.$$

Then, the soln.:

$$y(x) = \int_{\alpha}^{\beta} G(x, \tilde{x}) f(\tilde{x}) d\tilde{x}$$

$$= \left(\int_{\alpha}^x + \int_x^{\beta} \right) G(x, \tilde{x}) f(\tilde{x}) d\tilde{x}$$

$$= \int_{\alpha}^x \frac{y_1(\tilde{x}) y_2(x)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x}$$

$$+ \int_x^{\beta} \frac{y_2(\tilde{x}) y_1(x)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x}$$

Exercise.

$$\text{check } \underline{y(\alpha) = y(\beta) = 0.}$$

$$y(\alpha) = \int_{\alpha}^{\alpha} \dots + \int_{\alpha}^{\beta} \frac{y_2(\tilde{x}) y_1(\alpha)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x}$$

$$= 0$$

$$y(\beta) = \int_{\alpha}^{\beta} \frac{y_1(\tilde{x}) y_2(\beta)}{W(\tilde{x})} f(\tilde{x}) d\tilde{x} + \int_{\beta}^{\beta} \dots$$

$$= 0$$

④

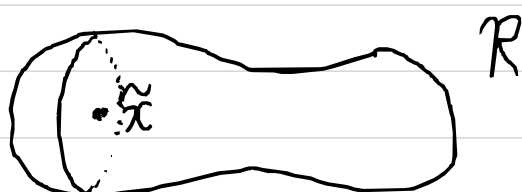
Green fns generalize to higher dim.s (to more variables), e.g., 3D Euclidean space (x_1, x_2, x_3) Cartesian coords.

$$\delta^{(3)}(\underline{x} - \underline{\tilde{x}}) = \delta(x_1 - \tilde{x}_1) \delta(x_2 - \tilde{x}_2) \delta(x_3 - \tilde{x}_3)$$

(3-D Dirac delta fn.)

pt.
 $\underline{\tilde{x}}$

$$\delta(\underline{x} - \underline{\tilde{x}}) = 0 \text{ unless } \underline{x} = \underline{\tilde{x}}$$



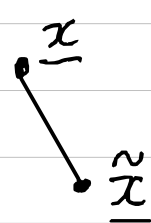
$$\int_R \delta(\underline{x} - \underline{\tilde{x}}) f(\underline{x}) d^3x = \begin{cases} f(\underline{\tilde{x}}) & \text{if } \underline{\tilde{x}} \in R \\ 0 & \text{if } \underline{\tilde{x}} \notin R \end{cases}$$

$$\nabla^2 = \underline{\nabla} \cdot \underline{\nabla} = \partial_i \partial_i = \nabla_i \nabla_i \quad : \text{ the Laplacian}$$

Poisson's Eq.: $\nabla^2 \Phi(\underline{x}) = 4\pi G \rho(\underline{x})$

gravitational potl.

mass density



A Green fn. for ∇^2 :

$$G(\underline{x}, \underline{\tilde{x}}) = -\frac{1}{4\pi} \frac{1}{|\underline{x} - \underline{\tilde{x}}|}$$

satisfies $\nabla^2 G(\underline{x} - \underline{\tilde{x}}) = \delta(\underline{x} - \underline{\tilde{x}})$

How to prove this? We must prove:

$$\int_R \delta(\underline{x} - \underline{\tilde{x}}) f(\underline{x}) d^3x = \begin{cases} f(\underline{\tilde{x}}) & \text{if } \underline{\tilde{x}} \in R \\ 0 & \text{if } \underline{\tilde{x}} \notin R \end{cases}$$

(1) Show that $\nabla_x^2 G(\underline{x}, \underline{\tilde{x}}) = 0$ when $\underline{x} \neq \underline{\tilde{x}}$.

(OR $\nabla_x^2 \frac{1}{|\underline{x} - \underline{\tilde{x}}|} = 0$)

(Hint: use polar coords w/ $\underline{\tilde{x}}$ as the origin.)

$$\int_R \nabla^2 \left(-\frac{1}{4\pi} \frac{1}{|\underline{x} - \underline{\tilde{x}}|} \right) f(\underline{\tilde{x}}) d^3\tilde{x} = 0 \text{ when } \underline{x} \notin R.$$

Use the divergence thm to show

$$\int_R \nabla^2 \left(-\frac{1}{4\pi} \frac{1}{|\underline{x} - \underline{\tilde{x}}|} \right) f(\underline{\tilde{x}}) d^3\tilde{x} = f(\underline{x}) \text{ when } \underline{x} \in R.$$

$$\nabla^2 \left(-\frac{1}{4\pi} \frac{1}{|\underline{x} - \underline{\tilde{x}}|} \right) = \delta^{(3)}(\underline{x} - \underline{\tilde{x}})$$

$$\int_R \delta^{(3)}(\underline{x} - \underline{\tilde{x}}) f(\underline{\tilde{x}}) d^3\tilde{x} = f(\underline{x})$$

Dir. thm: $\nabla^2 \phi = \nabla \cdot \nabla \phi$

w/ $\phi = -\frac{1}{4\pi |\underline{x} - \underline{\tilde{x}}|}$

(5)

$$\int \nabla^2 \varphi f(\underline{\tilde{x}}) d^3 \underline{\tilde{x}} = \int \underline{\nabla} \cdot \underline{\nabla} \varphi f(\underline{\tilde{x}}) d^3 \underline{\tilde{x}}$$

By the div. thm. :

$$\int \underline{\nabla} \cdot \underline{\nabla} \varphi d^3 \underline{\tilde{x}} = \oint_{\partial V} (\underline{\nabla} \varphi) \cdot \hat{n} dS$$

∂V : boundary of the volume V .

For the Grn's fn. $\varphi = -\frac{1}{4\pi |\underline{x} - \underline{\tilde{x}}|}$:

(1) at $\underline{\tilde{x}} \rightarrow \underline{\tilde{x}}$, $\nabla^2 \varphi = \delta^{(3)}(\underline{x} - \underline{\tilde{x}})$.

(2) at the boundary $|\underline{x} - \underline{\tilde{x}}| \rightarrow \infty$,

the fn. decays as $|\underline{x} - \underline{\tilde{x}}|^{-2}$,

& the boundary term vanishes.

$$\int \nabla^2 \varphi f(\underline{\tilde{x}}) d^3 \underline{\tilde{x}} = \int \delta^{(3)}(\underline{x} - \underline{\tilde{x}}) f(\underline{\tilde{x}}) d^3 \underline{\tilde{x}} = f(\underline{x})$$

Ch.3. Hilbert Spaces

§3.4.1. Definitions

Def.: A Hilbert sp. is an infinite dimensional complex vec. sp. w/ an inner product $\langle \cdot, \cdot \rangle$

& an infinite orthonormal basis

$\{u_1, u_2, u_3, \dots\}$. We'll study Hilbert spaces of complex valued fn.s $y(x)$ of 1 real variable $x \in [a, b]$ w/

(1) inner product: $\langle f, g \rangle = \int_a^b f^*(x)g(x) dx$

Note: $\langle \alpha f, g \rangle = \alpha^* \langle f, g \rangle$ for $\alpha \in \mathbb{C}$

$\{u_1, u_2, u_3, \dots\}$ is orthonormal, which means that $\langle u_i, u_j \rangle = \delta_{ij}$.

(2) a countable basis $\{y_1, y_2, y_3, \dots\}$ s.t.

any fn. $f \in \mathcal{H}$ can be expanded as a unique linear combination:

$$f(x) = \sum_{i=1}^{\infty} c_i y_i(x), \quad c_i \in \mathbb{C}$$

§3.5. Sturm-Liouville Theory

The theory of inhomogeneous eq-s of form

$$\mathcal{L}y = f \quad \text{on } x \in [a, b]$$

2nd-order linear differential operator, that is Self-adjoint (S-A).

§3.5.2. Self-adjoint operators

Let



$$\mathcal{L}_x = -\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + r(x) \quad \text{--- (*)}$$

$$\mathcal{L}_x y = -(py')' + \sigma y$$

p & σ = Real & $p > 0$ for $\forall x \in [a, b]$

(2)

We say $\mathcal{L}$ in form (*) is in S-A form.

(not necessarily S-A)

"In Self-adjoint form" is not equal to being "self-adjoint".

DEF. A 2nd-order lin. diff. op. D is S-A on a Hilbert sp. of fn.s on $[a, b]$ if

$$\langle u, Dv \rangle = \langle v, Du \rangle^*$$

$$\forall u, v \in \mathcal{H}$$

$$\langle u, Dv \rangle = \langle Du, v \rangle$$

: def. of self-adj.

D .

Consider $\mathcal{L}$ in form (*):

$$\text{LHS: } \langle u, \mathcal{L}v \rangle = \int_a^b u^*(x) \left(-(pv')' + \sigma v \right) dx$$

$$= \left[-(pv')u^* \right]_a^b - \int_a^b \left[(-pv')u^{*'} + u^*\sigma v \right] dx$$

$$= \left[(-pv')u^* \right]_a^b + \left[v(pu^{*'})' \right]_a^b$$

$$- \int_a^b \left[v(pu^{*'})' - u^*\sigma v \right] dx$$

$$= \left[v pu^{*'} - p v' u^* \right]_a^b + \int_a^b (\mathcal{L}u)^* v dx$$

$$= \left[p v u^{*'} - p v' u^* \right]_a^b + \langle \mathcal{L}u, v \rangle$$

$$\text{So, if } \left[p v u^{*'} - p v' u^* \right]_a^b = 0, \quad \forall u, v \in \mathcal{H},$$

then $\mathcal{L}$ is S-A.

.... (**)

§3.5.3. Ex.s of B.C.s

If the B.C.s on fn.s in $\mathcal{H}$ make (**) hold,

then $\mathcal{L}$ is S-A in $\mathcal{H}$.

(3)

(1) [1st kind of B.C.s] Homogeneous B.C.s

$$\left. \begin{aligned} c_1 u(a) + c_2 u'(a) &= 0 \\ d_1 u(b) + d_2 u'(b) &= 0 \end{aligned} \right\} \begin{aligned} &\text{for } \forall u \in \mathcal{H} \\ &\& c_i, d_i \text{ are real.} \\ &\quad i=1,2 \quad (\in \mathbb{R}) \end{aligned}$$

(2) [2nd kind of B.C.s]

$$\text{If } p(a) = p(b) = 0$$

& u & v are finite at a & b , then $(**)$ holds.

$$\left[p v u^{*'} - p v' u^* \right]_a^b = 0,$$

(3) [3rd kind of B.C.s]

If $p(a) = p(b)$ & $u(a) = u(b) \quad \forall u \in \mathcal{H}$,
then $(**)$ holds.

§3.5.4. Weight fn.s

Consider

$$\tilde{\mathcal{L}} = - \frac{d}{dx} \left(A(x) \frac{d}{dx} \right) - B(x) \frac{d}{dx} + C(x)$$

..... (***)

w/ $A(x) > 0$ on $x \in [a, b]$.Let $\omega(x) > 0$ on $[a, b]$ & define $\mathcal{L} = \omega \tilde{\mathcal{L}}$ & solve for ω s.t. $\mathcal{L}$ is in S-A form.

(4)

$$\mathcal{L} = \omega \left(-\frac{d}{dx} \left(A \frac{d}{dx} \right) - B \frac{d}{dx} + C \right)$$

$$= -\frac{d}{dx} \left(\omega A \frac{d}{dx} \right) + \frac{d\omega}{dx} A \frac{d}{dx} - B\omega \frac{d}{dx} + C\omega$$

$$= -\frac{d}{dx} \left(\omega A \frac{d}{dx} \right) + \left(\frac{d\omega}{dx} A - B\omega \right) \frac{d}{dx} + C\omega$$

Choose $\omega' A - B\omega = 0$

& then $\mathcal{L}$ is in S-A form w/ $\begin{cases} \rho = \omega A \\ \sigma = \omega C \end{cases}$

$$\frac{\omega'}{\omega} = \frac{B}{A} \longrightarrow \omega(x) = \exp \left(\int_a^x \frac{B(\tilde{x})}{A(\tilde{x})} d\tilde{x} \right)$$

a fixes $\omega(a) = 1$.

Note:

$$\omega(x) > 0 \text{ for } x \in [a, b]$$

Given $\omega(x) > 0$, we can define a new inner product $\langle \cdot, \cdot \rangle_\omega$ by

$$\begin{aligned} \langle f, g \rangle_\omega &= \int f^*(\tilde{x}) g(\tilde{x}) \omega(\tilde{x}) d\tilde{x} \\ &= \langle f, \omega g \rangle = \langle \omega f, g \rangle \end{aligned}$$

Satisfies all properties of an Inner product.

§3.5.5. Eigenfn.s & Eigenval.s

$$\tilde{\mathcal{L}} y = \lambda y$$

Eigenvalue

Eigenfunction y

We can find $\omega(x)$, s.t., $\omega \tilde{\mathcal{L}}$ is of S-A form.

Then,

$$\mathcal{L} y = \lambda \omega y.$$

The eigenvals of self-adjoint^(S-A) ops are real
& the eigenfunc.s (e-func.s) of distinct
eigenvalues (e-val.s) are orthogonal w.r.t.
 $\langle \cdot, \cdot \rangle_w$.

Proof. Let e-func.s be $y_1, y_2, \dots$
w/ e-val.s are $\lambda_1, \lambda_2, \lambda_3, \dots$

i.e.

$$\mathcal{L} y_i = \lambda_i \omega y_i \quad (\text{no summation})$$

for $\forall i$

$$\rightarrow \langle y_j, \mathcal{L} y_i \rangle = \lambda_i \langle y_j, \omega y_i \rangle \quad \dots \text{Eq. (1)}$$

$$\rightarrow \langle y_j, \mathcal{L} y_i \rangle^* = \lambda_i^* \langle y_j, \omega y_i \rangle^*$$

$$\rightarrow \langle y_i, \mathcal{L} y_j \rangle = \lambda_j^* \langle \omega y_i, y_j \rangle$$

because $\mathcal{L}$ is S-A.

$$\rightarrow \langle y_i, \mathcal{L} y_j \rangle = \lambda_i^* \langle y_i, \omega y_j \rangle$$

$$\text{From Eq. (1), } \langle y_i, \mathcal{L} y_j \rangle = \lambda_j \langle y_i, \omega y_j \rangle$$

$$(\lambda_i^* - \lambda_j) \langle y_i, \omega y_j \rangle = 0$$

$$\text{Let } i=j \rightarrow (\lambda_i^* - \lambda_i) \langle y_i, \omega y_i \rangle = 0$$

$$\lambda_i^* = \lambda_i$$

so λ_i is real.

$$\text{Let } \lambda_i \neq \lambda_j. \rightarrow (\lambda_i - \lambda_j) \langle y_i, \omega y_j \rangle = 0$$

$$\hookrightarrow \langle y_i, y_j \rangle_w = 0$$

Always choose a basis.

Big Thm. The e-val.s of a S-A op. on $\mathcal{H}$

w/ weight ω form a discrete infinite sequence

$\lambda_1, \lambda_2, \lambda_3, \dots$, s.t. $|\lambda_n| \rightarrow \infty$ as $n \rightarrow \infty$.

& the e-func.s (w/ weight ω) $y_1, y_2, y_3, \dots$

form an orthonormal basis of $\mathcal{H}$:

$$\langle y_n, y_m \rangle_w = \delta_{nm}$$

(2)

Any fn. in $\mathcal{H}$ can be expanded:

$$u(x) = \sum_{n=1}^{\infty} u_n y_n(x)$$

$$\begin{aligned} \langle y_m, u \rangle_{\omega} &= \left\langle y_m, \sum_{n=1}^{\infty} u_n y_n(x) \right\rangle \\ &= \sum_{n=1}^{\infty} u_n \langle y_m, y_n \rangle_{\omega} = \sum_{n=1}^{\infty} u_n \delta_{mn} = u_m \end{aligned}$$

$$\therefore u_m = \langle y_m, u \rangle_{\omega} = \int_a^b y_m^*(x) \omega(x) u(x) dx$$

$$\begin{aligned} u(x) &= \sum_{n=1}^{\infty} \int_a^b y_n^*(\tilde{x}) \omega(\tilde{x}) u(\tilde{x}) y_n(x) dx \\ &= \int_a^b d\tilde{x} \left[u(\tilde{x}) \omega(\tilde{x}) \sum_{n=1}^{\infty} y_n^*(\tilde{x}) y_n(x) \right] \end{aligned}$$

$$\omega(\tilde{x}) \sum_{n=1}^{\infty} y_n^*(\tilde{x}) y_n(x) = \delta(\tilde{x} - x)$$

Lemma. (Parseval's) $\int_a^b |u|^2 \omega dx = \int_a^b \langle u, u \rangle_{\omega} \omega dx$
 $= \sum_{n=1}^{\infty} |u_n|^2$

Proof. LHS = $\langle u, u \rangle_{\omega} = \left\langle \sum_n u_n y_n(x), \sum_m u_m y_m(x) \right\rangle_{\omega}$

$$= \sum_n \sum_m u_n^* u_m \langle y_n, y_m \rangle_{\omega}$$

$$= \sum_{n,m} u_n^* u_m \delta_{nm} = \sum_n u_n^* u_n = \text{RHS} \quad \square$$

Exercise. If $u = \sum u_n y_n$ & $v = \sum v_n y_n$,

Show: $\langle u, v \rangle_{\omega} = \sum_n u_n^* v_n$

§3.5.8. Green Fn.s Revisited

(3)

$\mathcal{L}$ is S-A w/ spectrum $\{\lambda_n\}$ & basis of e-fn.s $\{y_n\}$, w/ weight w .

Claim. $G(x, \tilde{x}) = \sum_{n=1}^{\infty} \frac{y_n(x) y_n^*(\tilde{x})}{\lambda_n}$

is a Green fn. of $\mathcal{L}$, if $\lambda_n \neq 0$ for $\forall n$.

Proof. $\mathcal{L}_x G(x, \tilde{x}) = \mathcal{L}_x \sum_{n=1}^{\infty} \left(\frac{y_n(x) y_n^*(\tilde{x})}{\lambda_n} \right)$

$$= \sum_{n=1}^{\infty} \left(\frac{(\mathcal{L}_x y_n(x)) y_n^*(\tilde{x})}{\lambda_n} \right) = \sum_{n=1}^{\infty} \left(\frac{\cancel{\lambda_n} w(x) y_n(x) y_n^*(\tilde{x})}{\cancel{\lambda_n}} \right)$$

$$= w(x) \sum_{n=1}^{\infty} \left(y_n(x) y_n^*(\tilde{x}) \right) = \left[w(x) \sum_{n=1}^{\infty} \left(y_n(x)^* y_n(\tilde{x}) \right) \right]^*$$

c.c.'ed twice

$$= [\delta(\tilde{x} - x)]^* \Rightarrow \mathcal{L}_x G(x, \tilde{x}) = \delta(x - \tilde{x})$$

§3.5.9. Solving ODEs

EX.

e-val.s

$$\mathcal{L} y(x) - \nu y(x) = f(x)$$

$\nu = \text{const.}$

Assume $w=1 \rightarrow$ Have $\{y_n\}$ & $\{\lambda_n\}$

$$y = \sum a_n y_n \rightarrow f = \sum f_n y_n$$

$f_n = \langle y_n, f \rangle$ are known.

We want a_n 's.

$$\sum_n a_n \lambda_n y_n - \nu \sum_n a_n y_n = \sum_n f_n y_n$$

$$\sum_n a_n (\lambda_n - \nu) y_n = \sum_n f_n y_n$$

$\{y_n\}$: bases $\rightarrow$ i.e., linearly independent.

$$y_n = \sum_n \frac{f_n}{\lambda_n - \nu} y_n(x) \text{ is a soln.}$$

So, $a_n (\lambda_n - \nu) = f_n$ & $a_n = \frac{f_n}{\lambda_n - \nu}$

if $\lambda_n \neq \nu$

§3.6. Legendre Polynomials

④

$$\left\{ \begin{array}{l} (i) \\ (ii) \end{array} \right\} \mathcal{L} = -\frac{d^2}{dx^2} \quad (x \in [0, 2\pi R])$$

$$(i) \text{ B.C.s : } \underline{y(0) = y(2\pi R) = 0}$$

Homogeneous 2-pt. B.C.s

$$(ii) \text{ B.C.s : } \left[\begin{array}{l} y(0) = y(2\pi R) \\ \underline{y'(0) = y'(2\pi R)} \end{array} \right]$$

periodic B.C.s.

(i) & (ii) have different spectra of e-val.s for $\mathcal{L}$
& different e-func.s of $\mathcal{L}$.

Very Important for exams:

Green's Functions,
Fourier Transforms,
Laplace Transforms,
Contour Integrals.
(Bromwich)
Residue Theorem,
ODEs,
Complex Variables
(singularities, poles,
branch cut),
Jordan's Lemma

Recap.

$$\tilde{L} = -\frac{d}{dx} \left(A(x) \frac{d}{dx} \right) - B(x) \frac{d}{dx} + C(x)$$

$$A(x) > 0$$

 $\mathcal{L} = \omega \tilde{L}$ is in S-A form when

$$\omega(x) = \exp \left(\int_a^x \frac{B(\tilde{x})}{A(\tilde{x})} d\tilde{x} \right)$$

$$\mathcal{L} = -\frac{d}{dx} \left(p \frac{d}{dx} \right) + \sigma(x)$$

$$\omega / \begin{cases} p(x) = A\omega \\ \sigma(x) = C\omega \end{cases}$$

 $\langle \cdot, \cdot \rangle_\omega$ is an I.P.

$$\langle f, g \rangle_\omega \equiv \langle f, \omega g \rangle \equiv \langle \omega f, g \rangle \\ = \int_a^b f^*(x) \omega(x) g(x) dx$$

§3.5.5. Eigenfn.s & Eigenval.s

$$\mathcal{L}y(x) = \lambda y(x)$$

 $\lambda = \text{e-val.}$ $y = \text{e-fn.}$

$$\mathcal{L}y(x) = \lambda \omega(x) y(x)$$

In words,

" y is an eigenfn. of $\mathcal{L}$ w/ weight ω "

Assume : $\mathcal{L}$ is S-A.

Theorem. The eigenval.s of a S-A operator are real,
& the eigenfn.s of distinct eigenvals are
orthogonal w.r.t. the weighted inner product
(IP).
 $\langle \cdot, \cdot \rangle_\omega$

Proof.Let e-fn.s be $y_1, y_2, y_3, \dots$ w/ e-val.s $\lambda_1, \lambda_2, \lambda_3, \dots$, i.e.,

$$\mathcal{L}y_i = \lambda_i \omega y_i \quad \left(\begin{array}{l} \text{No summation} \\ \text{here} \end{array} \right)$$

$$\Rightarrow \langle y_j, \mathcal{L}y_i \rangle = \langle y_j, \lambda_i \omega y_i \rangle$$

$$\langle y_j, \mathcal{L}y_i \rangle = \lambda_i \langle y_j, \omega y_i \rangle \quad \dots (*)$$

$$\langle y_j, \mathcal{L}y_i \rangle^* = \lambda_i^* \langle y_i, \omega y_j \rangle^*$$

$$\langle y_i, \mathcal{L}y_j \rangle = \lambda_j^* \langle \omega y_i, y_j \rangle$$

bc $\mathcal{L}$ is S-A.

From (*),

(2)

$$\langle y_i, \mathcal{L} y_j \rangle = \lambda_j \langle y_i, \omega y_j \rangle$$

$$(\lambda_i^* - \lambda_j) \langle y_i, \omega y_j \rangle = 0$$

let $i=j$. $\rightarrow (\lambda_i^* - \lambda_i) \underbrace{\langle y_i, \omega y_j \rangle}_{\neq 0} = 0$

$$\rightarrow \lambda_i = \lambda_i^*$$

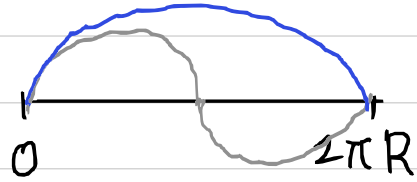
so λ_i is real.

let $i \neq j$. $\rightarrow (\lambda_i - \lambda_j) \langle y_i, \omega y_j \rangle = 0$

$$\rightarrow \langle y_i, y_j \rangle_\omega = 0$$

$$\mathcal{L} = -\frac{d^2}{dx^2}, \quad x \in [0, 2\pi R]$$

$$(1) \quad y(0) = y(2\pi R) = 0$$



$$-\frac{d^2}{dx^2} (y_n(x)) = \lambda_n y_n$$

$$y_n \equiv \sin\left(\frac{n}{2R}x\right) \quad (n=1, 2, \dots)$$

$$\lambda_n = \left(\frac{n}{2R}\right)^2$$

$$(2) \quad y(0) = y(2\pi R) \quad \& \quad y'(0) = y'(2\pi R)$$

$$\left. \begin{aligned} -y_n'' &= \lambda_n y_n \\ y_m &= e^{imx/R} \\ \lambda_m &= \left(\frac{m}{R}\right)^2 \end{aligned} \right\} \rightarrow \begin{aligned} y_0 &= 1, \quad \lambda_0 = 0 \\ &\quad \uparrow \\ &\text{zero mode} \\ m &\in \mathbb{Z} \\ m &= 0, \pm 1, \pm 2, \dots \end{aligned}$$

Alternatively,

$$y_m^C = \cos \frac{mx}{R}, \quad m=0, 1, 2, \dots$$

$$y_m^S = \sin \frac{mx}{R}, \quad m=1, 2, 3, \dots$$

2

(3) Legendre polynomials

$$(1-x^2)y'' - 2xy' + l(l+1)y = 0$$
$$-\left[(1-x^2)y'\right]' = \underline{l(l+1)y} \quad \text{where } l(l+1) = \lambda$$

where the weight fn. $w(x) = 1$.

$$x \in [-1, 1) \quad \& \quad l = 0, 1, 2, \dots$$

$$\rho(x) = 1 - x^2 \longrightarrow \rho(-1) = \rho(1) = 0$$

$\hookrightarrow y$ is finite at ± 1
means S-A.

(self-adjoint)

$\exists$ discrete spectrum
& the eigenfunctions (e-fns)
are polynomials.

Hermite polynomials $\longrightarrow$ orthogonal

$\uparrow$
weight $w(x) = 1$ for them.

MMP L10

C4. Integral Transform

§4.5-1. Def & Notation for Fourier Transforms

Given $f(x)$, $-\infty < x < \infty$,

& $\int_{-\infty}^{\infty} |f(x)| dx$ is finite, its

Fourier transform $\tilde{f}(k)$ is defined by

$$\rightarrow \tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

The inverse F.T. :

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk$$

"wave num."
(basis of plane waves)

complex
coeff.s
of the
basis
func.s.
 $f(x)$

§4.5.2. Dirac δ -func.

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x') e^{-ikx'} dx' \right] \times e^{ikx} dk$$

$$= \int_{-\infty}^{\infty} dx' f(x') \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-x')} dk$$

$$= \int_{-\infty}^{\infty} dx' f(x') \delta(x-x')$$

$$\longrightarrow \delta(x-x') = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ik(x-x')} dk$$

• If f is real & $f(x) = f(-x)$,
then $\tilde{f}(k)$ is real.

Proof. $\tilde{f}(k)^* = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{ikx} dx$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(-y) e^{-iky} (-dy) \quad \boxed{y = -x}$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(y) e^{-iky} (dy)$$

$$= \tilde{f}(k)$$

• If f is real & $f(x) = -f(-x)$,
then $\tilde{f}(k)$ is pure imaginary, i.e.,
 $\tilde{f}(k)^* = -\tilde{f}(k)$

• Deriv.s

$$\mathcal{F}\left[\frac{df}{dx}\right] = ik \tilde{f}(k)$$

Proof. LHS = $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{df}{dx} e^{-ikx} dx$

$$= \frac{1}{\sqrt{2\pi}} \left[f e^{-ikx} \right]_{-\infty}^{\infty} \rightarrow 0$$

$$- \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) (-ik) e^{-ikx} dx$$

$$= ik \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

$$= ik \tilde{f}(k)$$

$$\mathcal{F}\left[\frac{d^2f}{dx^2}\right] = (ik)^2 \tilde{f}(k) = -k^2 \tilde{f}(k)$$

$$\mathcal{F}[f^{(n)}(x)] = (ik)^n \tilde{f}(k)$$

$$\mathcal{F}[xf(x)] = i \frac{d}{dk} \tilde{f}(k)$$

$$\mathcal{F}[x^n f(x)] = \left(i \frac{d}{dk}\right)^n \tilde{f}(k)$$

Proof of $f(x)$: $LHS = \int_{-\infty}^{\infty} \frac{xf(x)e^{-ikx}}{\sqrt{2\pi}} dx$

$$RHS = i \frac{d}{dk} \int_{-\infty}^{\infty} \frac{f(x)e^{-ikx}}{\sqrt{2\pi}} dx$$

$$= i \int_{-\infty}^{\infty} \frac{f(x)(-ix)e^{-ikx}}{\sqrt{2\pi}} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} xf(x)e^{-ikx} dx$$

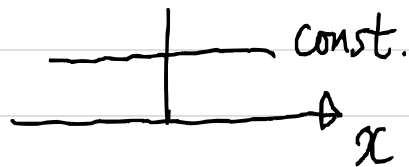
$$= LHS$$

§4.5.4. Ex.5

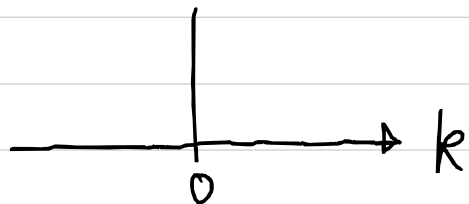
1. Constant $f(x) = 1$

$$\begin{aligned}\hat{f}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} dx \\ &= \sqrt{2\pi} \delta(x)\end{aligned}$$

The F.T. of constant f^n is a δ -fn peaked at 0.



F.T.
↓



2. Single-freq. mode

$$f(x) = e^{ik_0 x}$$

2. Single-freq. mode $f(x) = e^{ik_0 x}$

k_0 : fixed wv. num.

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int e^{ik_0 x} e^{-ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int e^{-i(k-k_0)x} dx$$

$$= \sqrt{2\pi} \delta(k-k_0)$$

peaked at
single wavenumber.

$|f(x)|$ is const. & spread out evenly
over all x

& its F.T. is peaked at a single
wv. num.

3. Dirac δ -func. $f(x) = \delta(x-x_0)$

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x-x_0) e^{-ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ikx_0}$$

$f(x) \rightarrow$ peaked & $|\tilde{f}(k)|$ is const.
 $\rightarrow$ spread over whole of k -space.

4. Gaussian func.

$$f(x) = \frac{1}{\sqrt{\sigma} (2\pi)^{1/4}} e^{-(x-x_0)/(4\sigma^2)}$$

①

MMP L11

§4.5.4.

4. Gaussian

$$f(x) = \frac{1}{\sigma^{1/2} (2\pi)^{1/4}} e^{-(x-x_0)^2/(4\sigma^2)}$$

σ : width of the Gaussian

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = 1$$

$$\sigma^{1/2} (2\pi)^{1/4} \tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x-x_0)^2/(4\sigma^2)} e^{-ikx} dx$$

Put the normalization here.

$$\text{Let } x' = x - x_0$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ik(x'+x_0)} e^{-(x')^2/(4\sigma^2)} dx'$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ikx_0} \int_{-\infty}^{\infty} e^{-ikx'} e^{-(x')^2/(4\sigma^2)} dx'$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ikx_0} \int_{-\infty}^{\infty} dx' e^{-(x' + 2i\sigma^2 k)^2/(4\sigma^2) - k^2\sigma^2}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ikx_0} e^{-k^2\sigma^2} \int_{-\infty}^{\infty} dx' e^{-\frac{(x' + 2i\sigma^2 k)^2}{4\sigma^2}}$$

$$x'' = x' + 2i\sigma^2 k$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ikx_0} e^{-k^2\sigma^2} \int_{-\infty}^{\infty} dx'' e^{-(x'')^2/(4\sigma^2)}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ikx_0} e^{-k^2\sigma^2} (2\sigma)\sqrt{\pi}$$

$$\therefore \tilde{f}(k) = \frac{\sqrt{2\sigma}}{(2\pi)^{1/4}} e^{-ikx_0} e^{-k^2\sigma^2}$$

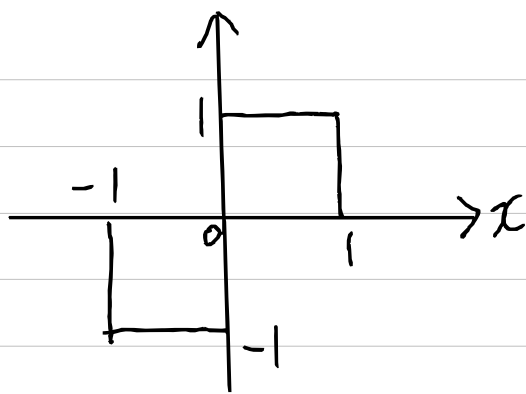
The intensity in x ("space") $|f|^2$ is a Gaussian w/ variance σ^2 .

The intensity in k ("wavenumber") $|\tilde{f}|^2$ is a Gaussian w/ variance $\frac{1}{4\sigma^2}$.

(2)

5. Square Waves

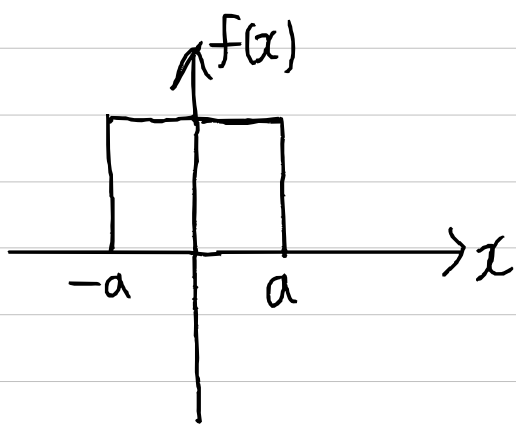
$$f(x) = \begin{cases} 1, & x \in [0, 1] \\ -1, & x \in [-1, 0] \\ 0, & |x| > 1 \end{cases}$$



$$\begin{aligned} \tilde{f}(k) &= \frac{1}{\sqrt{2\pi}} \left[\int_{-1}^0 (-1) e^{-ikx} dx + \int_0^1 e^{-ikx} dx \right] \\ &= \frac{1}{\sqrt{2\pi}} \left\{ - \left[-\frac{1}{ik} e^{-ikx} \right]_{-1}^0 + \left[-\frac{1}{ik} e^{-ikx} \right]_0^1 \right\} \\ &= \frac{1}{\sqrt{2\pi}} \frac{2}{ik} (1 - \cos k) = -\frac{4i}{\sqrt{2\pi} k} \sin^2\left(\frac{k}{2}\right) \end{aligned}$$

6. "Tophat" func.

$$f(x) = \begin{cases} 1, & |x| < a \\ 0, & |x| \geq a \end{cases}$$



$$\begin{aligned} \tilde{f}(k) &= \frac{1}{\sqrt{2\pi}} \int_{-a}^a e^{-ikx} dx \\ &= \sqrt{\frac{2}{\pi}} \frac{\sin ka}{k} \end{aligned}$$

§4.5.5. "Bandwidth Thm"

If a function has a "spread" Δx in x -sp., then its spread in k -sp. will be s.t. $\Delta x \sim \frac{1}{\Delta k}$.

where Δx and Δk are the sqrts of the variance of $|f(x)|^2$ and $|\tilde{f}(k)|^2$.

For Gaussians: $\Delta x \Delta k = \frac{1}{2}$ exactly

In general, $\Delta x \Delta k \geq \frac{1}{2}$

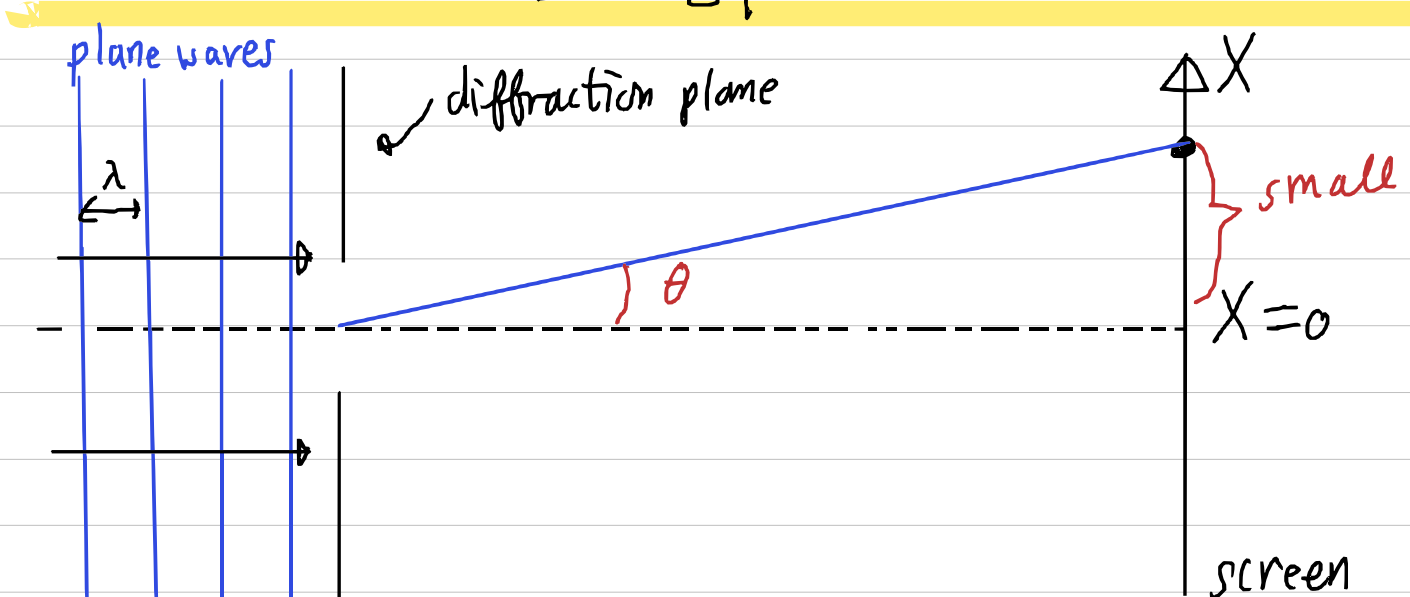
In QM, $p = \hbar k$ & $\Delta x \Delta p \geq \frac{\hbar}{2}$

Heisenberg's U. Relation

§4.6. Diffraction thru an aperture

The intensity $I(k)$ is the square of the abs. val. of the F.T. of the aperture func. $h(x)$

$$I(k) = |\mathcal{F}[h(x)]|^2$$



$$k = \frac{2\pi \sin \theta}{\lambda}$$

→ λ is the WL of the incident light

& θ is the angle of diffraction

We have $\sin \theta \approx \tan \theta \approx \theta = \frac{X}{D}$

so $k \approx \frac{2\pi X}{\lambda D}$ i.e. $k \propto X$

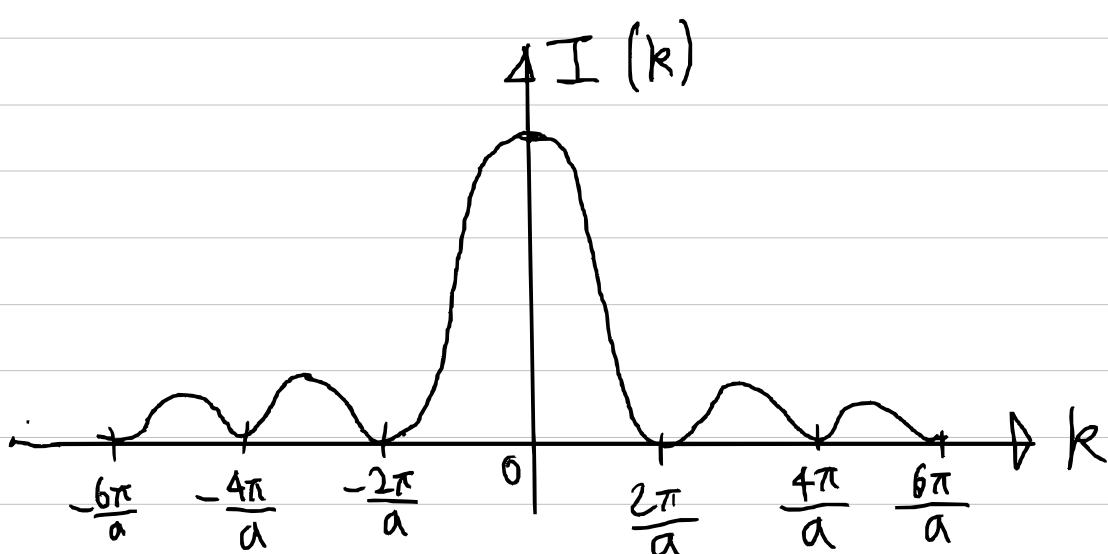
Consider a single slit of width a .

$$h(x) = \begin{cases} 1 & , x \in [-\frac{a}{2}, \frac{a}{2}] \\ 0 & , \text{otherwise} \end{cases}$$

$$\tilde{h}(k) = \sqrt{\frac{2}{\pi}} \frac{\sin(ka/2)}{k} = \frac{a \operatorname{sinc}(ka/2)}{\sqrt{2\pi}}$$

$$\operatorname{sinc}(x) = \frac{\sin x}{x} \quad (\text{"sine cardinal"})$$

→ $I(k) = \frac{a^2 \operatorname{sinc}^2(ka/2)}{2\pi}$



§4.6.1. Parseval's Thm for FTs

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |\tilde{f}(k)|^2 dk$$

Proof.

$$\text{LHS} = \int_{-\infty}^{\infty} f(x)^* f(x) dx$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dx \left[\int_{-\infty}^{\infty} \tilde{f}(k)^* e^{-ikx} dk \right] \left[\int_{-\infty}^{\infty} \tilde{f}(k') e^{+ik'x} dk' \right]$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} dk \int_{-\infty}^{\infty} dk' \tilde{f}(k)^* \tilde{f}(k') \underbrace{\int_{-\infty}^{\infty} dx e^{-ix(k-k')}}_{\delta(k-k')}$$

$$= \int_{-\infty}^{\infty} dk \int_{-\infty}^{\infty} dk' \tilde{f}(k)^* \tilde{f}(k') \delta(k-k')$$

$$= \int_{-\infty}^{\infty} dk \tilde{f}(k)^* \tilde{f}(k) = \text{RHS} \quad \square$$

Exercise. Prove that

$$\langle g(x), f(x) \rangle = \int_{-\infty}^{\infty} f(x) g(x)^* dx = \int_{-\infty}^{\infty} \tilde{f}(k) \tilde{g}(k)^* dk$$

Verify that Parseval's identity holds for the Gaussian & its Fourier Transform.

§4.6.2. Convolution Thm of FTS

The convolution of $f(x)$ & $g(x)$:

$$(f * g)(x) = \int_{-\infty}^{\infty} f(y) g(x-y) dy$$

Exercise.

Prove: $f * g = g * f$

★ 2 equivalent statements of the convolution thm. :

$$1. \quad \mathcal{F}[f * g] = \sqrt{2\pi} \tilde{f}(k) \tilde{g}(k)$$

$$2. \quad \mathcal{F}[fg] = \frac{1}{\sqrt{2\pi}} (\tilde{f} * \tilde{g})(k)$$

Proof of 1.

$$\text{LHS} = \mathcal{F}[f * g] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f * g(x) e^{-ikx} dx$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(y) g(x-y) dy e^{-ikx} dx$$

$$= \int_{-\infty}^{\infty} f(y) \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x-y) e^{-ikx} dx \right] dy$$

$$= \int_{-\infty}^{\infty} f(y) e^{-iky} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x-y) e^{-ik(x-y)} dx \right] dy$$

$$x' = x - y$$

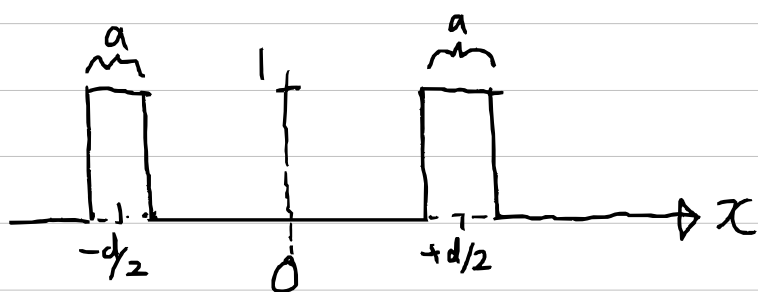
$$= \int_{-\infty}^{\infty} f(y) e^{-iky} \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(x') e^{-ikx'} dx' \right] dy$$

$$I(k) = |\mathcal{F}[h]|^2 \quad \longrightarrow \quad h \text{ is the op. func.}$$

$k \propto x$ position on screen

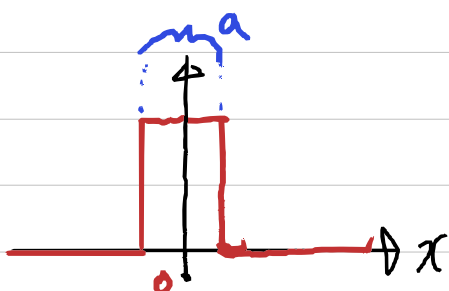
Double slit diffraction & interference

$$h(x) =$$



$$= \begin{cases} 1 & , & |x - d/2| < a/2 \\ 1 & , & |x + d/2| < a/2 \\ 0 & , & \text{otherwise} \end{cases}$$

$$h(x) = f * g(x)$$



$$f(x) = \delta(x - d/2) + \delta(x + d/2)$$

(5)

$$\mathcal{F}[f] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left[\delta\left(x - \frac{d}{2}\right) + \delta\left(x + \frac{d}{2}\right) \right]$$

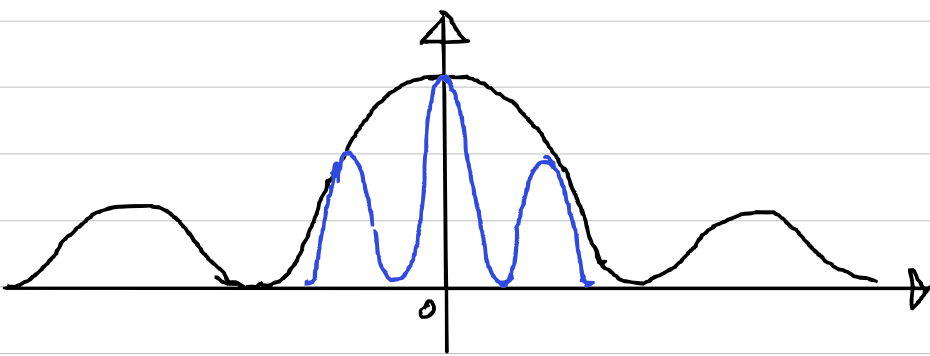
$$= \frac{1}{\sqrt{2\pi}} \left[e^{-ikd/2} + e^{ikd/2} \right]$$

$$= \sqrt{\frac{2}{\pi}} \cos\left(\frac{kd}{2}\right)$$

$$\therefore \mathcal{F}[h] = \sqrt{2\pi} \mathcal{F}(f) \mathcal{F}(g)$$

$$I(k) = \frac{2a^2}{\pi} \underbrace{\text{sinc}^2\left(\frac{ka}{2}\right)}_{\text{broad envelope}} \underbrace{\cos^2\left(\frac{kd}{2}\right)}_{\text{interference fringes}}$$

when $d > 0$.



Solving DEs using F.T.s

$$\mathcal{F}[f'] = ik \tilde{f}(k)$$

Classic example diffusion

$\theta(x, t)$ is the temp. at position x of an only conducting bar at time

$$\otimes \frac{\partial \theta}{\partial t} = D \frac{\partial^2 \theta}{\partial x^2}$$

θ_1 & θ_2 are soln.s $\rightarrow a_1 \theta_1 + a_2 \theta_2$ is a soln.

$$\theta(\pm \infty, t) = 0$$

Consider the eqn. w/ I.C.s.

$$\theta(x, t=0) = f(x) = \int_{-\infty}^{\infty} \delta(x - \tilde{x}) f(\tilde{x}) d\tilde{x}$$

If we find soln. of * w/ I.C.

$\theta(x, t=0) = \delta(x - \tilde{x})$, say $\theta_{\text{fundam}}(x, t)$,

then the soln. w/ I.C. $f(x)$ is

$$\int \theta_{\text{fundam}}(\tilde{x}, t) f(\tilde{x}) d\tilde{x}$$

$$\frac{\partial \theta}{\partial t} = D \frac{\partial^2 \theta}{\partial x^2}$$

Take F.T. w.r.t. x : $\tilde{\theta}(k, t)$

$$\left. \begin{aligned} \frac{\partial \tilde{\theta}}{\partial t} &= D (ik)^2 \tilde{\theta} \\ \frac{\partial \tilde{\theta}}{\partial t} &= -Dk^2 \tilde{\theta} \end{aligned} \right\} \rightarrow \tilde{\theta}(k, t) = A(k) e^{-Dk^2 t}$$

$$A(k) = \tilde{\theta}(k, t=0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ikx} dx \delta(x - \tilde{x})$$

$$= \frac{1}{\sqrt{2\pi}} e^{-ik\tilde{x}}$$

$$\tilde{\theta}(k, t) = \frac{1}{\sqrt{2\pi}} e^{-ik\tilde{x}} e^{-Dk^2 t}$$

$\theta(x, t)$: soln. w/ I.C. $\theta(x=t=0) = \delta(x - \tilde{x})$

$$\theta(x, t) = \mathcal{F}^{-1}[\tilde{\theta}(k, t)]$$

(6)

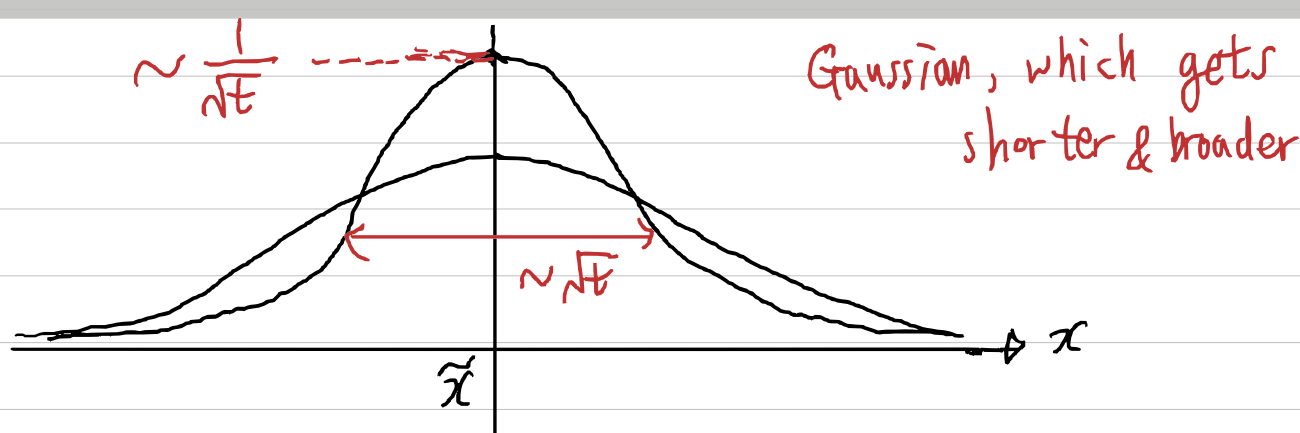
Exercise

Show that

$$\mathcal{F}^{-1}[\tilde{\theta}(k, t)] = \frac{1}{2\pi} \sqrt{\frac{\pi}{Dt}} e^{-(x-\tilde{x})^2/(4Dt)}$$

As $t \uparrow$, Height of the Gaussian
(Incr.) $\downarrow$ (Decr.)

Width of the Gaussian $\uparrow$ (Incr.)



§4.7. Laplace Transforms

The Laplace transform of a fn. $f(t)$ is

$$\mathcal{L}[f(t)] = \hat{f}(s) = \int_0^{\infty} e^{-st} f(t) dt$$

$s = \sigma + i\omega$ is a complex variable & σ has to be +ve. ($\sigma > 0$)

$\sigma > 0$ guarantees the convergence of the integral.

§4.7.1. Properties

- $\mathcal{L}[a f_1(t) + b f_2(t)] = a \mathcal{L}[f_1] + b \mathcal{L}[f_2]$
- $\mathcal{L}\left[\frac{df}{dt}\right] = s \hat{f}(s) - f(0)$

Proof: LHS = $\int_0^{\infty} \frac{df}{dt} e^{-st} dt$

$$= [f e^{-st}]_0^{\infty} - \int_0^{\infty} f(-s) e^{-st} dt$$

$$= -f(0) + \int_0^{\infty} s f(t) e^{-st} dt$$

$$= -f(0) + s \mathcal{L}[f]$$

$$= \text{RHS} \quad \square$$

- $\mathcal{L}[f^{(n)}(t)] = s^n \hat{f}(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$

Prove this by induction:

- $\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} (\hat{f}(s))$

Proof:

$$\text{RHS} = (-1)^n \frac{d^n}{ds^n} \int_0^{\infty} e^{-st} f(t) dt$$

$$= (-1)^n \int_0^{\infty} (-t)^n e^{-st} f(t) dt$$

$$= \int_0^{\infty} t^n e^{-st} f(t) dt = \text{LHS}$$

§4.7.2 - Examples

- $\mathcal{L}[1] = \frac{1}{s}$

Proof. $\mathcal{L}[1] = \int_0^{\infty} e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^{\infty}$
 $= (-) \left(-\frac{1}{s} \right) = \frac{1}{s}$

$$\mathcal{L}[e^{-at}] = \frac{1}{s+a}$$

$$\mathcal{L}[\cos \omega t] = \frac{s}{s^2 + \omega^2}$$

Proof:

$$\begin{aligned} \text{LHS} &= \int_0^{\infty} (\cos \omega t)(e^{-st}) dt \\ &= \frac{1}{2} \int_0^{\infty} (e^{+i\omega t} + e^{-i\omega t}) e^{-st} dt \\ &= \frac{1}{2} \left[\frac{1}{i\omega - s} e^{(i\omega - s)t} + \frac{1}{-i\omega - s} e^{(-i\omega - s)t} \right]_0^{\infty} \\ &= -\frac{1}{2} \left[\frac{1}{i\omega - s} + \frac{1}{-i\omega - s} \right] \\ &= -\frac{1}{2} \frac{-\cancel{i\omega} - s + \cancel{i\omega} - s}{(i\omega - s)(-i\omega - s)} \\ &= -\frac{1}{2} \frac{-2s}{(i\omega - s)(-i\omega - s)} = \frac{s}{(s^2 + \omega^2)} \end{aligned}$$

- $\mathcal{L}[\sin \omega t] = \frac{\omega}{s^2 + \omega^2}$

(7)

*

$$\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$$

Proof.

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} \hat{f}(s)$$

$$f(t)=1 \quad \& \quad \hat{f}(s)=\frac{1}{s}$$

$$\begin{aligned} \mathcal{L}[t^n] &= (-1)^n \frac{d^n}{ds^n} (s^{-1}) = (-1)^n (-1)(-2) \cdots (-n) s^{-(n+1)} \\ &= \frac{n!}{s^{n+1}} \end{aligned}$$

□

$$\mathcal{L}[te^{-at}] = \frac{1}{(s+a)^2}$$

$$\mathcal{L}[f(at)] = \frac{1}{a} \hat{f}\left(\underset{\substack{\uparrow \\ s=\sigma+i\omega}}{s/a}\right) \quad \text{for } a>0$$

$$\mathcal{L}[e^{at} f(t)] = \hat{f}(s-a)$$

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(Laplace Transform)

§4.7.3. Ex. of LT to some differential eq.s

$$\ddot{f} + 5\dot{f} + 6f = 1, \quad f(0) = \dot{f}(0) = 0$$

↳ Eq. (**)

$$\mathcal{L}[\ddot{f} + 5\dot{f} + 6f] = \mathcal{L}[1]$$

$$s^2 \hat{f} - s \cancel{f(0)} - \cancel{\dot{f}(0)} + 5[s \hat{f} - \cancel{f(0)}] + 6\hat{f} = \frac{1}{s}$$

$$(s^2 + 5s + 6) \hat{f} = (s+2)(s+3) \hat{f} = \frac{1}{s}$$

$$\therefore \hat{f}(s) = \frac{1}{s(s+2)(s+3)} = \frac{1}{6s} - \frac{1}{2(s+2)} + \frac{1}{3(s+3)}$$

$$\begin{aligned} \rightarrow f(t) &= \frac{1}{6} \cdot 1 - \frac{1}{2} e^{-2t} \cdot 1 + \frac{1}{3} e^{-3t} \cdot 1 \\ &= \frac{1}{6} - \frac{1}{2} e^{-2t} + \frac{1}{3} e^{-3t} \end{aligned}$$

Check this $f(t)$ satisfies (**) & the B.C.s.

§4.7.4. Convolution Thm.

$$f_1 * f_2 = \int_{-\infty}^{\infty} f_1(t') f_2(t-t') dt'$$

If f_1 & f_2 vanish for $t < 0$, then

$$f_1 * f_2 = \int_0^t f_1(t') f_2(t-t') dt'$$

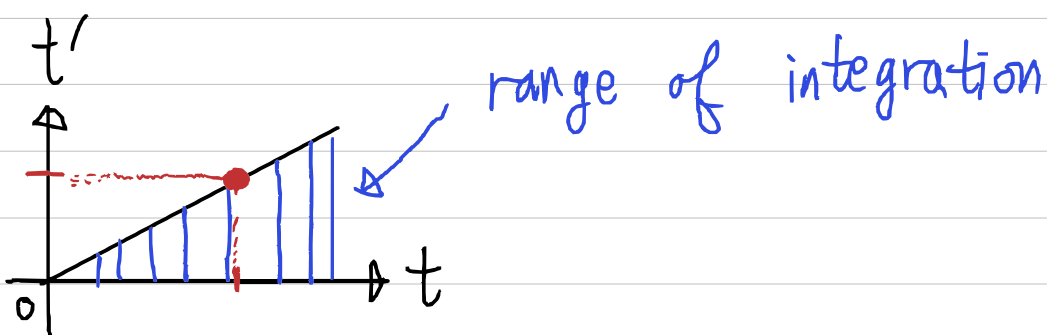
Thm.

If $g(t) = f_1 * f_2(t)$, then $\hat{g}(s) = \hat{f}_1(s) \hat{f}_2(s)$.

Proof.

$$\hat{g}(s) = \int_0^{\infty} dt e^{-st} \int_0^t f_1(t') f_2(t-t') dt'$$

$$= \int_0^{\infty} dt \int_0^t dt' f_1(t') f_2(t-t') e^{-st}$$



$$\hat{g}(s) = \int_0^{\infty} dt' \int_{t'}^{\infty} dt f_1(t') f_2(t-t') e^{-st}$$

Let $t - t' = t''$

$$= \int_0^{\infty} dt' f_1(t') \int_0^{\infty} dt'' f_2(t'') e^{-st' + t''}$$

$$= \int_0^{\infty} dt' f_1(t') e^{-st'} \int_0^{\infty} dt'' f_2(t'') e^{-st''}$$

$$= \hat{f}_1(s) \hat{f}_2(s)$$

□

(10)

EX,

Evaluate $J = \int_0^1 y^n (1-y)^m dy$

We consider

$$g(t) = t^n * t^m = \int_0^t \tau^n (t-\tau)^m d\tau$$

$$g(1) = J$$

Substituting $\tau = st$
 $d\tau = t ds$

$$g(t) = \int_0^1 s^n t^n (t-ts)^m t ds = t^{n+m+1} \underbrace{\int_0^1 s^n (1-s)^m ds}_{=J}$$

$$= t^{n+m+1} J$$

$$\hat{g}(s) = J \frac{(n+m+1)!}{s^{n+m+2}}$$

Using the convolution thm,

$$\hat{g}(s) = \frac{n!}{s^{n+1}} \frac{m!}{s^{m+1}} = J(n+m+1)!$$

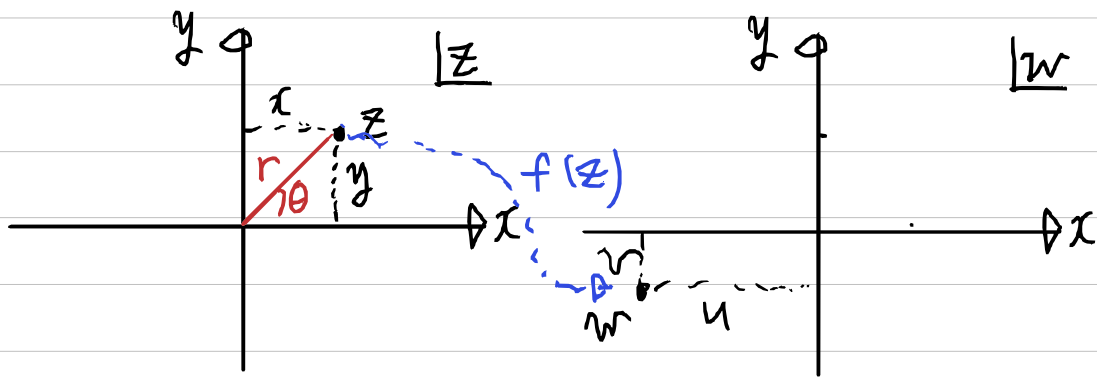
MMP L13

①
11/11/2024

§1. Complex Fns of a complex variable

$$f: D \rightarrow \mathbb{C}$$

domain complex



$$z = x + iy = re^{i\theta}$$

$$\text{e.g., } f(z) = \frac{1}{z}$$

$$f(z) = u + iv = u(x, y) + i v(x, y)$$

$$u, v \in \mathbb{R}$$

$$z = re^{i\theta} = re^{i\theta + 2\pi ni}$$

$$n \in \mathbb{Z}$$

$$r = |z| \quad \& \quad \theta = \text{Arg}(z) \xrightarrow{\theta \text{ is ambiguous}} \text{"argument"}$$

(2)

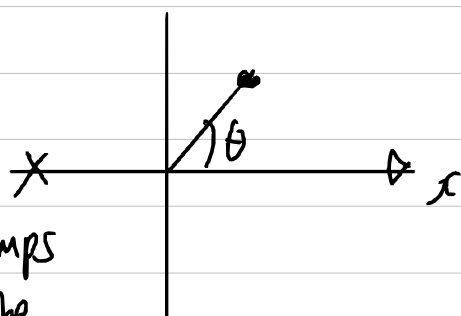
Fix ambiguity by a choice of branch.

Define $-\pi < \theta \leq \pi$

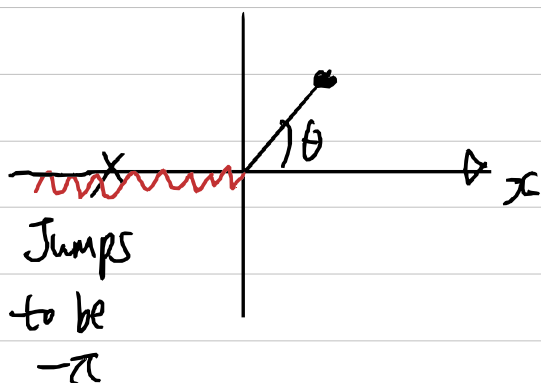
$$(-\pi < \text{Arg}(z) \leq \pi)$$

(The principal branch of argument of z)

Jumps to be $-\pi$



$\text{Arg}(z)$ is discontinuous across the neg x axis.



EX 5 (i) $f(z) = |z|$

(ii) $f(z) = \text{Arg}(z)$

(a) $f(z) = z^3 = (x+iy)^3$
 $= x^3 - 3xy^2 + i(3x^2y - y^3)$
 $= u(x,y) + i v(x,y)$

(b) $g(z) = z^{1/3} (= f^{-1})$

$$= \begin{cases} (|z| e^{i\theta})^{1/3} \\ (|z| e^{i\theta+2\pi i})^{1/3} \\ (|z| e^{i\theta+4\pi i})^{1/3} \end{cases}$$

where $\theta = \text{Arg}(z)$

$\downarrow$

$$= \begin{cases} |z|^{1/3} e^{i\theta/3} \\ |z|^{1/3} e^{(i\theta+2\pi i)/3} \\ |z|^{1/3} e^{(i\theta+4\pi i)/3} \end{cases} \quad \rightarrow 3 \text{ diff. val-s for } z^{1/3}$$

(c)

$$\left. \begin{aligned} e^z &= 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \\ \cos z &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} + \dots \\ \ln(z+1) &= z - \frac{z^2}{2} + \frac{z^3}{3} + \dots \end{aligned} \right\} \begin{aligned} &\forall z \in \mathbb{C} \\ &\rightarrow |z| < 1 \end{aligned}$$

Can't assume familiar properties from real fn's.

ex $e^{i\pi} = -1 < 0$

(3)

§2. Continuity, Differentiability, & Analyticity

When does $f'(z) = \frac{df}{dz}$ exist?

§2.1. Definitions

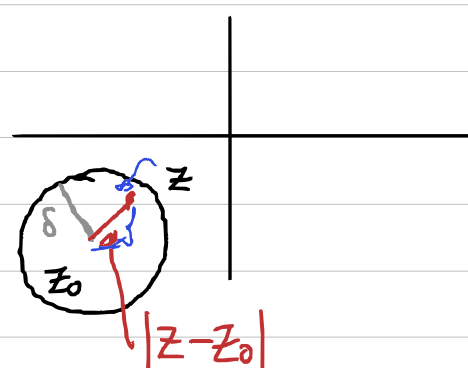
Eq. (2.3)

$f(z)$ is continuous at $z_0 \in D$ if, for every $\epsilon > 0$,
 $\exists \delta > 0$, s.t.

$$|z - z_0| < \delta \longrightarrow |f(z) - f(z_0)| < \epsilon$$

When this is so, we write

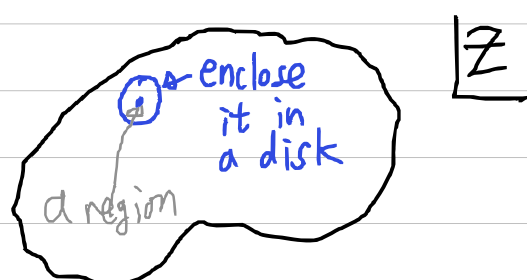
$$\lim_{z \rightarrow z_0} f(z) = f(z_0)$$



Def:

 $f(z)$ is differentiable at $z = z_0$ if $\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$ exists.& then we define the limit to be $\left. \frac{df}{dz} \right|_{z=z_0}$.Def. [Open subset]

A subset D of $\mathbb{R}$ is open if for each $z \in D$,
 $\exists$ a disk centered at z s.t. the whole disk
 is in D .



Def: A fn. f is analytic (or holomorphic)
 at $z_0 \in D$ if it's differentiable at every pt.
 in some open neighborhood of z_0 .

Domain $\longrightarrow$ open subset

A fn. is analytic in a domain D if it's
 analytic at every pt. z in D .

Excn Replacing $f(z) \longleftrightarrow z^2$, prove: when $f(z) = z^2$
 $f'(z) = 2z$ for $\forall z \in \mathbb{C}$

Def: a fn. analytic in $\mathbb{C}$ is called entire.

If $f(z)$ & $g(z)$ are analytic at z_0 , then
 $f(z)g(z)$ is analytic at z_0

& $\frac{f(z)}{g(z)}$ is analytic at z_0 if $g(z_0) \neq 0$.

Any polynomial in z is entire. e^z in z is entire. $\cos z$ in z is entire.

Ex of non-analytic fn.

$f(z) = z^*$
is not analytic anywhere.

§2.2. Cauchy-Riemann Conditions

When is a fn. $f(z)$ analytic in D ?

$$f(z) = u(x, y) + i v(x, y)$$

We assume: u_x, u_y, v_x, v_y all exist.
(thru out D)

$$\uparrow$$

$$\frac{\partial u}{\partial x}$$

Suppose $f(z)$ is differentiable at $z = z_0$

$$f' = \frac{df}{dz} \text{ exists at } z_0.$$

$$z = x + iy$$

$$\frac{\partial f}{\partial x} = \frac{df}{dz} \frac{\partial z}{\partial x} = f' = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{df}{dz} \frac{\partial z}{\partial y} = i f' = \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y}$$

$$i(u_x + i v_x) = u_y + i v_y$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Cauchy-Riemann Eqs

at $z = z_0$

It can be proved that $f(z)$ is differentiable at $z = z_0$

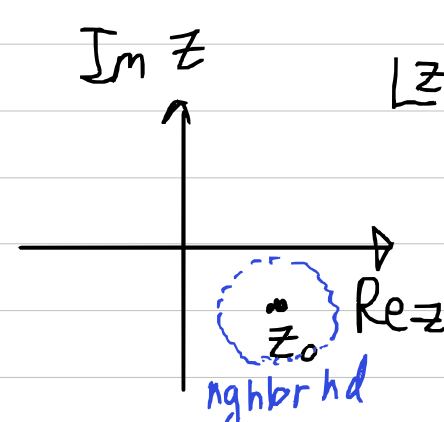
iff C-R eqs hold there.

→ A fn. f is analytic in domain D if
C-REs hold at every $z \in D$.

$$z = x + iy$$

$$f(z) = u(x, y) + i v(x, y)$$

$$\text{CR eq.s: } \left. \begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x} \end{aligned} \right\}$$



CR eq.s hold at $z = z_0 \iff f$ is differentiable at z_0

CR eq.s hold in a neighborhood of z_0

$\iff f$ is analytic at z_0

Exs. (a) $f(z) = z^2 = \underbrace{x^2 - y^2}_u + i \underbrace{(2xy)}_v = u + iv$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 2x \quad \checkmark$$

$$\frac{\partial u}{\partial y} = -2y, \quad -\frac{\partial v}{\partial x} = -2y \quad \checkmark$$

(b) $f(z) = z^* = x - iy$

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1 \longrightarrow \text{CRX}$$

not differentiable anywhere $\rightarrow$ Not analytic

(c) $f(z) = \text{Re}(z) = \frac{1}{2}(z + z^*)$

$\rightarrow$ not analytic

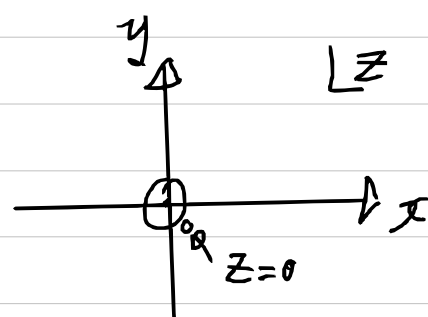
(d) $f(z) = |z|^2 = x^2 + y^2 = z z^*$

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial u}{\partial y} = 2y, \quad -\frac{\partial v}{\partial x} = 0$$

CREs hold only at $z=0$

so not analytic at $z=0$



We've a huge store of analytic fn's, e.g., $f(z) = z^n$
 $n \in \mathbb{N}$

$$f(z) = z^{-1} = \frac{1}{z} \rightarrow \text{analytic everywhere except at } z=0$$

$$f(z) = e^z \rightarrow \text{analytic}$$

$$f(z) = e^{z^2} \rightarrow \text{analytic}$$

$$f(z) = \frac{e^{z^2}}{z-5} \rightarrow \text{analytic except at } z=5$$

Thm. $f(z)$ is analytic at $z = z_0$ iff

f has a convergent power series expansion in some neighborhood of z_0 .

$$f(z) = c_0 + c_1(z - z_0) + c_2(z - z_0)^2 + \dots$$

$$= \sum_{n=0}^{\infty} c_n (z - z_0)^n$$

$\therefore$ Taylor Series



Thm. If the above holds then

$$C_n = \frac{1}{n!} f^{(n)}(z_0)$$

$\uparrow$
 $\frac{d^{(n)}f}{dz^{(n)}}$

analyticity $\rightarrow$ stringent
 $\rightarrow$ restricted

$\rightarrow$ In physics, there's debate whether using analytic func.s are good. (too restricted)

Results $e^z, \cos z, \cosh z, \ln(1+z)$ are all analytic. ($|z| < 1$)

$$\frac{d}{dz}(e^z) = e^z, \quad \frac{d}{dz}(\cos z) = -\sin z, \dots$$

§2.4. Harmonic fns

Def. A fn $w(x,y)$ is **harmonic** if

$$\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} = 0 = \nabla^2 w = 0$$

where $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right)$

Claim. If $f = u + iv$ is analytic, u & v are **harmonic**.

check C-RE.
Inverse: not sufficient

Prf:

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial u}{\partial x} \stackrel{CR}{=} \frac{\partial}{\partial x} \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \frac{\partial v}{\partial x}$$
$$\stackrel{CR}{=} \frac{\partial}{\partial y} \left(-\frac{\partial u}{\partial y}\right) = -\frac{\partial^2 u}{\partial y^2}$$
$$\nabla_x^2 u = -\nabla_y^2 u \quad u: \text{harmonic}$$

Ex. $\frac{\partial^2 v}{\partial x^2} = -\frac{\partial}{\partial x} \frac{\partial u}{\partial y} = -\frac{\partial}{\partial y} \frac{\partial u}{\partial x} = -\frac{\partial}{\partial y} \frac{\partial v}{\partial y}$

$v \rightarrow \text{harmonic} \quad \nabla_x^2 v = -\nabla_y^2 v$

Ex. Let $u(x,y) = \cos x \cosh y$.

(a) Show u is real part of a analytic fn. f & find f .

C-RE $\frac{\partial^2 u}{\partial x^2} = -\frac{\partial^2 u}{\partial y^2}$

(b) $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y} = -\cos x \sinh y$

$$\hookrightarrow v = \int (-\cos x) \sinh y \, dx + C_1(y)$$
$$= -\sin x \sinh y + C_1(y)$$

$$\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x} = -\sin x \cosh y$$

$$v = -\sin x \sinh y + C_2(x)$$

$$C_1(y) = C_2(x) = \text{both constant} = C$$

$$v = -\sin x \sinh y + C$$

$\therefore f = \cos x \cosh y - i \sin x \sinh y + C$
 $= u + iv$

$\hookrightarrow$ Write in terms of z .
(task)

§3. Multi-valued fns & Riemann surfaces

③

§3-1. Branches

Ex. $f(z) = z^{1/3}$ = "cube root of z "

choices (1) $f(z) = |z|^{1/3} e^{i\theta/3}$

$\theta = \text{Arg}(z)$: principal branch

$$(2) f(z) = |z|^{1/3} e^{i\theta/3 + 2\pi i/3}$$

$$(3) f(z) = |z|^{1/3} e^{i\theta/3 + 4\pi i/3} \quad (-\pi < \theta \leq \pi)$$

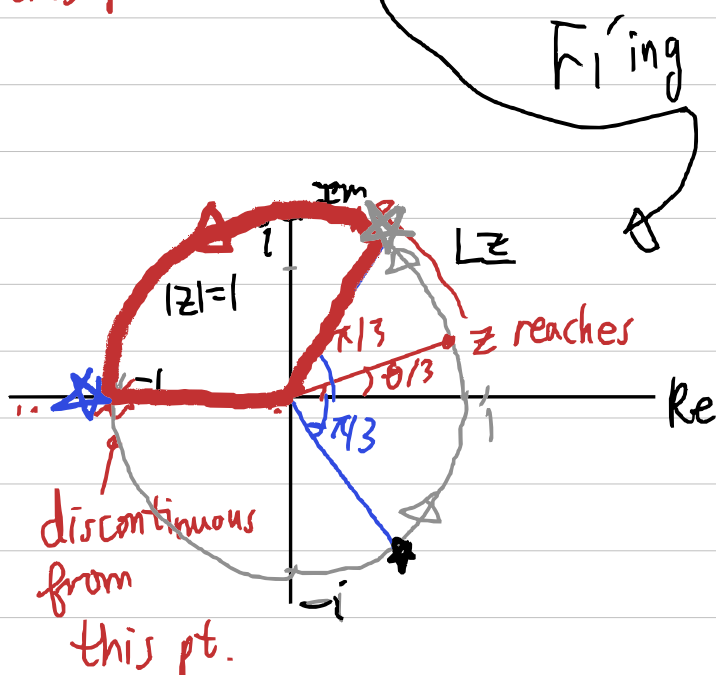
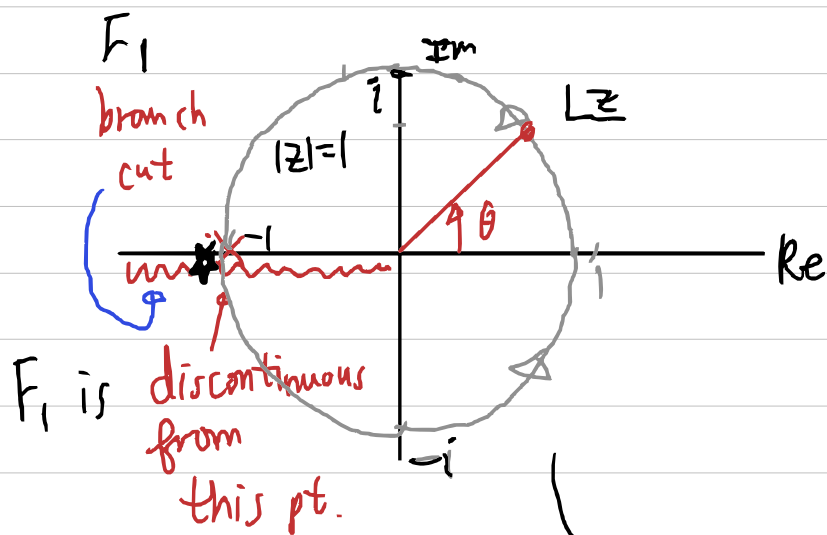
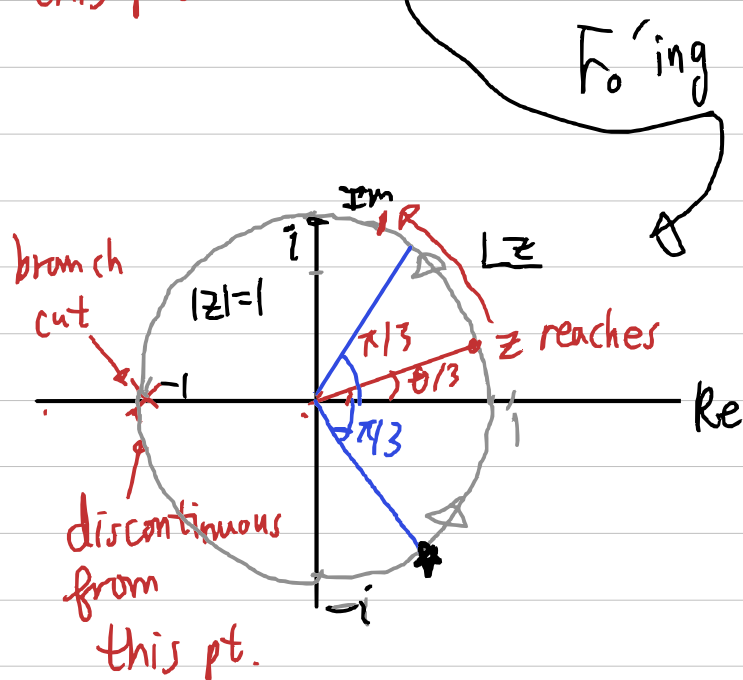
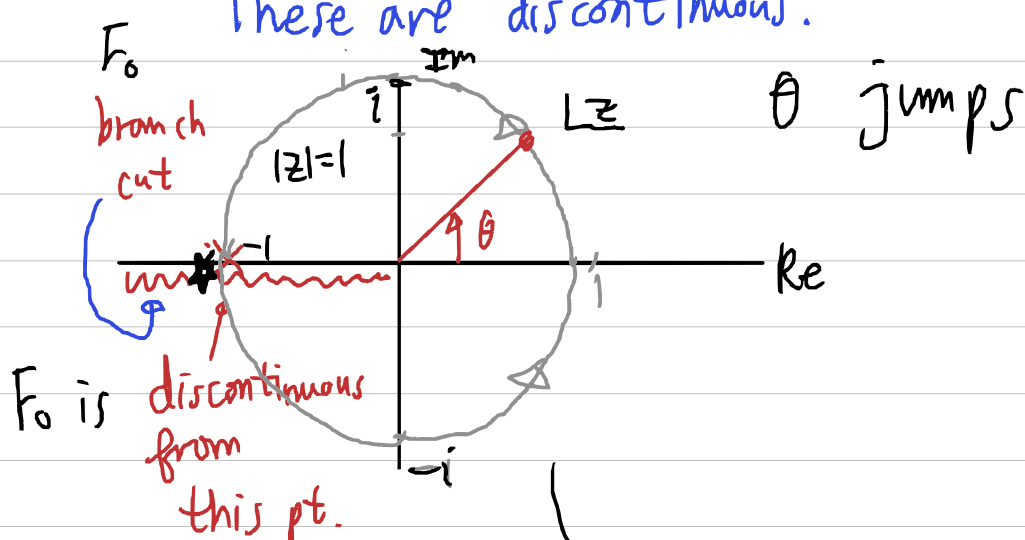
these are Ambiguous fns $\rightarrow$ well-defined fns are better to do maths!

3 branches : $F_0(z) = |z|^{1/3} e^{i\theta/3} e^{0 \cdot 2\pi i/3}$

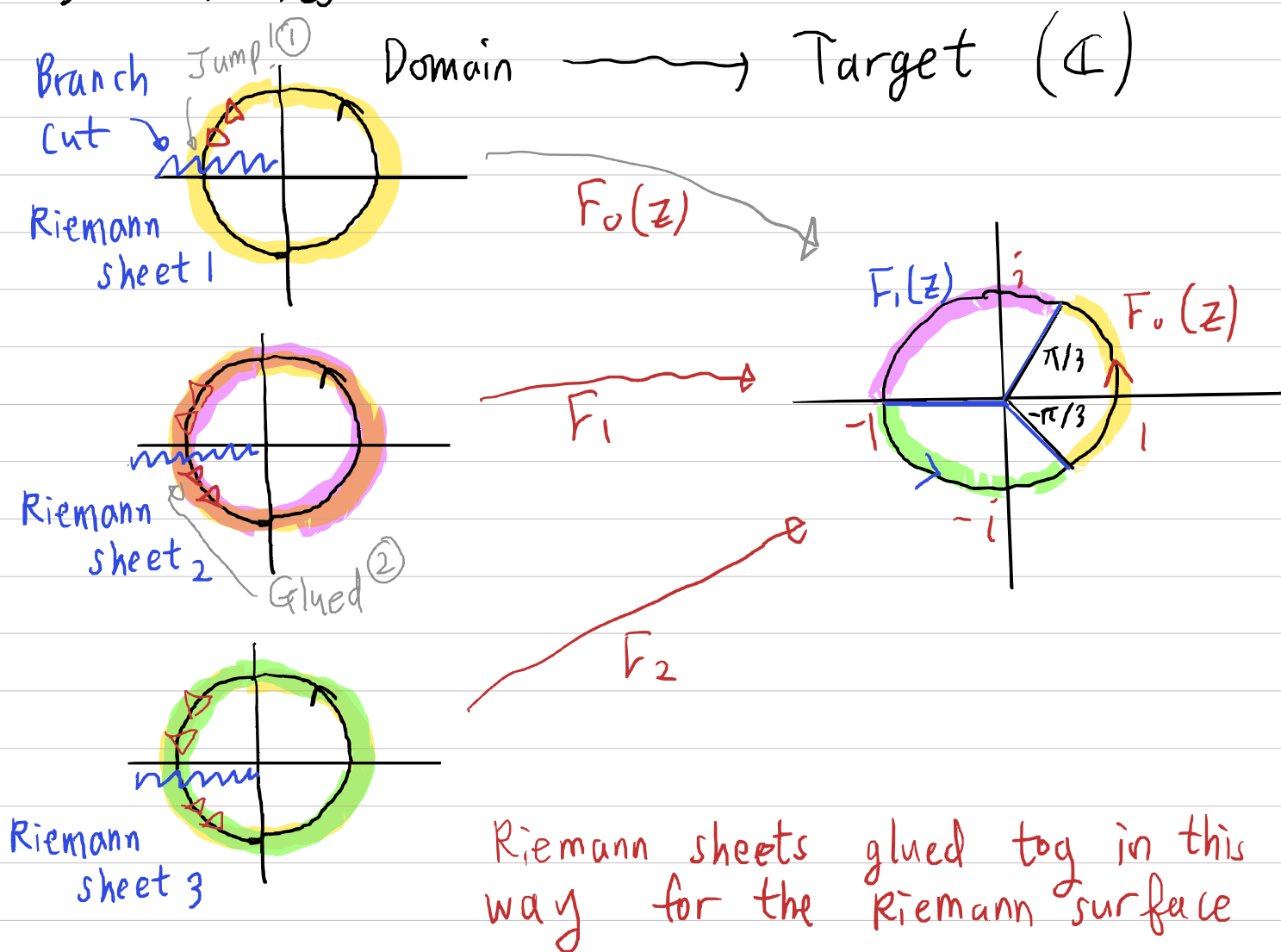
$$F_1(z) = |z|^{1/3} e^{i\theta/3 + 2\pi i/3}$$

$$F_2(z) = |z|^{1/3} e^{i\theta/3 + 4\pi i/3}$$

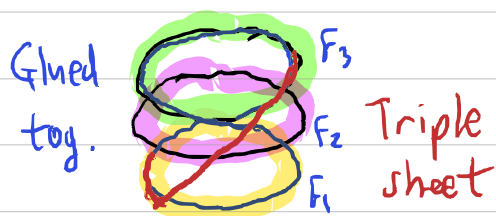
These are discontinuous.



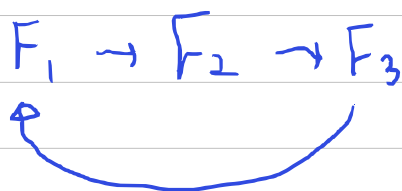
§ 3.1. Branches



Riemann sheets glued tog in this way for the Riemann surface for $f(z) = z^{1/3}$.



f is now single-valued & continuous.



$$f(z) = z^{1/2}$$

double-sheeted Riemann surf.

$$f(z) = z^{1/n}$$

n -sheeted Riemann surf.

②

EX 2. $f(z) = \ln z = \ln(|z|e^{i\theta + 2\pi ni})$

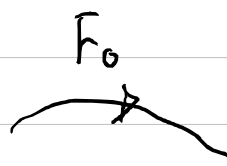
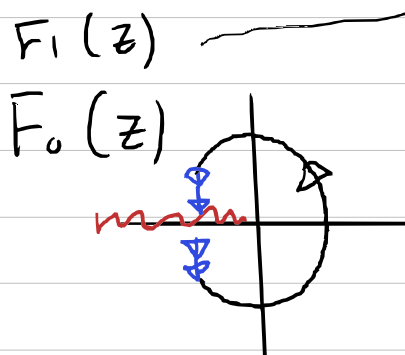
$z \neq 0$

$-\pi < \theta \leq \pi$

$n \in \mathbb{Z}$

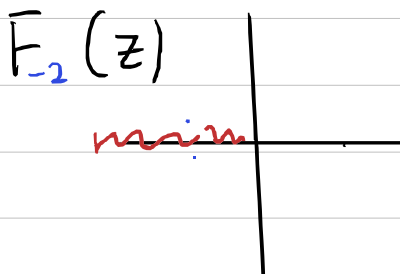
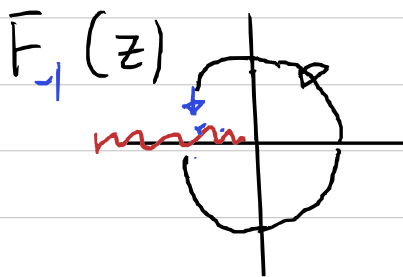
$f(z) = \ln|z| + i\theta + 2\pi ni = F_n(z)$

$F_n(z)$: infinitely many branches



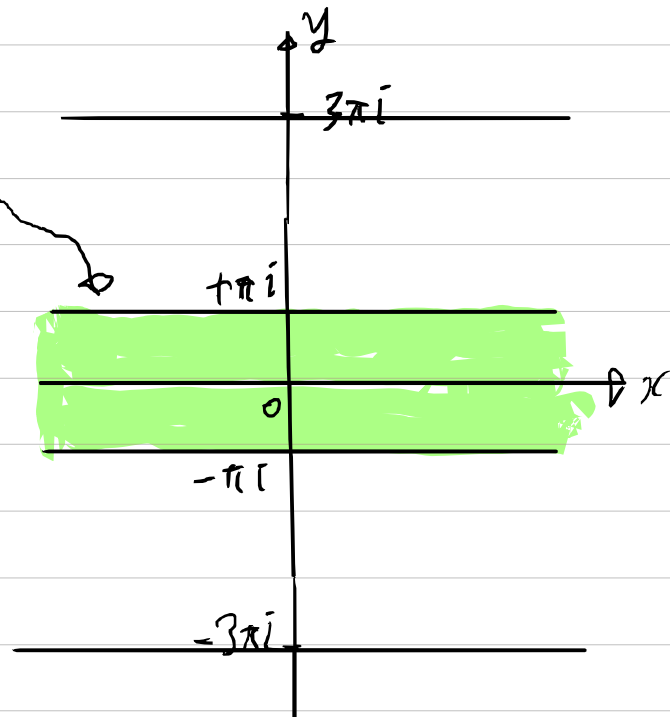
For $n=0$,

$F_0(z) = \ln|z| + i\theta$



∞ by many sheets in

Riemann surf. for $\ln z$.



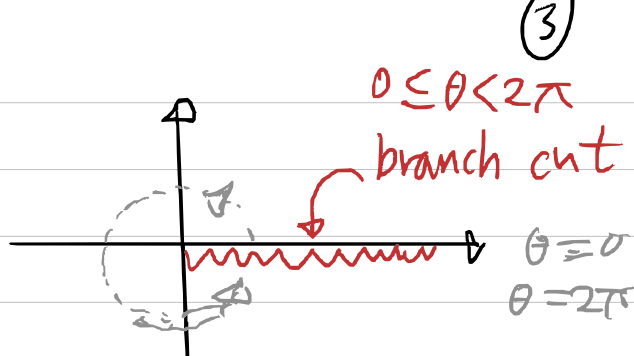
$\ln(z)$ is discontinuous at the branch cut.

$f(0) = 0^{1/3} = 0 \rightarrow 0 = \text{a branch pt.}$

(end to a branch cut)
(where a branch cut ends.)

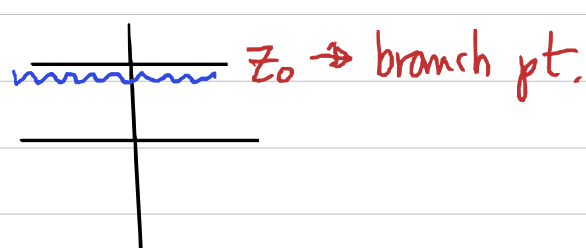
$\ln|z| + i\theta$
(Real) (Im)

★ Choices of branch cuts

$0 \leq \theta < 2\pi$ $\longleftrightarrow$ 

$$f(z) = z^{1/3}$$

EX $f(z) = (z - z_0)^{1/3}$ & $z = z_0 + |z - z_0| e^{i\theta}$



EX $f(z) = (z-a)^{1/2} (z+a)^{1/2}$ $a: \text{real}$

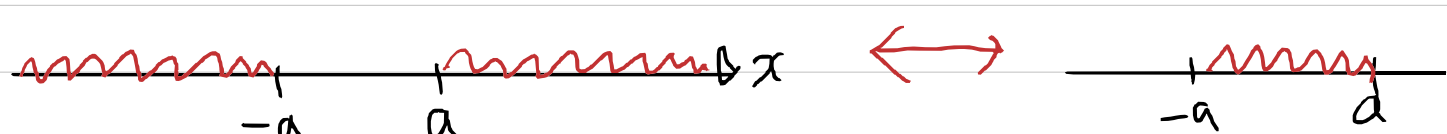


Fig. 3.4.

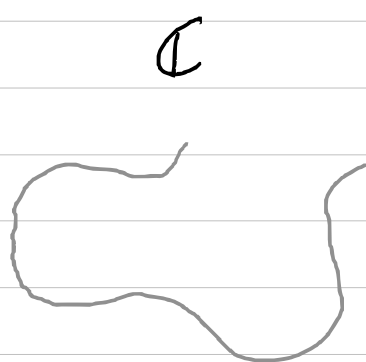
C4. Integration of complex func.s

§4.1. Contours

$$\oint_C f(z) dz$$

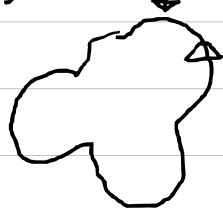
Contour:
a continuous
1D curve
in $\mathbb{C}$

& with a direction



some books write

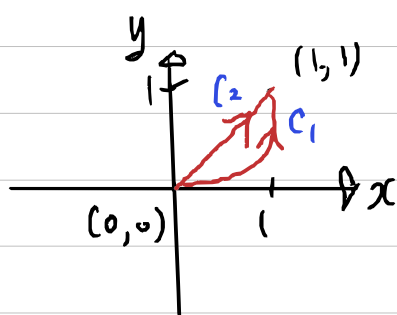
$\oint$: closed



but Prof. Dowker's notation

$\oint$: any contour
(open & closed)

EX. $f(z) = z$



C_1 is $y = x^2$
 C_2 is $y = x$

$$I_1 = \oint_{C_1} z dz \quad \& \quad I_2 = \oint_{C_2} z dz$$

$$z = x + iy \quad \& \quad dz = dx + i dy$$

$$I_j = \oint_{C_j} (x + iy)(dx + i dy)$$

$$= \oint_{C_j} (x dx - y dy + i y dx + i x dy)$$

$(j=1,2)$

$$\therefore I_1 : dy = 2x dx$$

Can do
 $y(x)$
 $\downarrow$

$$I_1 = \int_0^1 (x dx - 2xy dx + i y dx + i(2x^2) dx)$$

$$= \frac{1}{2} x^2 \Big|_0^1 - x^2 y \Big|_0^1 + i y x \Big|_0^1 + \frac{i}{3} x^3 \Big|_0^1$$

$$= \frac{1}{2} - y + i y + \frac{i}{3}$$

EXCR:

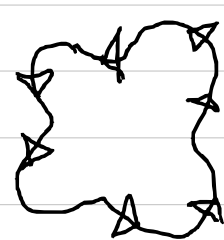
show $I_1 = I_2$

§4.2. Cauchy's Theorem

If $f(z)$ is analytic inside & on a closed contour C , then

$$\oint_C f(z) dz = 0$$

N.B.: It's conventional to integrate around a closed contour CCW



Proof.

$$f(z) = u(x, y) + i v(x, y)$$

$$z = x + iy$$

$$dz = dx + i dy$$

$$\begin{aligned} \oint_C (u + i v) (dx + i dy) &= \oint_C (u dx - v dy) \\ &+ i \oint_C (v dx + u dy) \end{aligned}$$

Green's fn. in the plane (version of Stokes' Theorem)



boundary of R is contour C

Stokes

$$\oint_C (P(x, y) dx + Q(x, y) dy) = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$

① $v = P, u = -Q$ $\xrightarrow{\text{Green's}}$ $\iint \left(-\frac{\partial u}{\partial y} - \frac{\partial v}{\partial x} \right) dx dy$

② $u = P, v = Q$ $\xrightarrow{\quad}$ $+i \iint \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy$

$$= 0 \text{ by C-REs}$$

§4.3. Consequences

§4.3.1. Path Independence

Let C_1 & C_2 be contours from z_a to z_b ,

s.t. f is analytic on C_1 & C_2

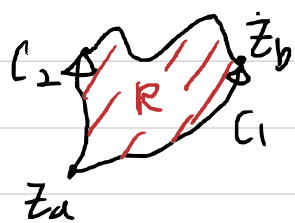
& can deform C_1 into C_2

a region in which f is analytic.

Then,

$$\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$$

Proof.



Let $C = C_1 - C_2$

go on C_1

& then C_2 backwards.

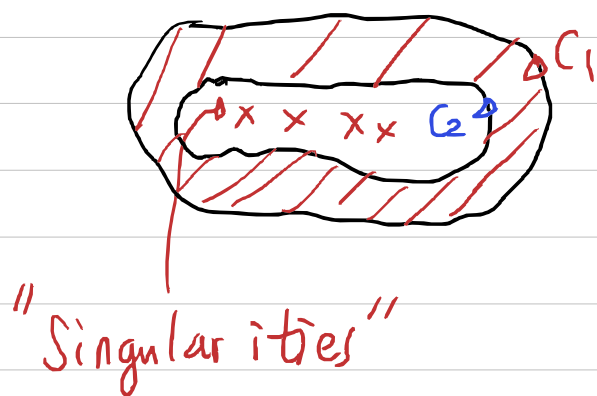
C is closed, f is analytic on & inside C ,

& so by Cauchy's Thm,

$$0 = \oint_C f(z) dz = \oint_{C_1} f(z) dz - \oint_{C_2} f(z) dz$$

§4.3.2. Closed contour deformation thm

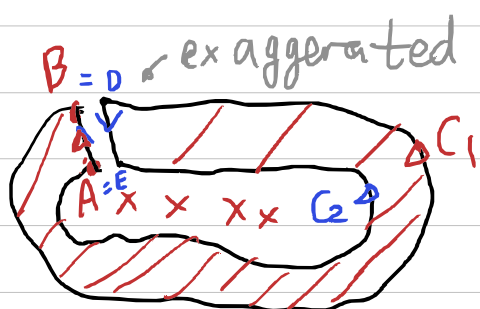
(2)



If C_1 & C_2 are deformed
& f is analytic on C_1 & C_2
& between them, so C_1 can

be deformed into C_2 , then
a region in which f is
analytic, then

$$\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz.$$



$A \rightarrow B \rightarrow D \rightarrow E \rightarrow A$
Closing

Let C : closed contour.

$$C = C_1 \text{ (from } B \text{ to } D) + \overline{DE} - C_2 \text{ (from } E \text{ to } A) + \overline{AB}$$

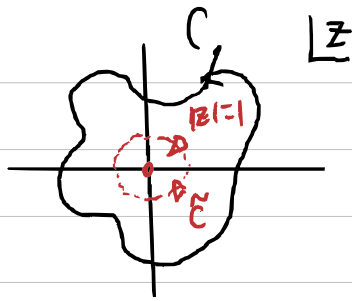
f is analytic in & on C , so by Cauchy's Thm:

$$0 = \oint_C f dz = \left(\oint_{C_1} + \overbrace{\oint_{\overline{DE}} - \oint_{C_2}}^{\text{cancel}} + \int_{\overline{AB}} \right) f dz$$

$$= \oint_{C_1} f dz - \oint_{C_2} f dz = 0$$

□

(3)



$\tilde{C}$ is a unit circle

$$\oint_C \frac{1}{z} dz$$

$$= \oint_{\tilde{C}} \frac{1}{z} dz$$

$$= \int_0^{2\pi} \cancel{e^{-i\theta}} i \cancel{e^{i\theta}} d\theta$$

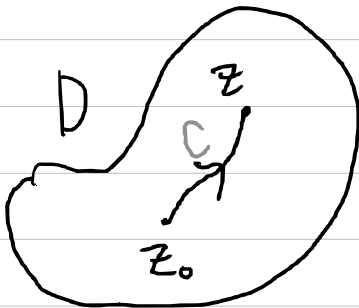
$$= 2\pi i$$

↑

$$z = e^{i\theta}$$

$$dz = ie^{i\theta} d\theta$$

§4.3.3. Fundam. Thm of Calc.



f is analytic on D .

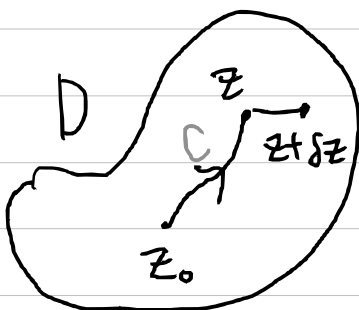
$$z_0 \in D$$

Define

$$F(z) = \int_{z_0}^z f(\tilde{z}) d\tilde{z}$$

along
any contour C in D

Thm. $F(z)$ is analytic & $f(z) = F'(z)$



§4.3.4. Cauchy's Integral Thm

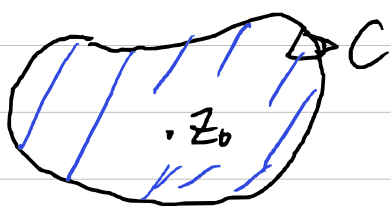
④

Thm. If f is analytic in & on a closed contour C
& z_0 is in the interior of C , then

$$f(z_0) = \frac{1}{2\pi i} \oint_C \boxed{\frac{f(z)}{z-z_0}} dz$$

→ The integrand

at $z=z_0$, there's a pole.



↓
The fn. knows the val. because it's analytic.

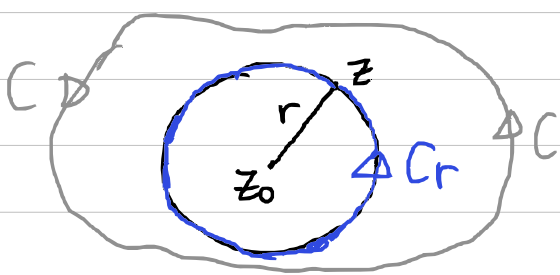
Proof

Consider small circles C_r of rad. r ,
($r \rightarrow 0$) around z_0 .

$$\text{Define: } I(r) = \oint_{C_r} \frac{f(z)}{z-z_0} dz = \int_0^{2\pi} \frac{f(z_0 + re^{i\theta})}{\cancel{re^{i\theta}} \cancel{ie^{i\theta} d\theta}} re^{i\theta} i d\theta$$

$$\text{On } C_r, \quad z = z_0 + re^{i\theta}$$

$$\rightarrow dz = re^{i\theta} i d\theta$$



$$I(r) = i \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta \quad \text{is independent of } r.$$

$$\therefore \oint_C \frac{f(z) dz}{z-z_0} = \lim_{r \rightarrow 0} I(r) = 2\pi i f(z_0) \quad \square$$

§4.3.5.

$$f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-z_0} dz$$

$$\rightarrow f'(z_0) = \frac{1!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^2} dz$$

$$\rightarrow f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^3} dz$$

⋮

$$\rightarrow f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz$$

④

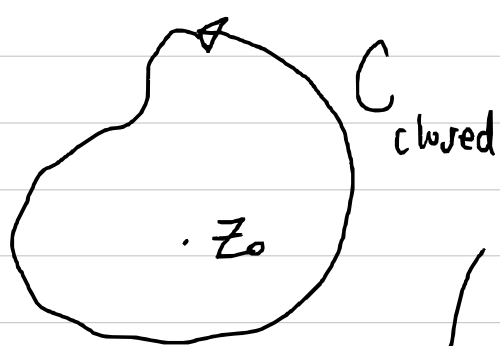
Let $f=1$.

$$1 = \frac{1}{2\pi i} \int \frac{1}{z-z_0} dz$$

$$\int \frac{1}{z-z_0} dz = 2\pi i$$

$$\int \frac{1}{(z-z_0)^n} dz = 0 \quad \text{for } n \geq 2.$$

Cauchy's Integral Thm

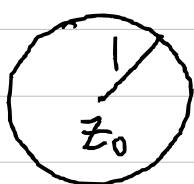


$f(z)$ analytic in & on C
 z_0 inside C

$$\oint_C \frac{1}{(z-z_0)} dz = 2\pi i \quad \left\{ \begin{array}{l} f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-z_0} dz \\ f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz \end{array} \right.$$

Not analytic at $z=z_0$

$$\oint_C \frac{1}{(z-z_0)^n} dz = 0 \quad \text{for } n \geq 2 \quad \rightarrow \text{Check.}$$



make C , unit circle around z_0 : $z = z_0 + e^{i\theta}$
 $dz = i d\theta e^{i\theta}$

§4.3.6. Liouville's Thm.

If $f(z)$ is entire & $\exists M > 0$
 s.t. $|f(z)| < M$, $\forall z \in \mathbb{C}$, then
 f is const.

Proof. Let $z_0 \in \mathbb{C}$. We'll show that
 $f'(z_0) = 0$

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^2} dz$$

where C is a circle of rad. r centered on z_0 .

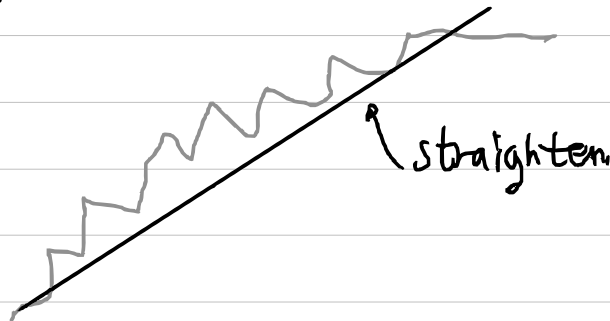
$$z = z_0 + r e^{i\theta} \quad \rightarrow \quad dz = i r e^{i\theta} d\theta$$

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z_0 + r e^{i\theta})}{r^2 e^{2i\theta}} i r e^{i\theta} d\theta$$

$$= \frac{1}{2\pi} \oint_C \frac{f(z_0 + r e^{i\theta})}{r e^{i\theta}} d\theta$$

$$= \frac{1}{2\pi r} \int_0^{2\pi} f(z) e^{-i\theta} d\theta$$

$$|f'(z_0)| \leq \frac{1}{2\pi r} \int_0^{2\pi} |f(z) e^{-i\theta}| d\theta$$



$$\frac{1}{2\pi r} \int_0^{2\pi} |f(z) e^{-i\theta}| d\theta = \frac{1}{2\pi r} M \underbrace{\int_0^{2\pi} d\theta}_{= 2\pi}$$

$$= M/r$$

$$\therefore |f'(z_0)| < \frac{M}{r}$$

$\rightarrow$ r can be chosen as big as we like.

$$\therefore |f'(z_0)| = 0$$

□

Fundam. Thm of Algebra Eq. (4.56)

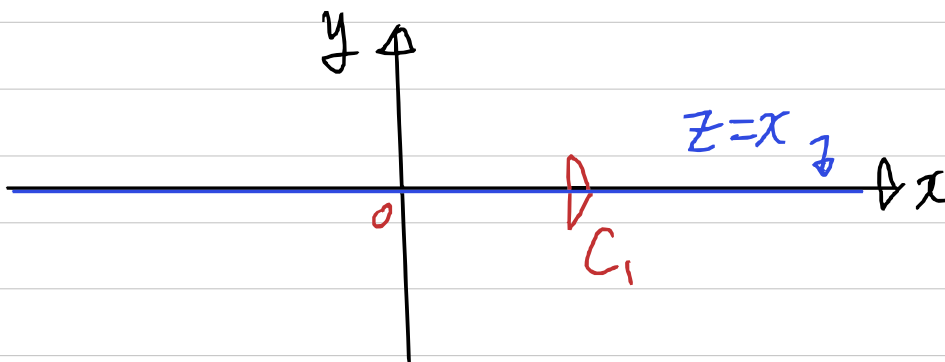
(2)

* F.T. of Gaussian : we used this claim

$$\int_{-\infty}^{\infty} e^{-(x'+ia)^2} dx' = \int_{-\infty}^{\infty} e^{-x^2} dx$$

Proof : Use Cauchy's Thm

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \oint_{C_1} e^{-z^2} dz$$



$$\int_{-\infty}^{\infty} e^{-(x'+ia)^2} dx' = \oint_{C_2} e^{-z^2} dz$$

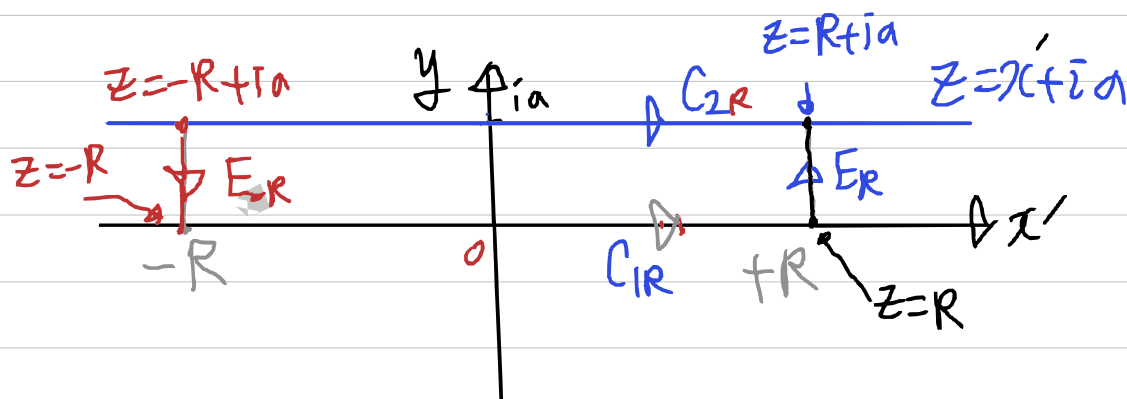


Prof's notation
 $\oint$ is not only closed
 $\oint_C \rightarrow$ contour integral

e^{-z^2} is entire $\rightarrow$ analytic everywhere.

Need closed contour — have to use a limiting argument.

(3)



$$C_R = C_{1R} + E_R - C_{2R} + E_{-R}$$

$$0 = \oint_C e^{-z^2} dz \quad \text{Eq. (4.60)}$$

$$= \left(\oint_{C_{1R}} - \oint_{C_{2R}} + \underbrace{\oint_{E_R} + \oint_{E_{-R}}}_{\text{need to show } \rightarrow 0!!} \right) e^{-z^2} dz$$

$$0 = \lim_{R \rightarrow \infty} \oint_C e^{-z^2} dz$$

$$\rightarrow 0 = \oint_{C_{1R}} e^{-z^2} dz + \lim_{R \rightarrow \infty} \left(\oint_{E_R} + \oint_{E_{-R}} \right) e^{-z^2} dz - \oint_{C_{2R}} e^{-z^2} dz$$

on E_R $z = R + iy$ ④

$$\oint_{E_R} e^{-z^2} dz = \int_0^a e^{-(R+iy)^2} i dy$$

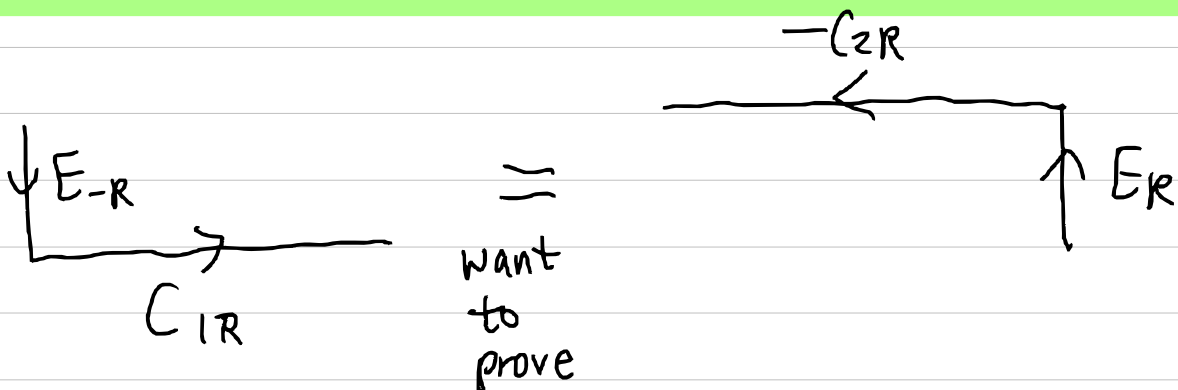
$$\left| \oint_{E_R} e^{-z^2} dz \right| \leq \left| \int_0^a e^{-(R+iy)^2} i dy \right|$$

all imaginary
terms
vanish

$$= e^{-R^2} \int_0^a e^{y^2} dy$$

$\rightarrow 0$ as $R \rightarrow \infty$

□



§5. Power Series

⑤

§5.1. A Taylor Series of an analytic fn. f

A Taylor Series of an analytic fn. f is a convergent power series:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

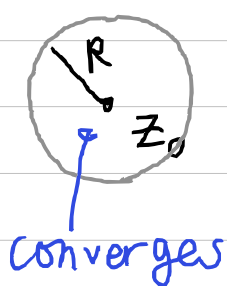
DEF: The rad. of convergence R around z_0

of a power series is the largest $R > 0$ s.t.

the power series converges at every pt. z

s.t. $|z - z_0| < R$.

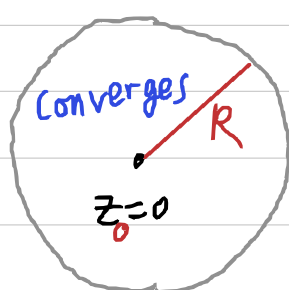
R is the MAX



Diverges at some pt.s
 $|z - z_0| = R$.

"does not converge"

EX. $\frac{1}{1-z} = 1 + z + z^2 + \dots = \sum_{n=0}^{\infty} z^n$



$R=1$ around $z_0=0$

diverges

$$\ln(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \dots$$

has rad. of convergence $R=1$ around $z_0=0$.

If $z=1$,

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

$\rightarrow$ converges for $z=1$

Thm.

Let $f(z)$ be analytic in the interior of disk of rad.

R centered on z_0 , which has a singularity on $|z - z_0| = R$.

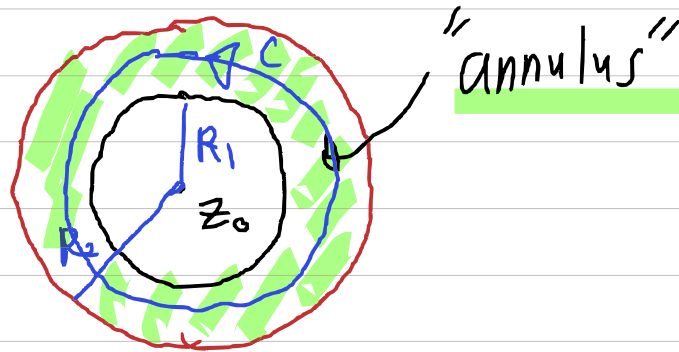
Then, f has a Taylor series in the disk:

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

§ 5.3 Laurent Series

(6)

Thm. Let f analytic b/t 2 circles of rad. R_1 & R_2 centered on z_0 :



Then, for $\forall z$ s.t. $R_1 < |z - z_0| < R_2$,

$\exists$ a convergent power series for f :

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z - z_0)^n \quad : \text{Laurent Series}$$

$$\text{where } A_n = \frac{1}{2\pi i} \oint_C \frac{f(z')}{(z' - z_0)^{n+1}} dz'$$

$$n \in \mathbb{Z}$$

& C is any contour in the annulus,
circles z_0 once.

N.B.

⑦

◦ If f is analytic inside the large circle
then

$$A_n = 0 \quad \text{for } n < 0 \\ \text{(-ve } n)$$

& Laurent series is just a Taylor series

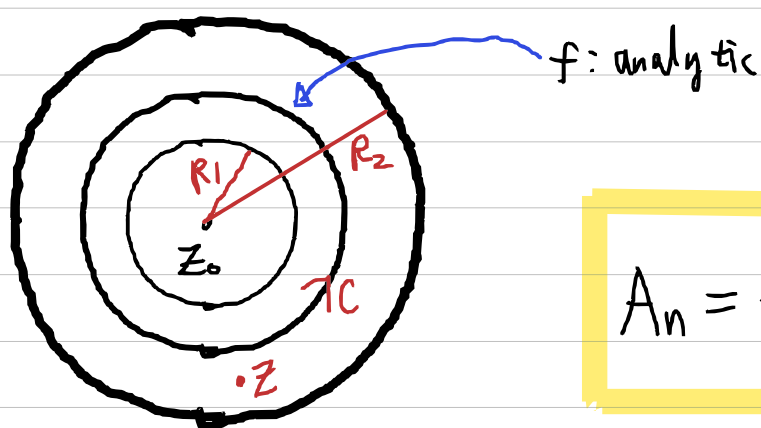
◦ If any $A_n \neq 0$ for $n < 0$,

then f is not analytic inside

$$|z - z_0| < R_1.$$

§5.3. Laurent Series

Laurent Series for analytic fns is a generalization of the Taylor series, when a fn is an annulus



$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(z')}{(z' - z_0)^{n+1}} dz'$$

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z - z_0)^n$$

$$A_n = \frac{1}{2\pi i} \oint_C \frac{f(z')}{(z' - z_0)^{n+1}} dz', \quad n=0, \pm 1, \pm 2, \dots$$

§5.4. Singularities

Consider f is analytic except at some isolated pts.

Apply Laurent series

in the case $R_1 = 0$,

so the fn. is analytic around z_0 for $0 < |z - z_0| < R_2$.

$$\rightarrow f(z) = \sum_{n=-\infty}^{\infty} A_n (z - z_0)^n$$

for $0 < |z - z_0| < R_2$.

Defn. If the negative powers terminate, i.e.,

$$f(z) = \sum_{n=-m}^{\infty} A_n (z - z_0)^n$$

then, we say $f(z)$ has a pole of order $m > 0$ at z_0 .

$$\begin{aligned} f(z) &= A_{-m}(z - z_0)^{-m} + A_{-m+1}(z - z_0)^{-m+1} \\ &\quad + \dots \\ &= (z - z_0)^{-m} \left[A_{-m} + A_{-m+1}(z - z_0)^1 + A_{-m+2}(z - z_0)^2 + \dots \right] \end{aligned}$$

Taylor Series

[] is analytic, call it $g(z)$

$$f(z) = \frac{g(z)}{(z - z_0)^m} = \frac{\text{a fn. that is analytic at } z_0}{(z - z_0)^m}$$

If $m=1$, we call it a **simple pole**.

$$f(z) = (z - z_0)^{-1} [A_{-1} + A_0(z - z_0)^1 + A_1(z - z_0)^2 + \dots]$$

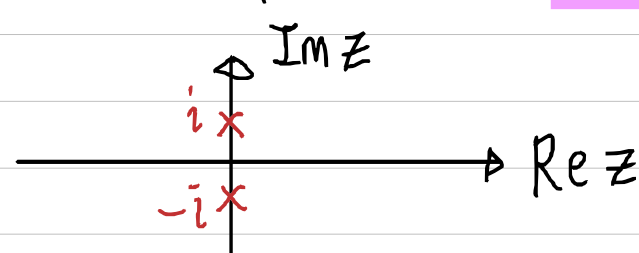
If $m=2$, we call it a **double pole**.

If $m=3$, we call it a **triple pole**.

(EX) $f(z) = \frac{e^z}{z-1}$ has a simple pole at $z=1$.

$$f(z) = \frac{\cos z}{z^2 + 1} = \frac{\cos z}{(z+i)(z-i)}$$

has 2 simple poles at $z = \pm i$.



$$f(z) = \frac{e^z}{(z - 5 + i)^3} \text{ has pole of order 3 (i.e., triple pole) at } z = 5 - i.$$

DEF.

If the series of neg. powers does NOT terminate, then f has an "essential" singularity at $z = z_0$.

$$f(z) = e^{1/z} \text{ at } z = 0.$$

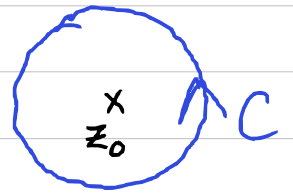
§6. Contour $\int$ using the Residue thm. (2)

§6.1. The Residue Thm.

DEF. Let f has an isolated singularity at $z=z_0$.

The residue of f at z_0 is

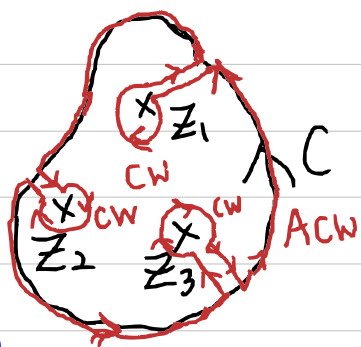
$$\text{Res}(z_0) = \frac{1}{2\pi i} \oint_C f(z) dz$$



Theorem. Let C be a closed contour w/
 f analytic on C & analytic inside C except at
 $z_1, z_2, \dots, z_N$.

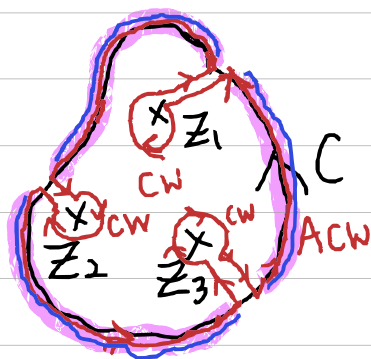
Then,

$$\oint_C f(z) dz = 2\pi i \sum_{k=1}^N \text{Res}(z_k)$$



By doing so, we have CW contours inside the
 ACW contour C .

PROOF. We construct C' , which is the contour shown



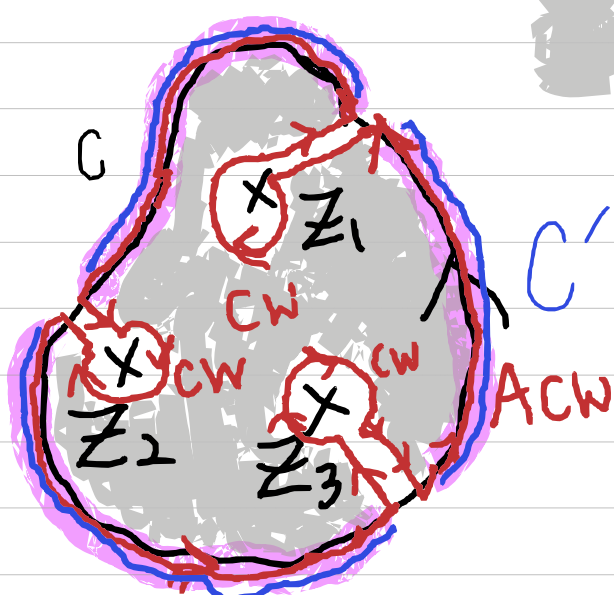
contour
 Pink : C'
 (or Blue lines)

$C' = C$ in pieces (yellow bits)

+ $2N$ line segments in opposite pairs.

one going
 the
 other
 way
 (2x)
 one
 going
 inwards

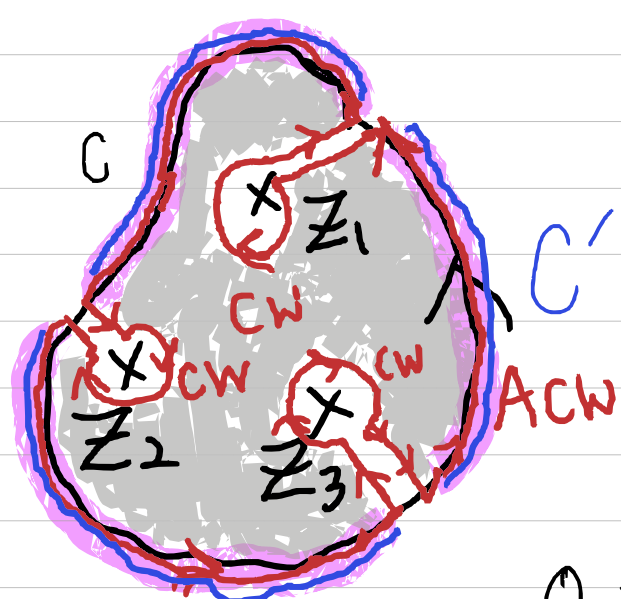
+ N circles around the singularities CW



The
 Gray area is the interior
 of C' .

All
 the singularities are
 outside of the
 interior of C' .
 (All z_k lie outside C')

(3)



f is analytic inside C'

so, $\oint_{C'} f(z) dz = 0.$

$$0 = \int_C f(z) dz + \int_{\text{line segments}} f(z) dz$$

$$- \sum_k \oint_{\text{circle surrounding } k} f(z) dz$$

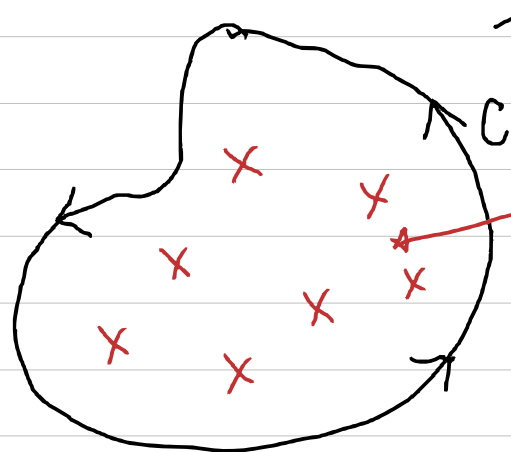
(ACW)
(or CCW)



$$\oint_{\text{circle surrounding } k} f(z) dz = 2\pi i \sum_{k=1}^N \text{Res}[z_k]$$

$$\left(\oint_C f(z) dz - 2\pi i \sum_{k=1}^N \text{Res}[z_k] \right) = 0$$

Some contour



$$\oint_C f(z) dz$$

Singul.s inside the contour.

§ 6.2. Residues

(a) If we know the Laurent series,

$$f(z) = \sum_{n=-\infty}^{\infty} A_n (z-z_0)^n$$

where $z=z_0$ is the singularity.

The Residue of $z_0 = A_{-1}$

$$= \text{Residue}(z_0)$$

$$= \text{Res}(z_0) = A_{-1}$$

$$(n=-1)$$

→ Easy to get the Residue once you know the Laurent series.

→ The Laurent series "picks out"

$$n=-1, \text{ s.t.}$$

$$\text{Res}(z_0) = A_{-1}$$

$$\& f(z) = \sum_{n=-\infty}^{\infty} A_n (z-z_0)^n$$

$$\oint \frac{1}{(z-z_0)^m} dz = \begin{cases} 2\pi i & \text{for } m=1 \\ 0 & \text{for } m \geq 2 \end{cases}$$

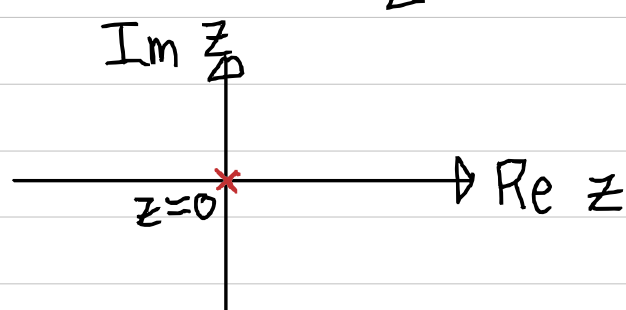
$$\oint (z-z_0)^m dz = 0 \quad \text{for } m \geq 0$$

(4)

EX. $f(z) = \frac{e^z}{z-1}$

$$\text{Res}(1) = \lim_{z \rightarrow 1} \left[\frac{e^z}{z-1} (z-1) \right] = e$$

EX. $f(z) = \frac{\sin z}{z^2}$ has a simple pole at $z=0$.



$$\lim_{z \rightarrow 0} \left(\frac{0}{0} \right) = 1$$

$$\text{Res}(0) = \lim_{z \rightarrow 0} \left[\frac{\sin z}{z^2} z \right] = \lim_{z \rightarrow 0} \left[\frac{\sin z}{z} \right] = 1$$

Pole of order m:

$$f(z) = \frac{A_{-m}}{(z-z_0)^m} + \frac{A_{-m+1}}{(z-z_0)^{m-1}} + \dots + \frac{A_{-1}}{z-z_0} + \sum_{n=0}^{\infty} A_n (z-z_0)^n$$

$$\lim_{z \rightarrow z_0} \left(\frac{d}{dz} \right)^{m-1} \left[(z-z_0)^m f(z) \right]$$

this term vanishes when we take the limit $z \rightarrow z_0$.

$$= (m-1)! A_{-1}$$



$$A_{-1} = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \left(\frac{d}{dz} \right)^{m-1} \left[(z-z_0)^m f(z) \right]$$

Just do this!

EX. $f(z) = \frac{e^z}{(z-1)^2}$

Plug all the things in!

$m=2$ double poles

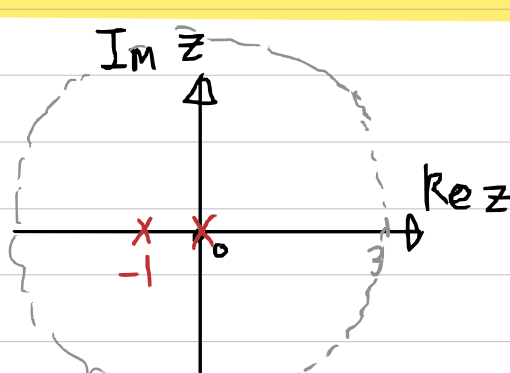
$$\text{Res}(1) = \lim_{z \rightarrow 1} \frac{1}{(2-1)!} \left(\frac{d}{dz} \right)^{2-1} \left[(z-z_0)^2 f(z) \right]$$

$$= \lim_{z \rightarrow 1} \frac{1}{1} \left(\frac{d}{dz} \right) \left[\cancel{(z-1)^2} \frac{e^z}{\cancel{(z-1)^2}} \right]$$

$$= \lim_{z \rightarrow 1} e^z = e$$

§ 6.3. EX.

$$I \equiv \oint_{\text{circle } |z|=3} \frac{3z^2+2}{z(z+1)} dz$$



$$= 2\pi i \left[\text{Res}(0) + \text{Res}(-1) \right]$$

$$= 2\pi i \left[2 + \left(-\frac{5}{1} \right) \right] = 2\pi i [2-5] = -6\pi i$$

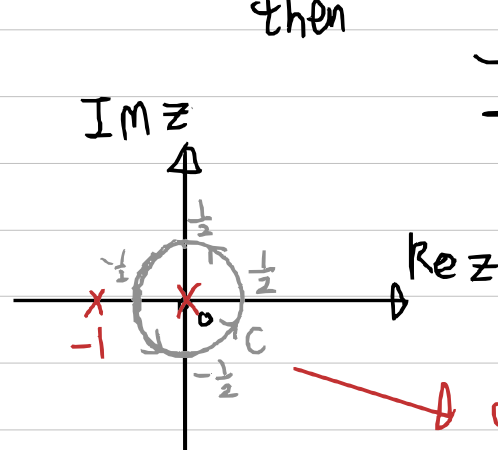
$$\therefore I = -6\pi i$$

If the singularity lies on the contour, we should use the principal value, s.t., we cut out the pt. $\text{pt.} - \epsilon < \text{pt.} < \text{pt.} + \epsilon$.

Then, take the limit $\epsilon \rightarrow 0$. Consider the contour starting from $(\text{pt.} - \epsilon)$ & ending at $(\text{pt.} + \epsilon)$.

The position of the contour is crucial.

If the contour were at $|z| = \frac{1}{2}$, then



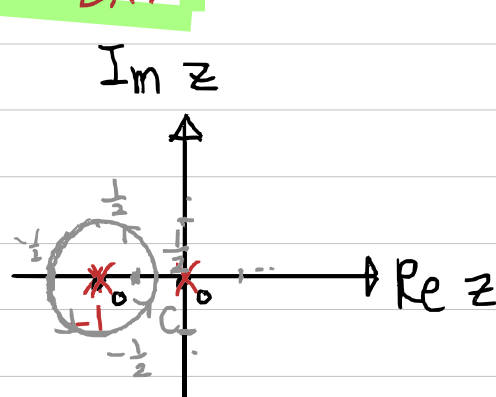
$$I = \oint_{\text{circle around } |z|=1/2} \frac{3z^2+2}{z(z+1)} dz$$

only one pole inside the contour.

$$I = 2\pi i \times \text{Res}(0)$$

$$= 4\pi i$$

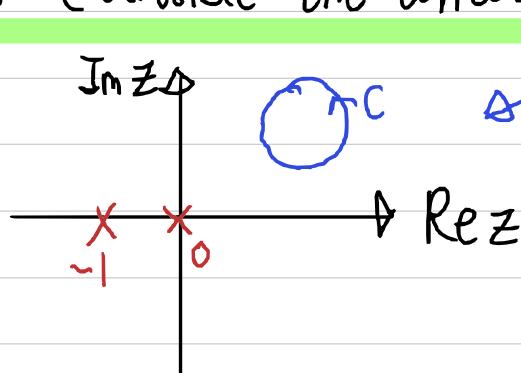
Another EX.



So, in the Residue Thm, we assume the pole is either inside the contour or outside the contour.

$$I = 2\pi i \times \text{Res}(-1) = -10\pi i$$

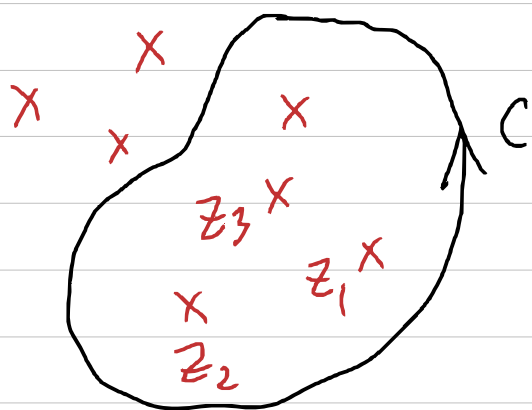
EX. (outside the contour)



In this case, by Cauchy's Thm, $I = 0$.

MMP L19

25/11/2024 ①



Isolated singularities

Residue Thm

$$\oint_C f(z) dz = 2\pi i \sum_{\substack{z_i \\ \text{inside } C}} \text{Res}(z_i)$$

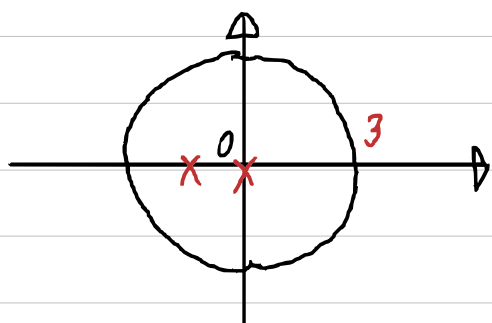
• Pole order m :

$$\text{Res}(z_0) = \frac{1}{(m-1)!} \left(\frac{d}{dz} \right)^{(m-1)} \left(f(z-z_0)^m \right)_{z \rightarrow z_0}$$

• Ex.s.

Ex. 6.3.2.

$$I = \oint_{|z|=3} \frac{3z^2+2}{z(z+1)} dz$$



$$I = 2\pi i (\text{Res}(0) + \text{Res}(-1))$$

$$= 2\pi i \left(2 + \frac{5}{-1} \right)$$

$$= -6\pi i$$

$$f(z) = \frac{e^z}{1+z^2} = \frac{e^z}{(z-i)(z+i)}$$

$$f(z) = \frac{1}{az^2+bz+c} = \frac{1}{a(z-z_+)(z-z_-)}$$

where $z_{\pm}$ are roots.

(2)

$$f(z) = \frac{e^z}{2z+i} = \frac{e^z}{2(z+\frac{i}{2})}$$

$$\text{Res}\left(z = -\frac{i}{2}\right) = \frac{e^{-i/2}}{2}$$

§6.4. Jordan's Lemma

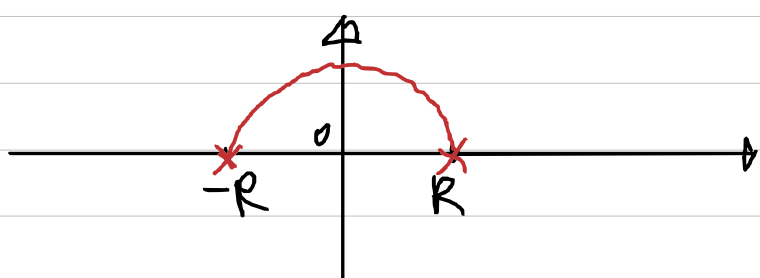
$$\text{Let } I(R) = \int_{S_R} e^{i\alpha z} f(z) dz$$

S_R is a semi-circle in upper $\alpha > 0$ (lower $\alpha < 0$) $\frac{1}{2}$ plane of radius R .

$$\text{Let } M(R) = \max_{S_R} |f(z)|$$

If $M(R) \rightarrow 0$ as $R \rightarrow \infty$, then

$$I(R) \rightarrow 0 \text{ as } R \rightarrow \infty.$$



N.B. :

$$\begin{aligned} |e^{i\alpha z}| &= |e^{i\alpha(x+iy)}| = |e^{i\alpha x} e^{-\alpha y}| \\ &= e^{-\alpha y} \quad (\text{since } |e^{i\alpha x}| = 1) \end{aligned}$$

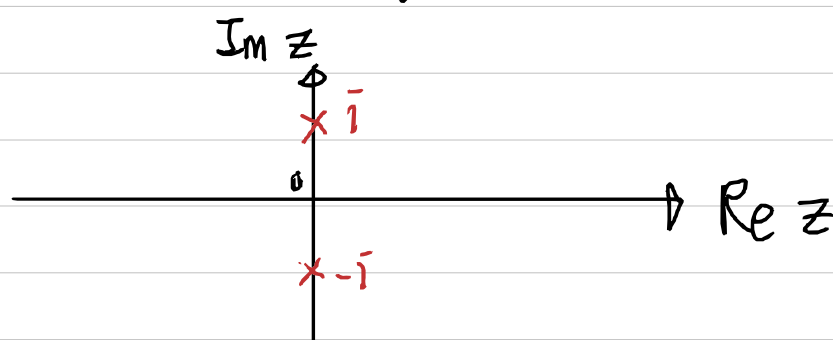
for $\alpha > 0$ (and $\alpha < 0$)

Thus, applies e.g. for $f(z) = \frac{1}{z}$.

EX. 6.3.3,

$$I = \int_{-\infty}^{\infty} \frac{\cos x}{x^2+1} dx$$

$$= \oint_{\text{Real axis}} \frac{\cos z}{1+z^2} dz$$



$$\cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$I = I_1 + I_2 = \frac{1}{2} \oint \frac{e^{iz}}{(z+i)(z-i)} dz + \frac{1}{2} \oint \frac{e^{-iz}}{(z+i)(z-i)} dz$$

(3)

Do I_1 first:

$$\text{Integrand of } I_1 = \frac{e^{iz}}{2(z+i)(z-i)}$$

Complete contour w/ a "semi-circle at ∞ "

By Jordan's lemma,

 $\int$ over semi-circle
at ∞ vanishes.
So, by residue thm., $I_1 = 2\pi i \operatorname{Res}(i)$

$$= 2\pi i \frac{e^{-1}}{2i} \cdot \frac{1}{2}$$

$$= 2\pi i \frac{e^{i \cdot i}}{i \cdot i} \cdot \frac{1}{2}$$

$$= \frac{\pi}{2} e^{-1}$$

Ex. Show $I_2 = \frac{1}{2} \pi e^{-1}$.

$$\text{So, } I = \pi e^{-1}$$

$$\text{Ex. } I = \int_{-\infty}^{\infty} \frac{\sin x}{x} dx = \int_{\text{Real axis}} \frac{\sin z}{z} dz$$

 $\downarrow$ analytic everywhere

$$= \int_{\text{R.A.}} \left(\frac{e^{iz}}{2iz} - \frac{e^{-iz}}{2iz} \right) dz$$

$$= I_1 - I_2$$



Real ax.

$$\oint_C \frac{\sin z}{z} dz$$

$$\text{Do } I_1: I_1 = \oint_C \frac{e^{iz}}{2iz} dz$$

Close contour w/ a semi-circle at ∞ in upper-half plane

by Jordan's Lemma


 $\int$ over the semi-circle
vanishes

$$\text{So, } I_1 = 0.$$

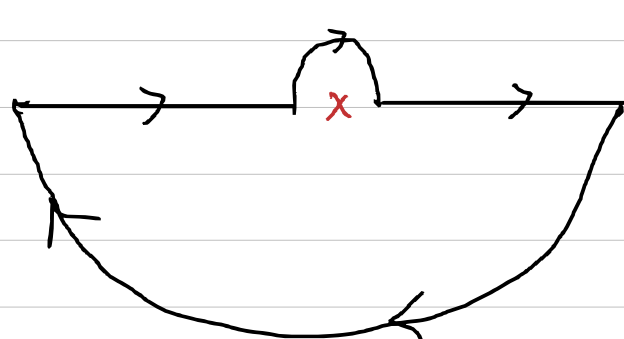
$$I_2 = \oint_C \frac{e^{-iz}}{2iz} dz$$

Close contour w/ a semi-circle at ∞ in the lower $-\frac{1}{2}$ plane:

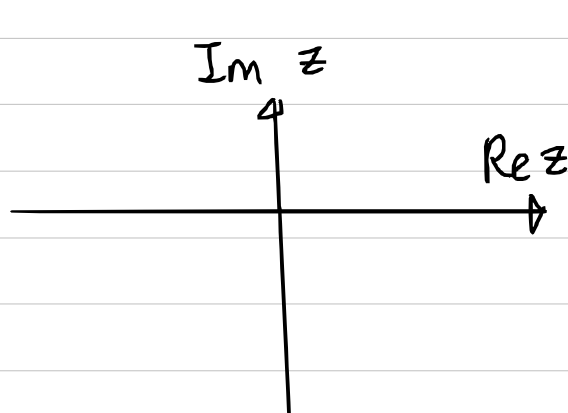
$$I_2 = -2\pi i \operatorname{Res}(0)$$

$$= -2\pi i \frac{1}{2i}$$

$$= -\pi$$

If CW, the $-$ ResidueIf CCW, the $+$ Residue

$$\begin{aligned} I &= I_1 - I_2 \\ &= 0 - (-\pi) \\ &= \pi \end{aligned}$$

For I_2 , e^{-iz} in the integrandso, the semi-circle at ∞ is in the lower $-\frac{1}{2}$ plane.

$$\oint_{C_1} f(z) dz = \oint_{C_2} f(z) dz$$

(If $f(z)$ is analytic.) $\frac{\sin z}{z} \rightarrow$ analytic everywhere

$$\lim_{z \rightarrow 0} \frac{\sin z}{z} = 1$$

§ 6.5.

(4)

Inverse Laplace Transforms

$$L[f(t)] = F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

As in
F.T.
 $e^{-j\omega t}$
 $\downarrow$
 $e^{+j\omega t}$

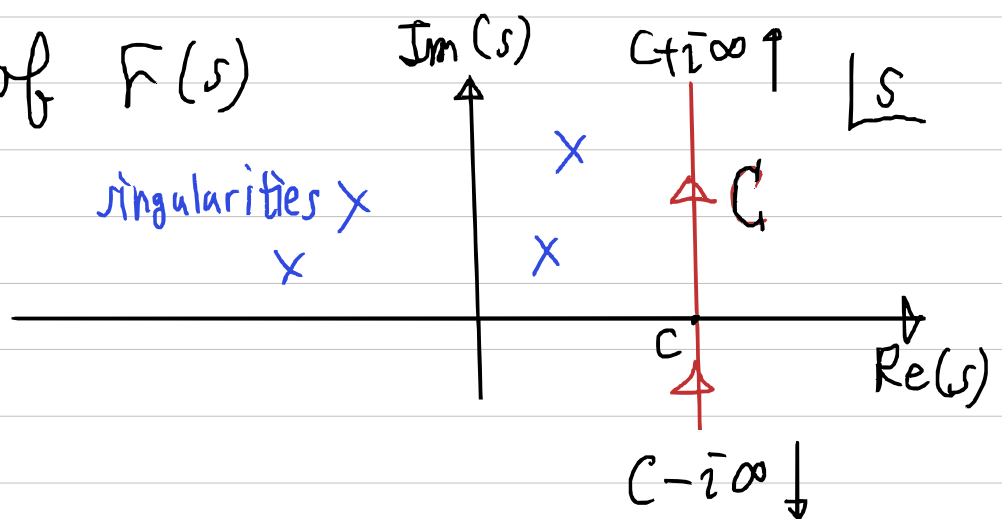
$$f(t) = L^{-1}[F(s)] = \frac{1}{2\pi i} \oint_C F(s) e^{st} ds$$

(Inv)

$$\oint_C = \int_{C-i\infty}^{C+i\infty}$$

"Bromwich Integral"

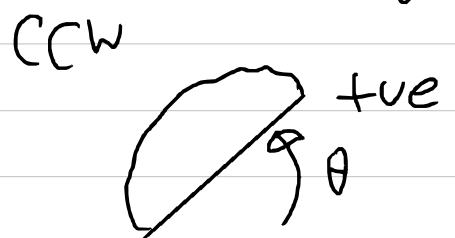
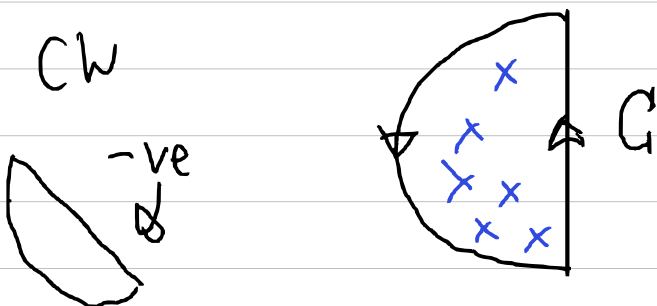
C is $\parallel$ to y -ax. & to the right of all singularities of $F(s)$



Do this $\int$ by closing contour, writing the Res thm & Jordan's lemma.

Close contour w/ ∞ semi-circle in left half of plane, so closed contour includes

ALL singul-s.



Can apply Jordan's lemma (rotated by 90°)

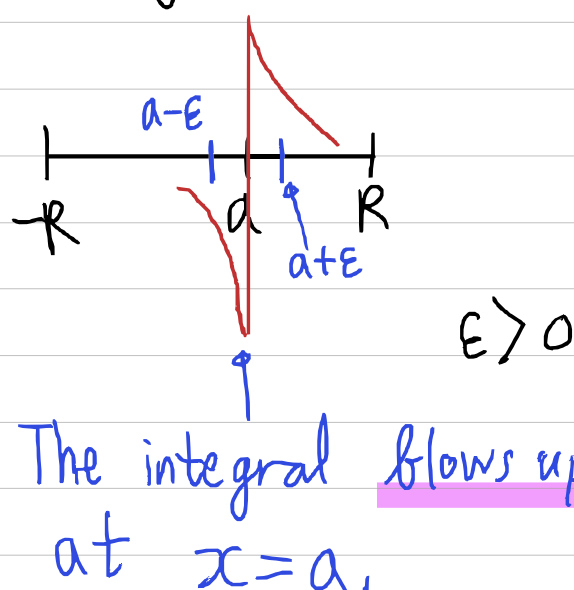
to show $\int$ around semi-circle vanishes.

$$\text{So, } f(t) = \frac{1}{2\pi i} \sum_{\substack{\text{ALL} \\ \text{singul.s } z_i \\ \text{of } F(s)}} + \text{Res}(z_i)$$

$$= \sum + \text{Res}(z_i)$$

§6.6. Principal Value (Complex Analysis)

EX. $I = \int_{-R}^R \frac{dx}{x-a}$

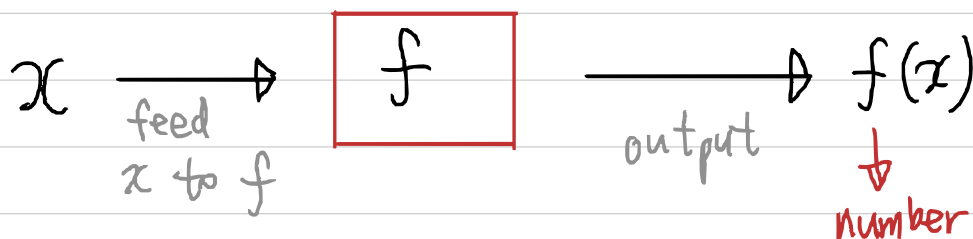
Principal Value P

$$\begin{aligned}
 P[I] &= \lim_{\epsilon \rightarrow 0} \left(\int_{-R}^{a-\epsilon} \frac{dx}{x-a} + \int_{a+\epsilon}^R \frac{dx}{x-a} \right) \\
 &= \lim_{\epsilon \rightarrow 0} \left(\left[\ln|x-a| \right]_{-R}^{a-\epsilon} + \left[\ln|x-a| \right]_{a+\epsilon}^R \right) \\
 &= \lim_{\epsilon \rightarrow 0} \left[\cancel{\ln \epsilon} - \ln|a+R| + \ln|a-R| - \cancel{\ln \epsilon} \right] \\
 &= \ln \left| \frac{a-R}{a+R} \right|
 \end{aligned}$$

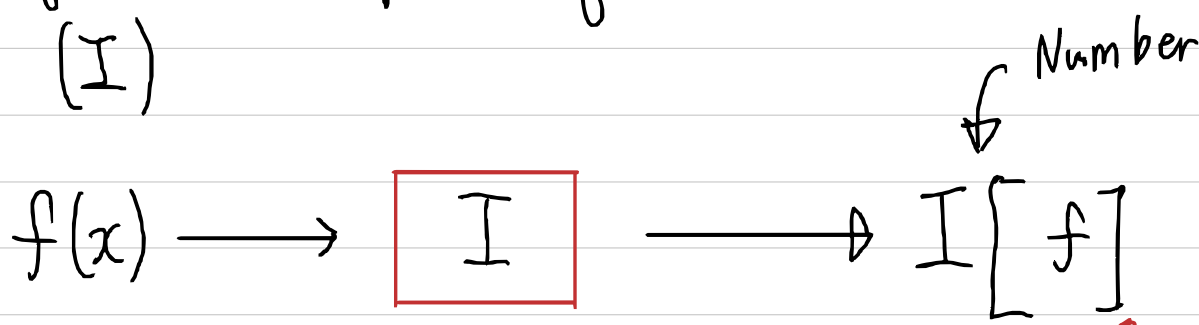
New
Chap.

Calculus of Variations

A fn. maps a # to another #.



A functional maps a fn. to a number (#)



Typically, functionals are integrals of f & its derivatives,

square brackets since f is a fn.

(EX) $y(x)$, $S[y] = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

$= \int_a^b \sqrt{(dx)^2 + (dy)^2}$

Arc length ds

(EX) $\rho(\underline{x})$,

$$U[\rho] = \frac{1}{2} \int \frac{\rho(\underline{x}) \rho(\underline{x}')}{4\pi \epsilon_0 |\underline{x} - \underline{x}'|} d^3 \underline{x} d^3 \underline{x}' \rightarrow \text{Number}$$

$\rho(\underline{x})$: Distribution of charges in space

§2. The Euler-Lagrange problem (2)

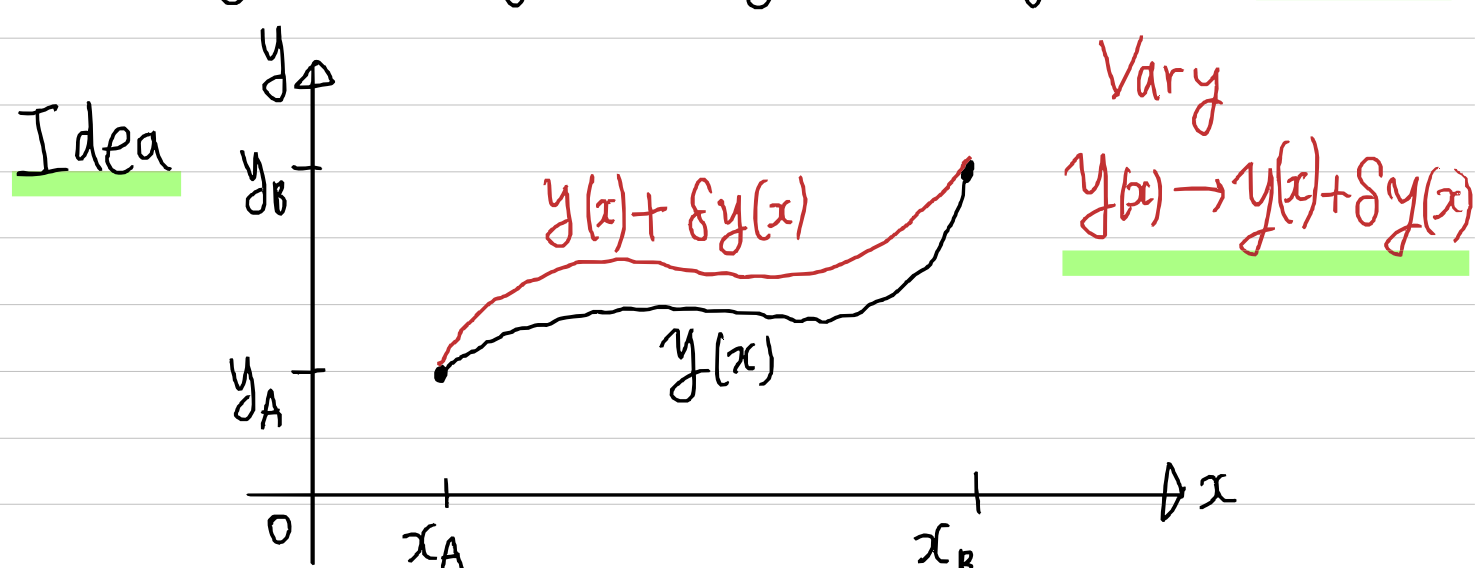
Let $I[y] = \int_{x_A}^{x_B} f(x, y(x), y'(x)) dx$,
(for $y(x)$)

where f has 3 arguments & $y' = \frac{dy}{dx}$.

Also, $y(x_A) = y_A$ & $y(x_B) = y_B$ are fixed.

The E-L (variational) problem is to find $y(x)$ that is a stationary fn. of $I[y]$ & that extremizes $I[y]$

s.t. $y(x_A) = y_A$ & $y(x_B) = y_B$ are fixed.



We want to find $y(x)$ s.t., under this change, I only changes by $O(\delta y^2)$.

(cf.) $h(x + \delta x) = h(x) + \delta x h'(x) + \frac{(\delta x)^2}{2} h''(x) + \dots$

h has stationary pt. at x if $h'(x) = 0$.

$$h(x + \delta x) - h(x) = O(\delta x^2)$$

We say I is stationary at y if

$$I[y + \delta y] - I[y] = O(\delta y^2)$$

$$\begin{aligned} I[y + \delta y] &= \int_{x_A}^{x_B} f(x, y + \delta y, y' + \delta y') dx \\ &= \int_{x_A}^{x_B} \left[f(x, y, y') + \delta y \frac{\partial f}{\partial y} + \delta y' \frac{\partial f}{\partial y'} \right] dx \\ &\quad + O(\delta y^2) \end{aligned}$$

$$I[y + \delta y] - I[y] = \int_{x_A}^{x_B} \left(\delta y \frac{\partial f}{\partial y} + \delta y' \frac{\partial f}{\partial y'} \right) dx + O(\delta y^2)$$

$$\begin{aligned} &= \int_{x_A}^{x_B} \left(\delta y \frac{\partial f}{\partial y} \right) dx + \left[\delta y \frac{\partial f}{\partial y'} \right]_{x_A}^{x_B} - \int_{x_A}^{x_B} \left[\delta y \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] dx + O(\delta y^2) \end{aligned}$$

By the I.B.P.,
 the boundary term vanishes since $\delta y(x_A) = \delta y(x_B) = 0$

$$= \int_{x_A}^{x_B} \left(\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right) \delta y dx + O(\delta y^2)$$

y extremizes I if this vanishes for any δy

So, $\iff \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$

$$\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

"E-L E"

Ex. $S[y] = \int_a^b \left[(1+x^2) y'^2 + y^4 \right] dx$

depends on $x, y, \& y'$, since $y = y(x)$, $y' = y'(x)$

E-L E: $4y^3 - \frac{d}{dx} \left[(1+x^2)(2y') \right] = 0$

$$4y^3 - (2x)(2y') - (1+x^2)(2y'') = 0$$

We can also get this result by varying $y \rightarrow y + \delta y$ in (*).

(3)

§2.1. The Beltrami Identity

(EX) $S[y] = \int_a^b (y'^2 - y^4) dy$

Suppose $I[y] = \int_a^b f(y, y') dx$

Then, $\frac{df}{dx} = \frac{dy}{dx} \frac{\partial f}{\partial y} + \frac{dy'}{dx} \frac{\partial f}{\partial y'} + \cancel{\frac{\partial f}{\partial x}}$

If $y(x)$ extremizes I , then

$$\frac{\partial f}{\partial y} = \frac{d}{dx} \left[\frac{\partial f}{\partial y'} \right]$$

$$\frac{df}{dx} - y' \frac{d}{dx} \left[\frac{\partial f}{\partial y'} \right] - \frac{dy'}{dx} \frac{\partial f}{\partial y'} = 0$$

$$\frac{df}{dx} - \frac{d}{dx} \left[y' \frac{\partial f}{\partial y'} \right] = 0$$

$$\frac{d}{dx} \left[f - y' \frac{\partial f}{\partial y'} \right] = 0 \Rightarrow f - y' \frac{\partial f}{\partial y'} = \text{const.}$$

Beltrami Id.

EX. $S[y] = \int_a^b (y'^2 - y^4) dx$

Beltrami $\rightarrow y'^2 - y^4 - y'(2y') = \text{const.}$
 $= -y'^2 - y^4$

1st deriv. : $y'^2 + y^4 = \text{const.}$

MMP L21

$$I[y] = \int_{x_A}^{x_B} f(x, y(x), y'(x)) dx$$

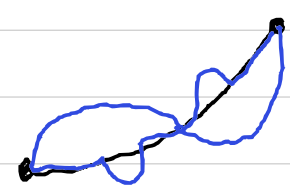
for y that extremizes $I[y]$

$$\text{E-L E} : \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) = 0$$

If f does NOT depend on x ($f = f(y, y')$)

& extremizes $I[y]$, then

$$f - y' \frac{\partial f}{\partial y'} = \text{const.} \quad (\text{Beltrami Identity})$$



$\delta y(x)$

$$\begin{aligned} \delta I &= I[y + \delta y] - I[y] = O(\delta y^2) \\ &= \int \left[\frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right) \right] \delta y(x) dx + O(\delta y^2) \end{aligned}$$

§2.2. Notation

Define functional deriv. of $I[y]$ w.r.t. $y(x)$ as

$$\frac{\delta I}{\delta y(x)} = \frac{\partial f}{\partial y} - \frac{d}{dx} \left(\frac{\partial f}{\partial y'} \right)$$

→ 3 diff. types of deriv.s ("dee's")

"E-L E" can be written : $\frac{\delta I}{\delta y(x)} = 0$.

§2.3. Lagrangian Mechanics

Single particle w/ position vec. $\underline{r}(t) = (x(t), y(t), z(t))$,

$$\begin{aligned} \text{or } \underline{r}(t) &= (x_1(t), x_2(t), x_3(t)) \\ &= \underline{x}(t) \end{aligned}$$

$$S[\underline{r}(t)] = \int_{t_A}^{t_B} L(t, \underline{r}(t), \dot{\underline{r}}(t)) dt$$

ACTION

$$L(t, \underline{r}, \dot{\underline{r}}) = T - V = \frac{1}{2} m |\dot{\underline{r}}|^2 - V(t, \underline{r})$$

Hamilton's principle : physical trajectory $\underline{r}(t)$

extremizes $S[\underline{r}(t)]$, i.e., E-LEs are

Newton's 2nd Law, $\|\dot{\underline{r}}\|^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2$

Let $V = V(\underline{r})$. There's an E-LE for x_1, x_2 , & x_3 .

$$\begin{aligned} x_1 : \frac{\partial L}{\partial x_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_1} \right) &= 0 \\ - \frac{\partial V}{\partial x_1} - \frac{d}{dt} \left(\frac{1}{2} m (2 \dot{x}_1) \right) &= 0 \end{aligned}$$

$$m \ddot{x}_i = - \frac{\partial V}{\partial x_i} \quad \text{vec. eq. } (\vec{i}=1, 2, 3)$$

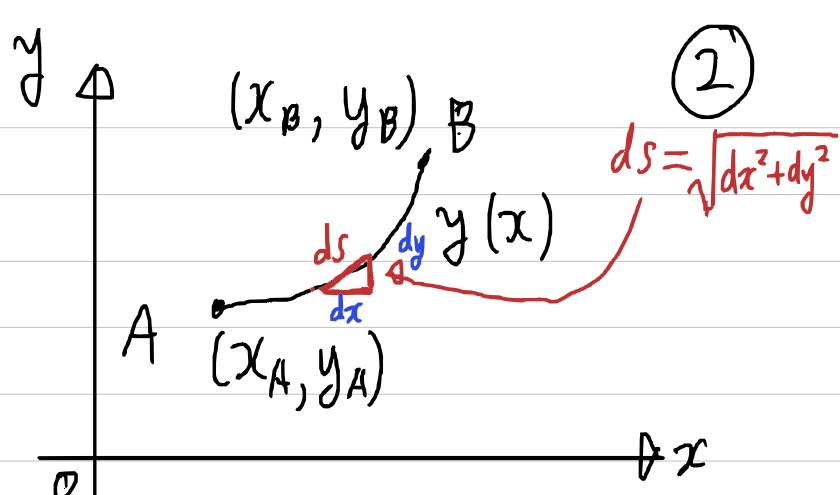
$$m \ddot{\underline{r}} = - \nabla V = \text{Force}$$

"2nd Law"

§2.4. Examples

(a) Shortest Path

2 Methods



(i) Curve from A to B is given by $y(x)$.

$$ds = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$I[y] = \int_A^B ds = \int_{x_A}^{x_B} \sqrt{1 + (y')^2} dx$$

The integrand doesn't depend on x & so

$$\sqrt{1 + (y')^2} - y' \frac{2y'}{2\sqrt{1 + (y')^2}} = C$$

$$1 + \cancel{(y')^2} - \cancel{(y')^2} = C \sqrt{1 + (y')^2}$$

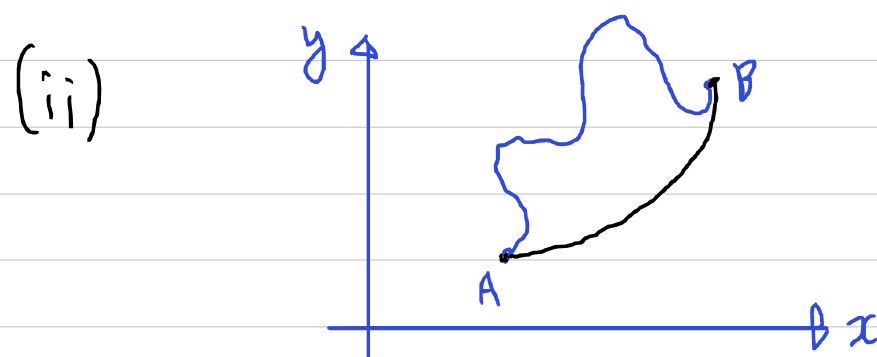
$$1 = C^2 (1 + (y')^2)$$

$$y' = D \cdot (\text{const.})$$

$$y = Dx + b$$

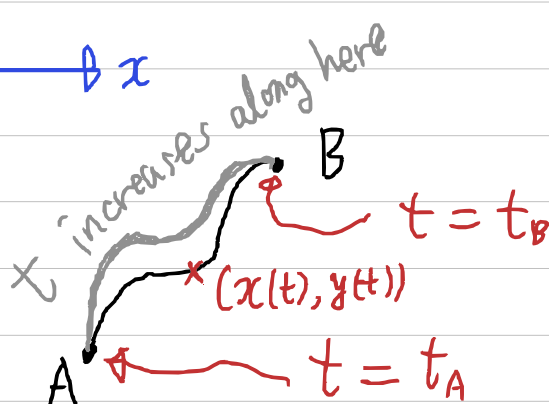
: a straight line

Solve for D & b using known endpoints.



Parameterized curve

$x(t)$ } determine
 $y(t)$ } the curve



$$I[x(t), y(t)] = \int_A^B ds$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{\dot{x}^2 + \dot{y}^2} dt$$

$$dx = \frac{dx}{dt} dt \quad \& \quad dy = \frac{dy}{dt} dt$$

$$I = \int_{t_A}^{t_B} \sqrt{\dot{x}^2 + \dot{y}^2} dt$$

$$\underline{x \text{ ELE}}: -\frac{d}{dt} \left(\frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right) = 0$$

$$\frac{\dot{x}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = A = \text{const.}$$

$$\underline{y \text{ ELE}}: \frac{\dot{y}}{\sqrt{\dot{x}^2 + \dot{y}^2}} = B = \text{const.}$$

$$\dot{x} = A \sqrt{\dot{x}^2 + \dot{y}^2} \quad \& \quad \dot{y} = B \sqrt{\dot{x}^2 + \dot{y}^2}$$

A & B cannot both be zero.

$$\text{Let } A \neq 0. \Rightarrow \frac{\dot{y}}{\dot{x}} = \frac{B}{A} \Rightarrow \frac{dy}{dx} = \frac{B}{A}$$

$$\text{Similar when } B \neq 0. \Rightarrow y = \frac{B}{A}x + C$$

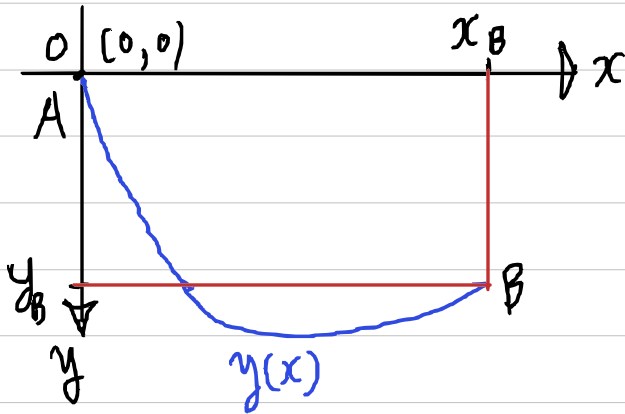
(straight line)

(3)

EX. (b) Least time or Brachistochrone

(historically important)

Galileo Galilei studied it.

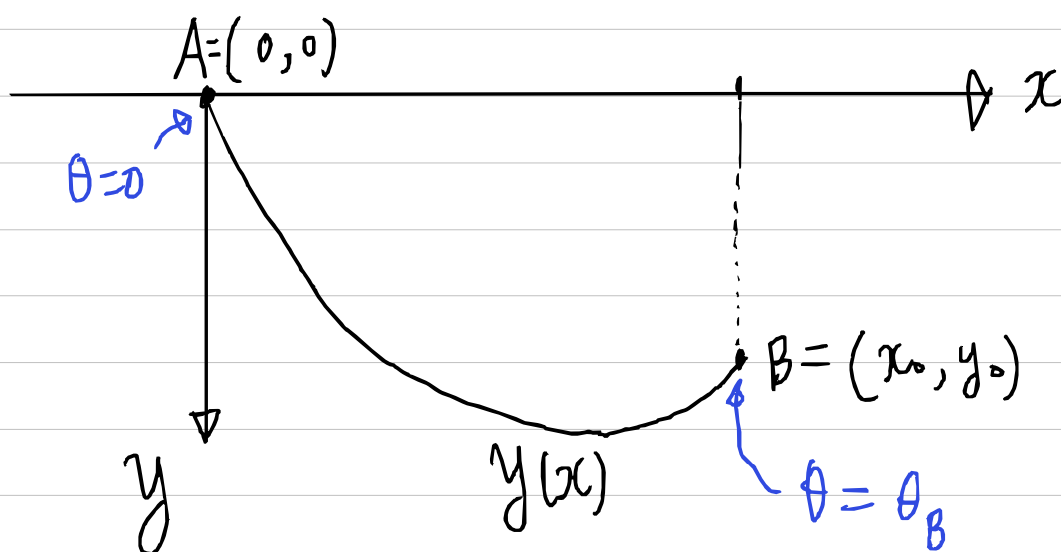


Consider a ball sliding down without any friction under gravity only.

(no other forces on it)

We know $\frac{1}{2}mv^2 = mgy$.

Q: What curve minimizes time for a particle (or a marble) to reach B from A in least time?

Brachistochrone

Find $y(x)$ that minimizes time taken by the particle that slides (without friction) from A to B. (rest)

Time taken $= I[y] = \int_A^B dt = \int_A^B \frac{ds}{v}$

$$v = \frac{ds}{dt} \longrightarrow dt = \frac{ds}{v}$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1 + y'^2} dx$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} : \text{Instantaneous Velocity}$$

→ Total En. is conserved.

$$\frac{1}{2} \cancel{mv^2} = \cancel{mgy}$$

K.E. gained = P.E. lost

$$\longrightarrow v = \sqrt{2gy}$$

$$I[y] = \frac{1}{\sqrt{2g}} \int_{x_A}^{x_B} \frac{\sqrt{1+y'^2}}{\sqrt{y}} dx$$

$\underbrace{\frac{\sqrt{1+y'^2}}{\sqrt{y}}}_{\text{doesn't depend on } x, \text{ so}} = f = \text{integrand}$

Beltrami Id. : $f - y' \frac{\partial f}{\partial y'} = C$

$$\sqrt{\frac{1+y'^2}{y}} - y' \frac{\cancel{2y'}}{\cancel{2}\sqrt{1+y'^2}\sqrt{y}} = C$$

$$\frac{(1+y'^2) - y'^2}{\sqrt{(1+y'^2)y}} \longrightarrow \frac{1}{C^2} = y(1+y'^2)$$

The Solution of in parameterized form is

$$\longrightarrow y'^2 = \frac{\alpha - y}{y}$$

$$\longrightarrow \alpha = \frac{1}{C^2}$$

$$\left. \begin{aligned} x &= a(\theta - \sin \theta) \\ y &= a(1 - \cos \theta) \end{aligned} \right\} \text{cycloid : } a = \frac{\alpha}{2}$$

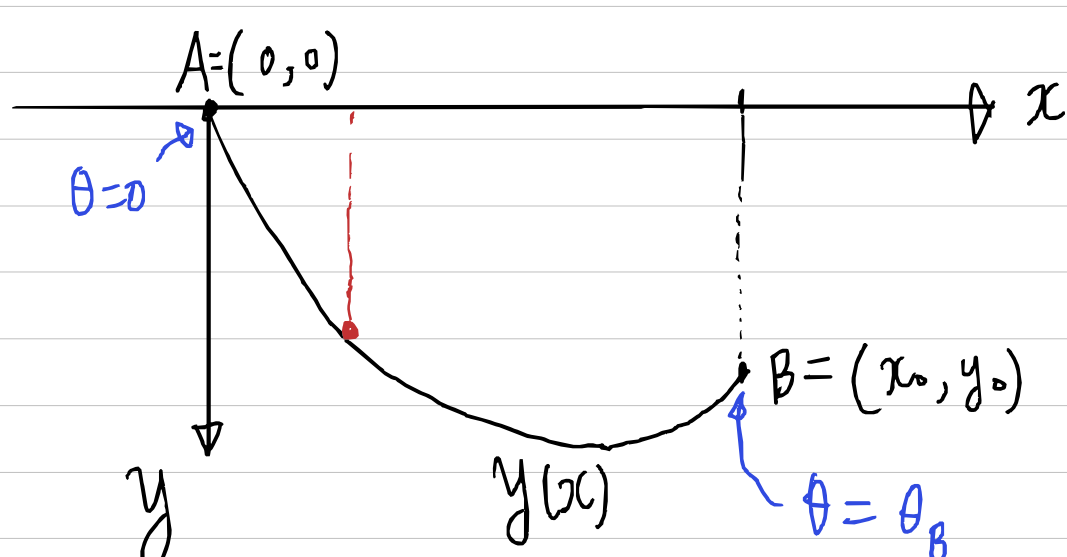
Excr. Check this is a soln.

(2)

We solve for a & θ_B by using bound. cond.s:

$$\left. \begin{aligned} a(\theta_B - \sin \theta_B) &= x_B \\ a(1 - \cos \theta_B) &= y_B \end{aligned} \right\} \rightarrow \text{fixed}$$

can only solve these numerically.



Note: y' diverges for $y=0$ & cycloid starts parallel to y -ax..

Excr. Show $I[y] = \sqrt{\frac{a}{g}} \theta_B$ is the total minimum time.

§2.5. Symmetries & Conservation Laws in Lagrangian Mechanics.

1D $x(t)$ & potential $V(x) \leftarrow t$ -indep.

$$L(x, \dot{x}) = T - V = \frac{1}{2} m \dot{x}^2 - V(x)$$

no t dependence

The sys. has a time translation symmetry.

$$\text{Beltrami} \rightarrow L - \dot{x} \frac{\partial L}{\partial \dot{x}} = \text{const.}$$

$$\frac{1}{2} m \dot{x}^2 - V - \dot{x}(m \dot{x}) = \text{const.}$$

$$\left(\frac{1}{2} m \dot{x}^2 + V \right) = \text{another const.}$$

Total En. = Hamiltonian H

3D

$$\underline{r}(t) = (x(t), y(t), z(t))$$

$$L = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - V(\underline{r})$$

Excr. Show: if L doesn't depend on time,

$$L - \dot{x} \frac{\partial L}{\partial \dot{x}} - \dot{y} \frac{\partial L}{\partial \dot{y}} - \dot{z} \frac{\partial L}{\partial \dot{z}} = \text{const.}$$

Generalized Beltrami Identity.

$$\begin{aligned} \frac{dL}{dt} &= \frac{\partial L}{\partial x} \frac{dx}{dt} + \frac{\partial L}{\partial y} \frac{dy}{dt} + \frac{\partial L}{\partial z} \frac{dz}{dt} \\ &\quad + \frac{\partial L}{\partial \dot{x}} \frac{d\dot{x}}{dt} + \frac{\partial L}{\partial \dot{y}} \frac{d\dot{y}}{dt} + \frac{\partial L}{\partial \dot{z}} \frac{d\dot{z}}{dt} \end{aligned}$$

$$\xrightarrow{T-V} L = \sum_{i=1}^3 \dot{x}_i \left(\frac{1}{2} m \right) (2 \dot{x}_i) = \text{const.} \quad (3)$$

$$\xrightarrow{} \frac{m}{2} \sum_i \dot{x}_i^2 + V(r) = \text{const.} \quad \text{another}$$

$$T = \frac{m}{2} \sum_i \dot{x}_i^2 \quad \text{Energy.}$$

If the mechanical sys. has N variables $q_1, \dots, q_N$ & $L \equiv L(q_i, \dot{q}_i)$.

Then, $H = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L$ is conserved.

Another ex. of symmetry $\rightarrow$ conservation

Suppose $V(\underline{q}, \underline{\dot{q}}, t)$ is indep. of q_k for some k , where $\underline{q} = (q_1, \dots, q_N)$

$$q_k \quad E-L \quad : \quad 0 = \frac{\partial L}{\partial q_k} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_k} \right)$$

$$\hookrightarrow \frac{\partial L}{\partial \dot{q}_k} = \text{const.}$$

1D EXAMPLE. In 1D, $L = \frac{1}{2} m \dot{x}^2 - V(x)$

$$\frac{\partial L}{\partial \dot{x}} = m \dot{x} \rightarrow \text{conserved.}$$

indep. of x .

In 3D, $L = \frac{m}{2} (\dot{x}^2 + \dot{y}^2 + \dot{z}^2) + V(y, z)$

$$\frac{\partial L}{\partial \dot{x}} = m \dot{x} \rightarrow \text{conserved.}$$

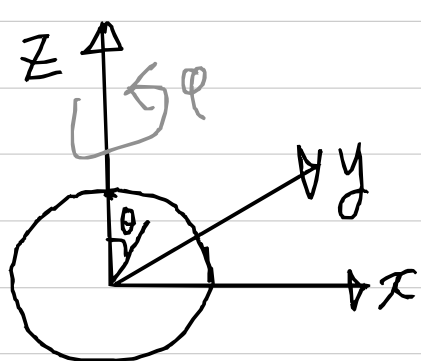
$m \dot{x} = p_x$: the x -mtm. is conserved.

Translation symmetry in x -dir. $\rightarrow$

the x -mtm. is conserved.

3D particle in polar coord.s

$$\underline{q}(t) = (r(t), \theta(t), \phi(t))$$



$$L(\underline{q}, \underline{\dot{q}}) = T - V$$

$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2 + r^2 \sin^2 \theta \dot{\phi}^2)$$

Assume $V(\underline{r}) = V(r, \theta)$ only. $\rightarrow$ indep. of ϕ .

$\rightarrow L = T - V$ is indep. of ϕ .

Rotational Symmetry around z -ax.

$$\rightarrow 0 = \frac{\partial L}{\partial \phi} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right)$$

$$\hookrightarrow \frac{\partial L}{\partial \dot{\phi}} = \text{const.}$$

const.

$$L_z = m r^2 \sin^2 \theta \dot{\phi} = \text{const.} \quad \text{ang. mtm. in } z\text{-dir.} \\ \underline{L} \text{ (vector) (confusing)}$$

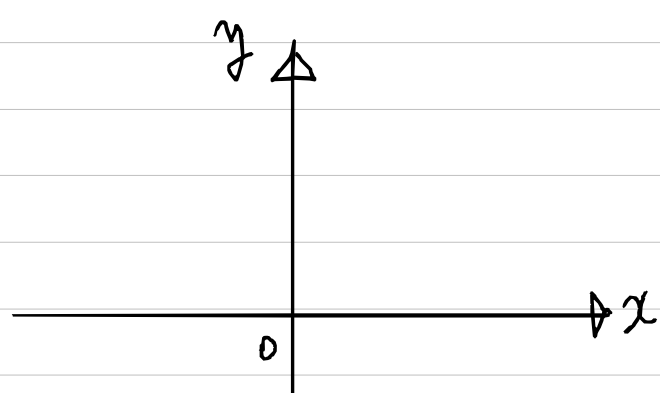
$$\underline{L} = \underline{r} \times \underline{p}$$

Symmetry $\rightarrow$ Conservation

If $V = V(r)$, then the sys. is symmetric under all rotations $\rightarrow \underline{L}$ is conserved.

§3. Constrained Optimization & Lagrange Multipliers

§3.1. Functions

Deal first with functions $f(x, y)$ 

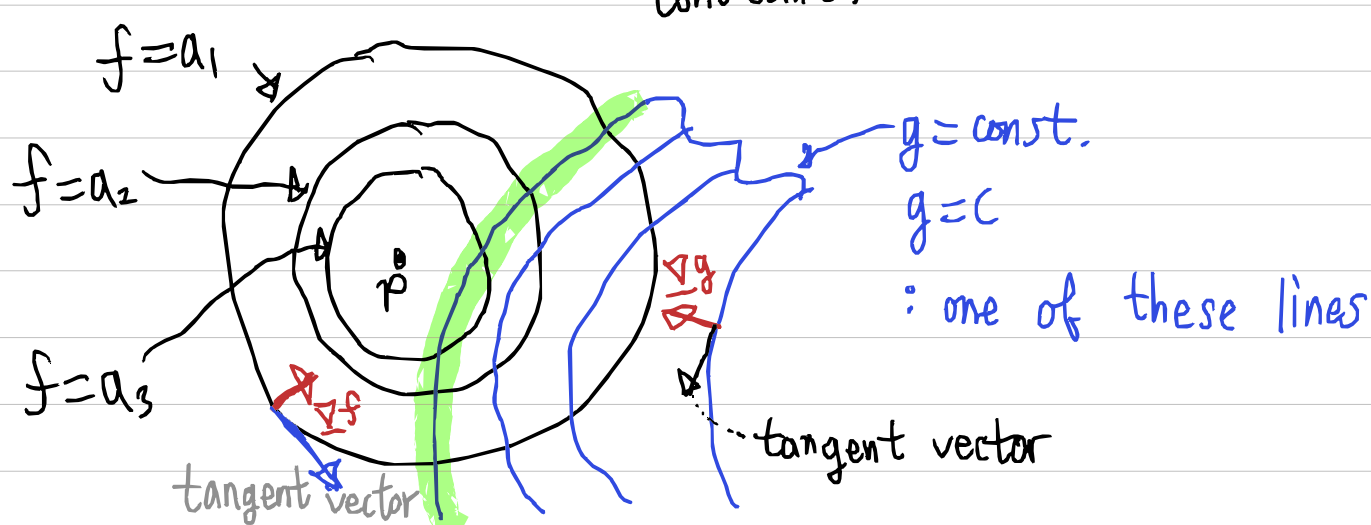
$$\nabla f(x_0, y_0) = 0$$

$$\nabla = (\partial_x, \partial_y)$$

Find stationary pt.s (x_0, y_0) of $f(x, y)$ subject to the constant

$$g(x, y) = C.$$

constant.

 f is extremized on $g=c$ at p

where

$$\nabla f = \lambda \nabla g$$

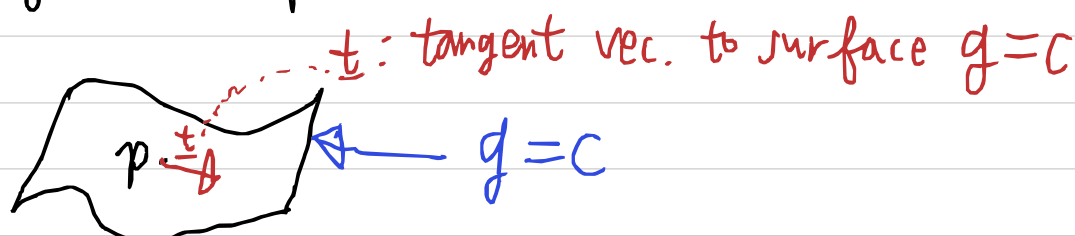
$$\nabla f - \lambda \nabla g = 0$$

 λ : Lagrange multiplier

	Variables	Eqn.s
Count	3	3

* Higher dim.s extremize $f(x_1, \dots, x_d)$ s.t.

$$g(x_1, \dots, x_d) = C.$$

→ defines a $(d-1)$ dimensional surface, Σ ,a.k.a., the constraint surface.Condition: $\underline{t} \cdot \nabla f|_p = 0$ for $\forall$ vec. $\underline{t}$ tangent to the surface $g=c$ at p .You want to find a pt. p s.t. if you take the directional deriv. of f , then

$$(\underline{t} \cdot \nabla f)$$

$$\underline{t} \cdot \nabla f|_p = 0.$$

(2)

Eqn. that one needs to solve:

If $\underline{\nabla} f|_p = \lambda \underline{\nabla} g|_p$ & $\underline{t}$ is tangent to Σ at p ,

then $\underline{t} \cdot \underline{\nabla} f|_p = \lambda \underline{t} \cdot \underline{\nabla} g|_p = 0$.

Generalize to multiple constraints.

If we want to extremize $f(x_1, \dots, x_d)$ s.t.

$$\left. \begin{array}{l} g_1(\underline{x}) = c_1 \\ g_k(\underline{x}) = c_k \end{array} \right\} \quad k \leq d-1$$
 → define a $(d-k)$ -dimensional surface Σ

Solve $\underline{\nabla} f - \lambda_1 \underline{\nabla} g_1 - \lambda_2 \underline{\nabla} g_2 - \dots - \lambda_k \underline{\nabla} g_k = 0$

Eq. (*)

	Variables	Eqn.s
Count	$d+k$	$d+k$

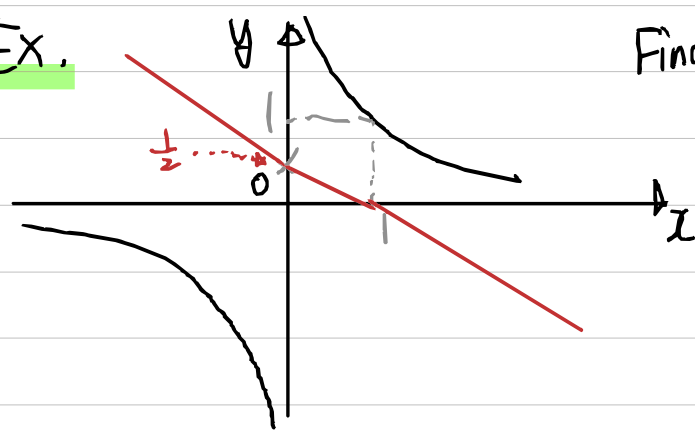
Why?

Want $\underline{t} \cdot \underline{\nabla} f|_p = 0 \quad \forall \underline{t} \text{ tangent to } \Sigma \text{ at } p.$

Eq. (**)

Claim. If (*) holds, then (**) holds.

Ex.



Find closest approach of

$$\begin{cases} xy = 1 \\ x + 2y = 1 \end{cases}$$

As close to a straight line as possible.

Seek x_1, y_1, x_2, y_2 s.t. $\begin{cases} x_1 y_1 = 1 \\ x_2 + 2y_2 = 1 \end{cases}$

Extremize $f(x_1, y_1, x_2, y_2) = (x_1 - x_2)^2 + (y_1 - y_2)^2$

s.t. $\begin{cases} x_1 y_1 = 1 \\ x_2 + 2y_2 = 1 \end{cases}$

Use Lagrange Multipliers:

$$\underline{\nabla} f - \lambda \underline{\nabla}(x_1 y_1) - \mu \underline{\nabla}(x_2 + 2y_2) = 0$$

Some $\underline{\nabla} u = 0$, $u = f - \lambda x_1 y_1 - \mu(x_2 + 2y_2)$

$d=4$ grad. operator

$$\begin{cases} x_1 y_1 = 1 \\ x_2 + 2y_2 = 1 \end{cases}$$

$$0 = \partial_{x_1} u, \quad 0 = \partial_{x_2} u$$

$$0 = \partial_{y_1} u, \quad 0 = \partial_{y_2} u$$

(3)

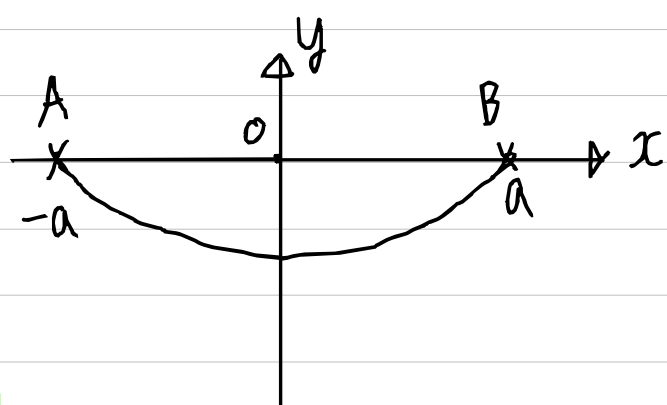
Show that $(x_1, y_1) = (\sqrt{2}, \frac{1}{\sqrt{2}})$

$$(x_2, y_2) = \left(\frac{1 + 3\sqrt{2}}{5}, \frac{4 - 3\sqrt{2}}{10} \right)$$

$$S_{\min} = \frac{1}{\sqrt{5}} (2\sqrt{2} - 1)$$

Minimum distance b/t them.

EX. We find shape of a heavy uniform rope forms when suspended b/t 2 pts at the same height.



Find $y(x)$ s.t. the

length is fixed $= L$,

& grav. PE of rope is

minimized.

mass per unit length $= \rho$

$$L = \int_A^B ds = \int_{-a}^a \sqrt{1 + (y')^2} dx$$

$$U(y) = \int_A^B g y \rho ds = g \rho \int_{-a}^a y \sqrt{1 + (y')^2} dx$$

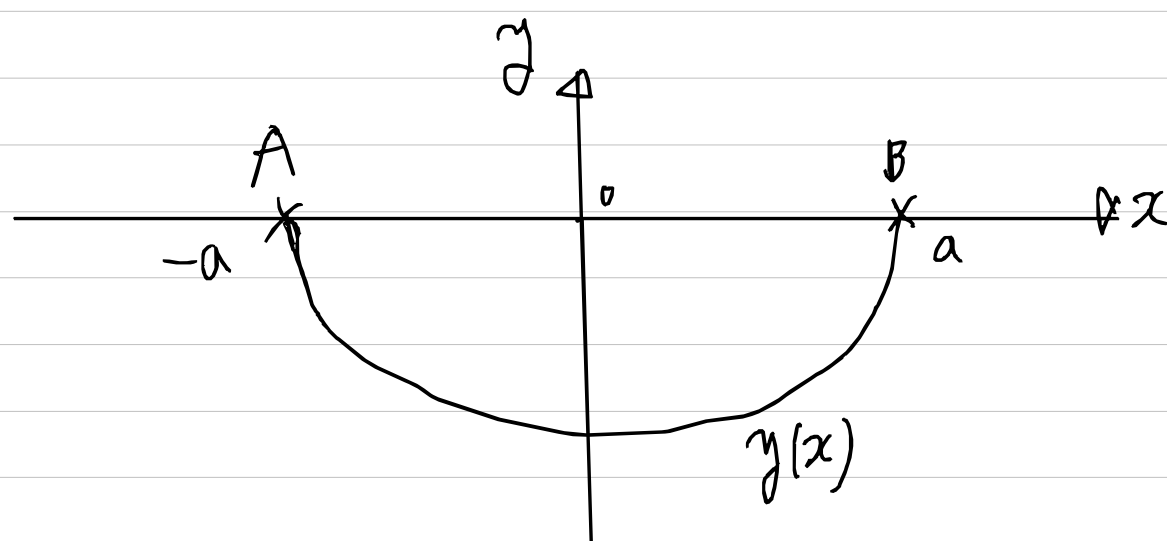
Extremize.

$$\begin{aligned} I[y] &= g \rho \int_{-a}^a y \sqrt{1 + (y')^2} dx - \lambda \int_{-a}^a \sqrt{1 + (y')^2} dx \\ &= \int_{-a}^a \mathcal{L}(y, y') dx \end{aligned}$$

Can use the Beltrami Identity

$$\mathcal{L} - y' \frac{\partial \mathcal{L}}{\partial y'} = 0$$

Example of constrained optimization



A rope of fixed length L

hangs b/t 2 fixed pt.s A & B.

*PE of the rope:

$$U[y] = \rho g \int_A^B y \, ds = \rho g \int_{-a}^a y \sqrt{1+y'^2} \, dx$$

$$L = L[y] = \int_A^B ds = \int_{-a}^a \sqrt{1+y'^2} \, dx$$

Find stationary point fn. y of

$$I[y] = U[y] - \lambda L[y]$$

↑
Lagrange Multiplier

$$I[y] = \int_{-a}^a \mathcal{L}(y, y') \, dx$$

$$\mathcal{L} = \rho g y \sqrt{1+y'^2} - \lambda \sqrt{1+y'^2}$$

doesn't depend on x ; we can use Beltrami:

$$\mathcal{L} - y' \frac{\partial \mathcal{L}}{\partial y'} = C$$

$$\Rightarrow \rho g y \sqrt{1+y'^2} - \lambda \sqrt{1+y'^2} - y' \left[\frac{\rho g y \cdot (2y')}{2\sqrt{1+y'^2}} - \frac{\lambda}{2\sqrt{1+y'^2}} \cdot 2y' \right] = C$$

$$\sqrt{1+y'^2} (\rho g y - \lambda) - \frac{(\rho g y - \lambda) y'^2}{\sqrt{1+y'^2}} = C$$

$$(1+y'^2 - y'^2) (\rho g y - \lambda) = C \sqrt{1+y'^2}$$

$$\frac{(\rho g y - \lambda)^2}{\eta(x)} = C^2 (1+y'^2)$$

$$\text{Let } \eta(x) = \rho g y(x) - \lambda,$$

$$\eta' = \rho g y'$$

$$\eta^2 = C^2 \left(1 + \frac{\eta'^2}{\rho^2 g^2} \right)$$

$$(\eta')^2 = \frac{\rho^2 g^2}{C^2} (\eta^2 - C^2)$$

$$\eta' = \pm \frac{\rho g}{C} \sqrt{\eta^2 - C^2}$$

$$= \pm \rho g \sqrt{\frac{\eta^2}{C^2} - 1}$$

(2)

$$x > 0 \rightarrow \eta' > 0$$

$$\int \frac{d\eta}{\sqrt{\frac{\eta^2}{c^2} - 1}} = \int e g \, dx$$

Let $\eta = C \cosh q$ $\rightarrow$ change variable

$$d\eta = C \sinh q \, dq$$

$$\int \frac{C \cancel{\sinh q} \, dq}{\cancel{\sinh q}} = \int e g \, dx$$

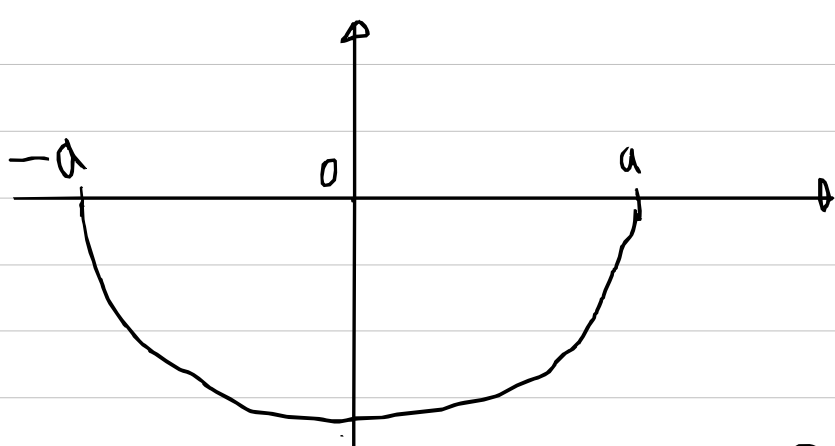
$$C q = e g x + d$$

$$\eta = C \cosh \left(\frac{e g x + d}{C} \right)$$

y is even $\rightarrow$ then, η is also even $\rightarrow$
 $d=0$

$$y = \frac{\eta + \lambda}{e g} = \frac{C}{e g} \cosh \left(\frac{e g x}{C} \right) + \frac{\lambda}{e g}$$

Solve for C & λ in terms of L & a .



$$\text{when } x=a: 0 = y(a) = \frac{C}{e g} \cosh \left(\frac{e g a}{C} \right) + \frac{\lambda}{e g}$$

$$L = \int_{-a}^a \sqrt{1 + y'^2} \, dx = \int_{-a}^a \left(\cosh \frac{e g x}{C} \right) dx$$

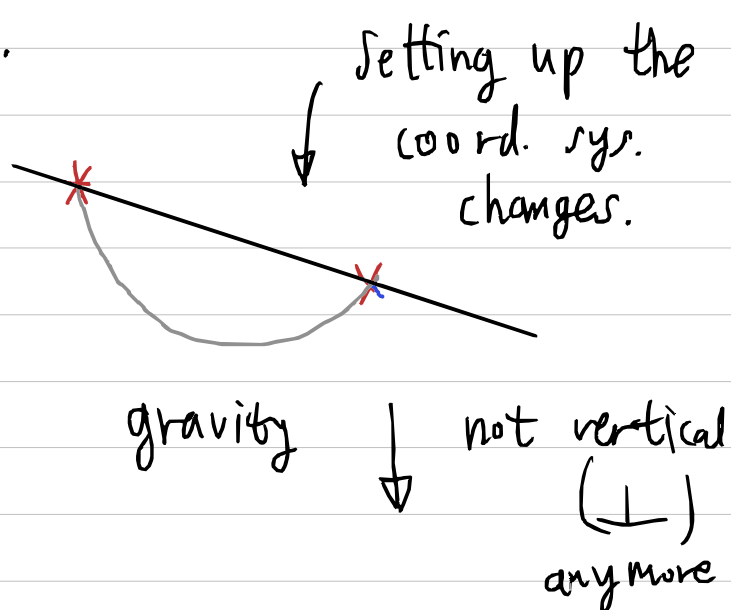
$$\boxed{y' = \sinh \left(\frac{e g x}{C} \right)} = \left[\frac{C}{e g} \sinh \left(\frac{e g x}{C} \right) \right]_{-a}^a$$

$$= \frac{2C}{e g} \sinh \left(\frac{e g a}{C} \right)$$

We solve $L = \frac{2C}{e g} \sinh \left(\frac{e g a}{C} \right)$ for C .

& solve $\otimes$ for λ .

(EX) $\rightarrow$



§ 4. Variational Methods for solving Schrödinger Eq.

③

§4.1. Variational formulation of Energy Eigenvalue Problem.

$$\hat{H} \Psi_i(q) = E_i \Psi_i(q)$$

Let's consider $E[\Psi] = \int_Q \Psi^* \hat{H} \Psi dq$

$$Q \leftrightarrow \mathbb{R}^{3 \times 10^{23}}$$

N.B. : $E[\Psi_i] = \int_Q \Psi_i^* E_i \Psi_i dq = E_i$

Vary $E[\Psi]$ subject to constraint that

$$N[\Psi] = 1 \quad \text{where} \quad N[\Psi] = \int_Q \Psi^* \Psi dq.$$

Define :

$$I[\Psi] = E[\Psi] - \epsilon N[\Psi]$$

Lagrange Multiplier

$$I[\Psi] = \int (\Psi^* \hat{H} \Psi - \epsilon \Psi^* \Psi) dq$$

$$\Psi \rightarrow \Psi + \delta \Psi$$

$$\begin{aligned} \delta I[\Psi, \delta \Psi] &= \int \left(\delta \Psi^* \hat{H} \Psi + \Psi^* \hat{H} \delta \Psi - \epsilon \delta \Psi^* \Psi - \epsilon \Psi^* \delta \Psi \right) dq \\ &\quad + o(\delta \Psi^2) \end{aligned}$$

$$= \int \delta \Psi^* (\hat{H} \Psi - \epsilon \Psi) dq + \int (\Psi^* \hat{H} \delta \Psi - \epsilon \Psi^* \delta \Psi) dq$$

$\hat{H}$ is S-A

$$\hookrightarrow \int \Psi^* \hat{H} \delta \Psi dq = \int (\hat{H} \Psi^*) \delta \Psi dq$$

$$\text{So, } \delta I = \int (\hat{H} \Psi - \epsilon \Psi) \delta \Psi^* dq$$

$$+ \int (\hat{H} \Psi^* - \epsilon \Psi^*) \delta \Psi dq$$

$$= 0 \quad \textcircled{0}$$

Make this vanish $\rightarrow E - E$

(4)

TWO PARTS: REAL & IMAGINARY

$$\delta I[\Psi, i\delta\Psi] = -i \int (\hat{H}\Psi - E\Psi) \delta\Psi^* dq + \int (\hat{H}\Psi^* - E\Psi^*) i\delta\Psi dq$$

= 0 (2)

$\downarrow$
 E-LE

$$i(2) + (1) \rightarrow \int (\hat{H}\Psi - E\Psi) \delta\Psi^* dq = 0$$

$$\hookrightarrow \hat{H}\Psi - E\Psi = 0$$

The stationary function of I is an Energy eigenfunction.

Make an Estimate $\rightarrow$ Accuracy needed
 EX $\hookrightarrow$ Suppose $\tilde{\Psi}_i$ is a guess at the i -th eigen-fn. Ψ_i :

$$\tilde{\Psi}_i = \Psi_i + \underbrace{\Delta\Psi_i}_{\text{Error}} \rightarrow \hat{H}\Psi_i = E_i\Psi_i$$

$$I[\tilde{\Psi}_i] = I[\Psi_i] + \underset{\text{Order}}{O(\Delta\Psi_i^2)}$$

$$\hookrightarrow E[\tilde{\Psi}_i] - \cancel{E_N[\tilde{\Psi}_i]} = E[\Psi_i] - \cancel{E_N[\Psi_i]} + O(\Delta\Psi_i^2)$$

$\parallel$ E_i

$$E[\tilde{\Psi}_i] = E_i + O(\Delta\Psi_i^2)$$